



CHAPTER III

Metals





3.1 Mining and processing of metals

Ore

The mineral deposit contains an economically recoverable amount of a metal.

Gangue

The waste material of the rock formation.



The yield for metal extracting is quite low; ores typically possess less than 1% of the desired metal.



Igneous rocks are formed from the solidification of molten mass following volcanic activity.

Granite, feldspar, and quartz (alkali, alkaline earths, gold, platinum and chromium)

Sedimentary rocks are formed through compaction of small grains deposited as sediment in a riverbed or sea.

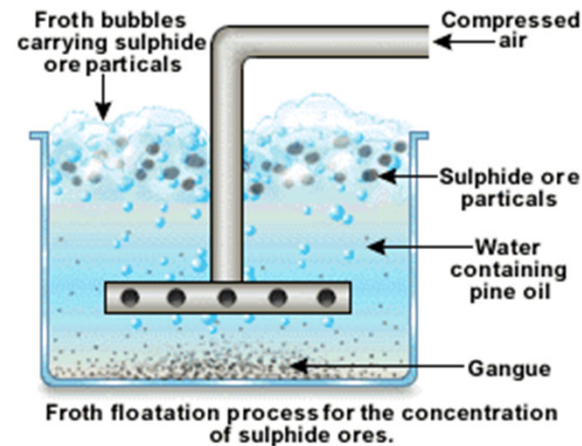
Limestone, sandstone and dolomite (copper, iron, zinc, lead, nickel, molybdenum and gold)

Metamorphic rocks are formations that have changed composition and morphology over time through the influence of temperature and pressure.

Marble, slate and gneiss (alkali, alkaline earths, gold, platinum and chromium)

Flotation Process

For sulphides ores of copper, zinc and lead concentration.



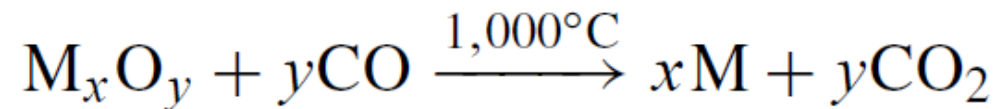
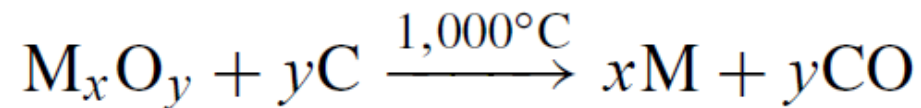
A strong current of air is blown through the suspension, producing a heavy froth or foam on the surface. The metal sulphide is wetted by the oil but the gangue is not and the sulphide-oil mixture is carried to the surface by films of oil. The froth is skimmed off, the gangue settles down on the bottom or remains underneath the froth. By this floatation method it is possible to concentrate over 90% of a sulphite ore to 1/10 of its original bulk.



Once the bulk ore is concentrated from its many impurities, the metal (M) is reduced from its natural oxides.

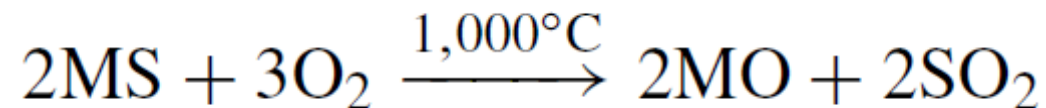
Smelting

The ore is heated in the presence of a reducing agent such as a carbonaceous material, or carbon monoxide



Pyrometallurgy

The treatment of ores at high temperature to convert ore minerals to raw metals, or intermediate compounds for further refining.

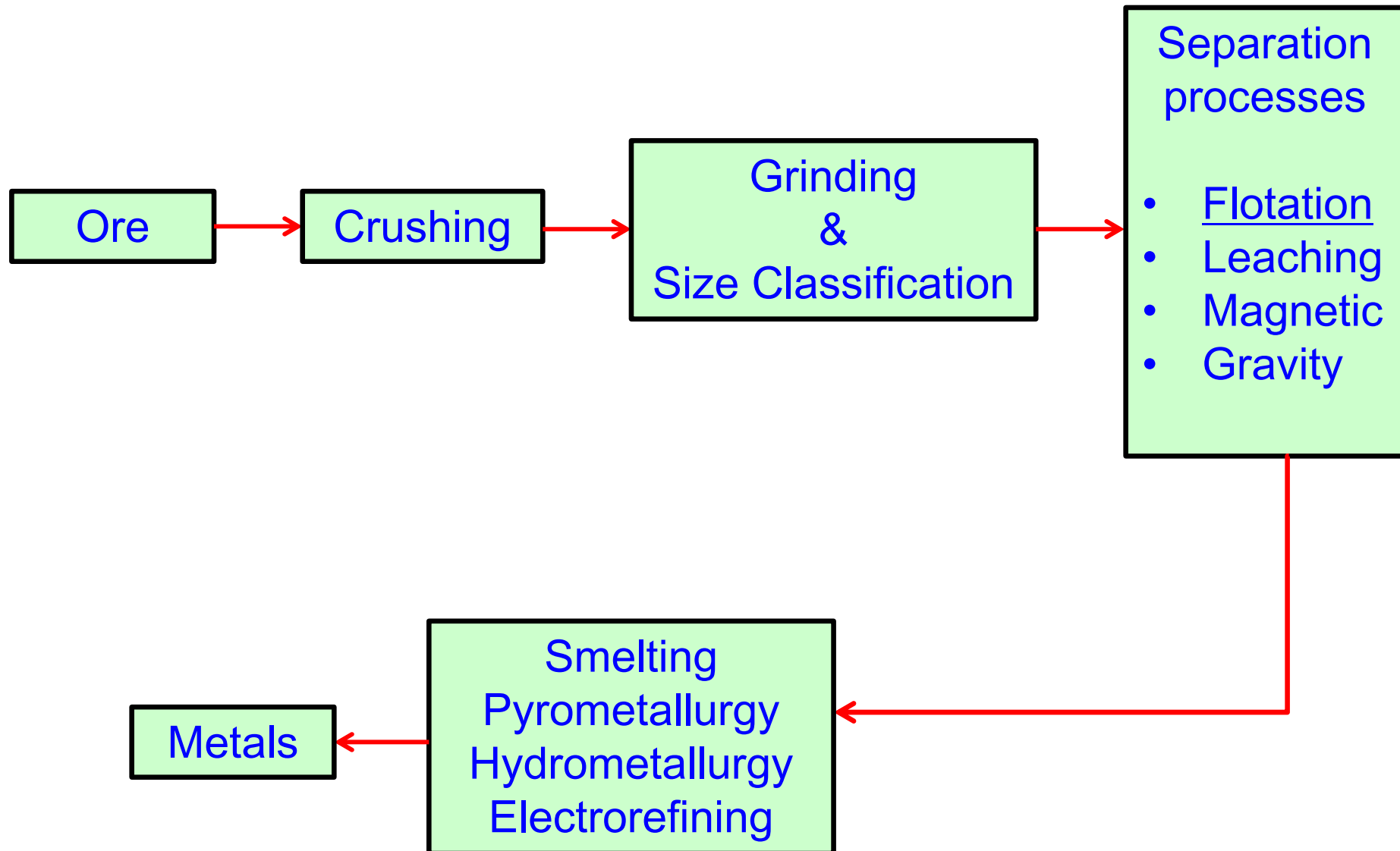


Hydrometallurgy or electrorefining

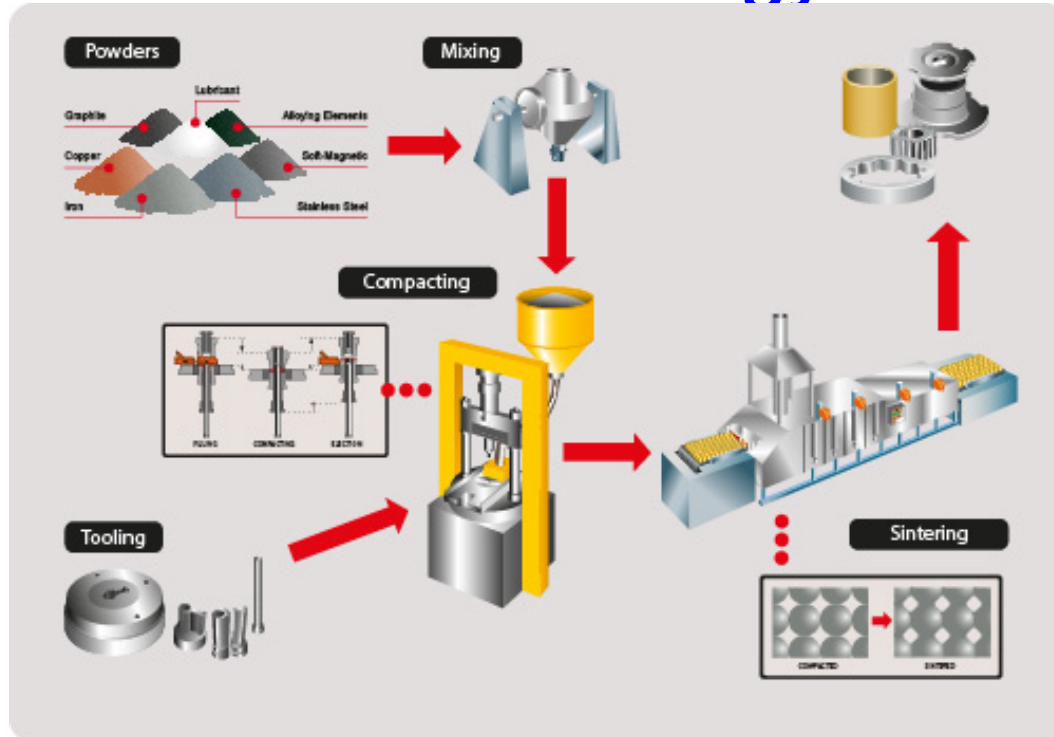
The processes use of aqueous solutions to extract metals or compounds from their ores.



Extractive metallurgy

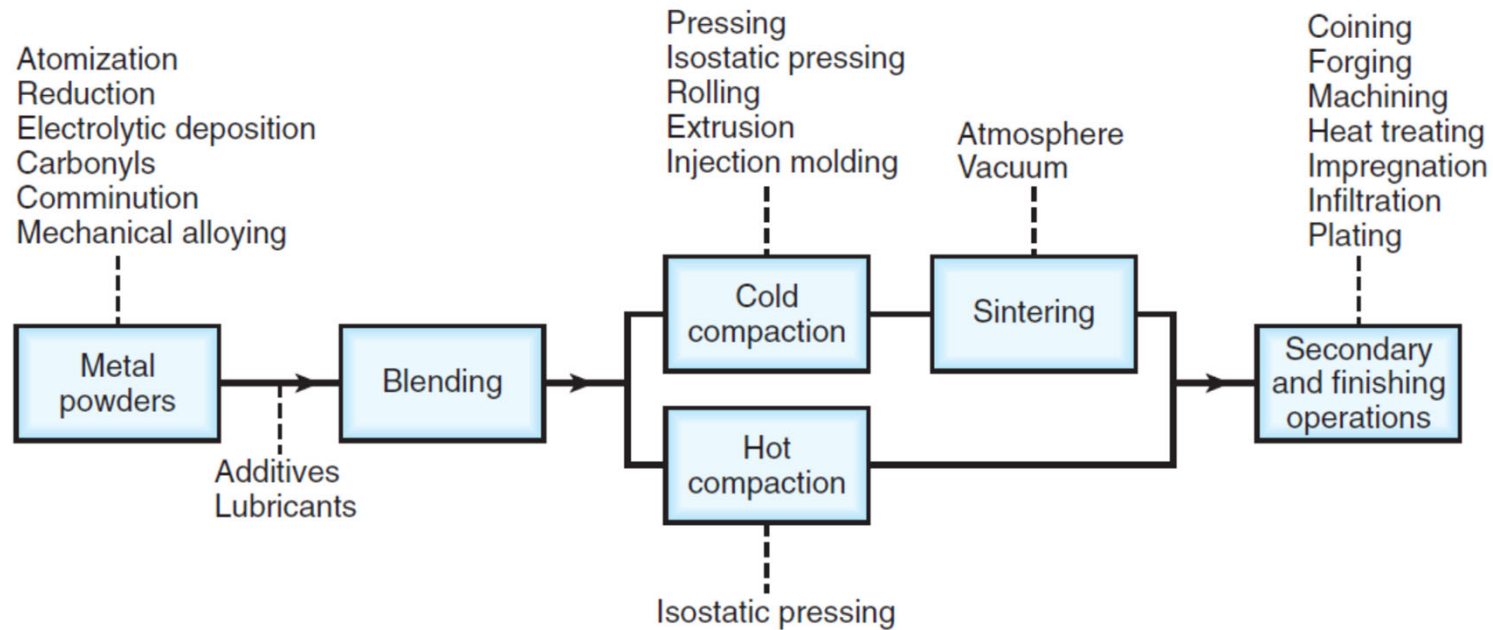


3.1.1 Powder Metallurgy



Powder metallurgy process involves metal powders compacted into desired shapes and sintered to form a solid piece

- Placing metal powder(s) into a closed metal cavity, or die, repeatedly compacting under high pressure (typically 200–300+MPa).
- Sintering in a furnace to yield a metal with desired porosity and hardness characteristics, which results in the welding together of powder particles to form a mechanically strong finished material.



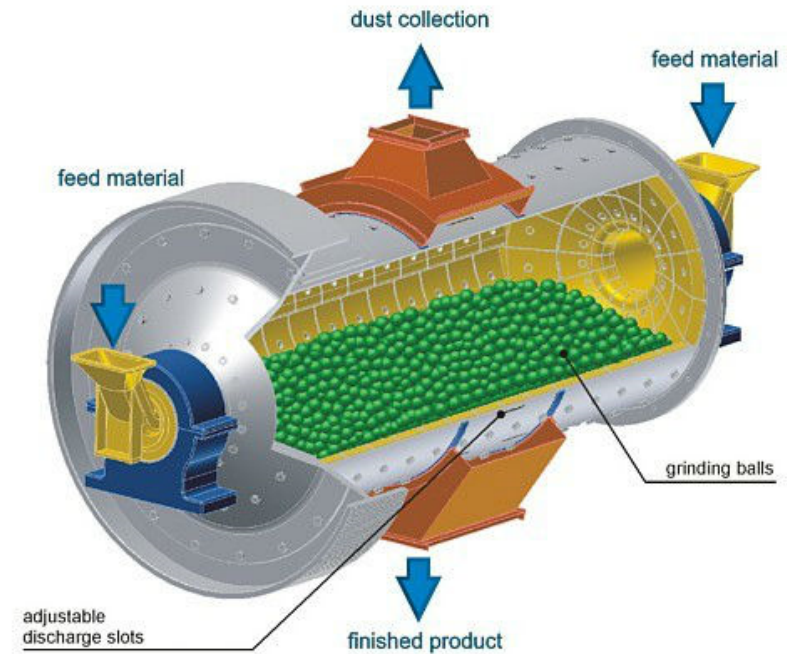
Powder-metallurgy process consists of:

- A. Powder production**
- B. Blending**
- C. Compaction**
- D. Sintering**
- E. Finishing operations**



A. Powder Production

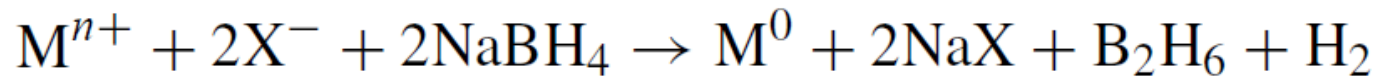
- Mechanical pulverization (粉碎)
 - For relatively brittle materials



Ball mills

- Chemical precipitation

-a reducing agent such as NaBH_4 is added to the metal ions in solution



- Thermolysis (carbonyl process)

-By passing CO gas over impure metals at a low temperature, $\text{M}(\text{CO})_4$ gas is formed, leaving the impurities behind. The carbonyl decomposes upon heating to a high temperature, forming the pure metal powder.

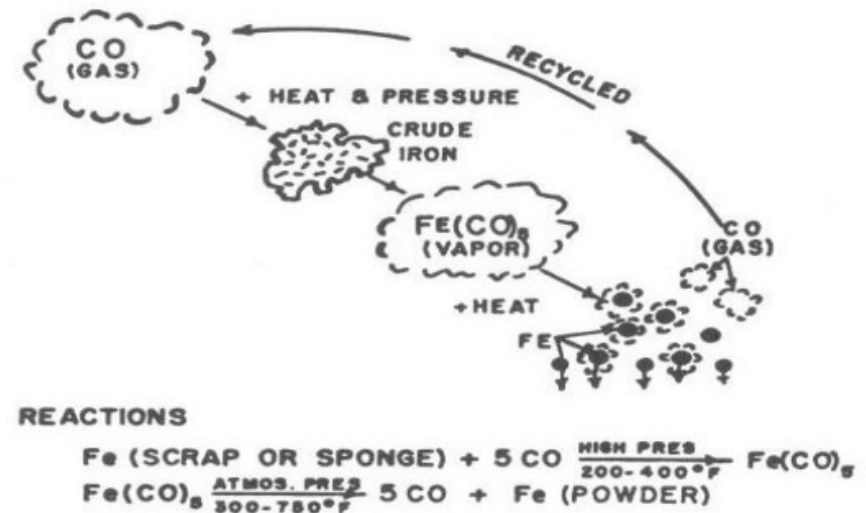
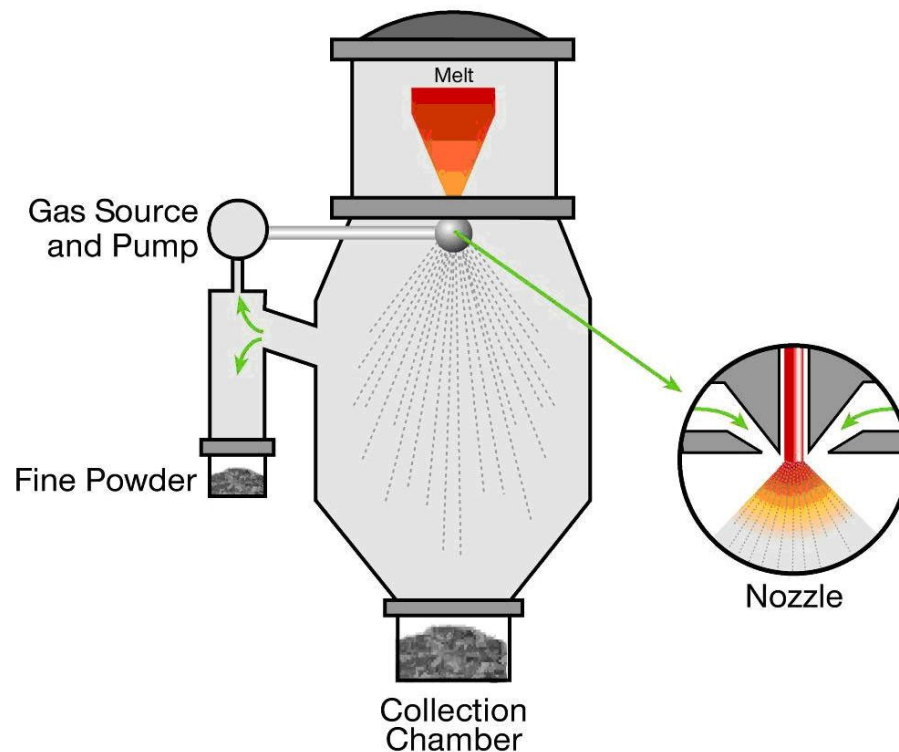


Fig. 2.6. Schematic drawing of decomposition using carbonyl process

- **Electrolysis**

- to produce metallic powders through redox reactions at electrode surfaces

- **Atomization**



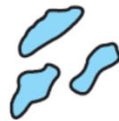
- This method involves the conversion of molten metal into a spray of droplets that solidify into powders. It is the most versatile and popular method for producing metal powders today, applicable to almost all metals, alloys as well as pure metals.

Particle Shape

- The choice of producing metal powders depends on the requirements of the end product.
- The microstructure, bulk and surface properties, chemical purity, porosity, shape, and size distribution of the particles depend on the particular process used.



Acicular (chemical decomposition)



Irregular rodlike (chemical decomposition, mechanical comminution)

(a) One-dimensional

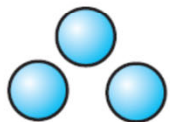


Flake (mechanical comminution)



Dendritic (electrolytic)

(b) Two-dimensional



Spherical (atomization, carbonyl (Fe), precipitation from a liquid)



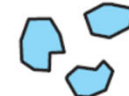
Irregular (atomization, chemical decomposition)



Rounded (atomization, chemical decomposition)



Porous (reduction of oxides)



Angular (mechanical disintegration, carbonyl (Ni))

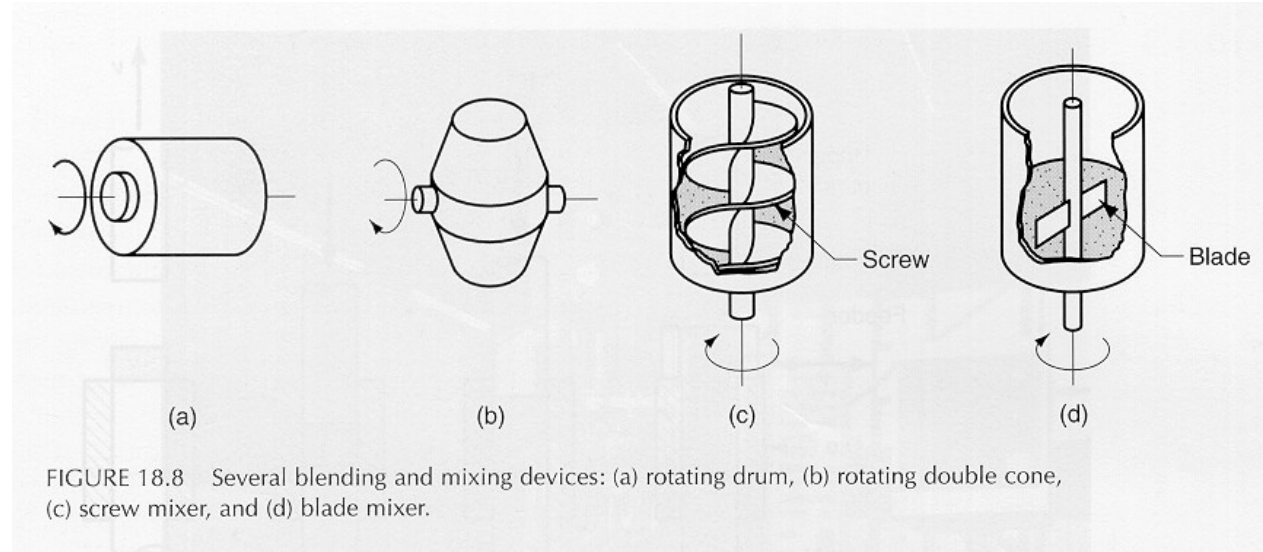
(c) Three-dimensional

Properties of metal powders that affect their behavior in processing are:

- Flow properties
- Compressibility
- Density

B. Blending

Blending (mixing) powders is the next step in powder-metallurgy processing



Hazards

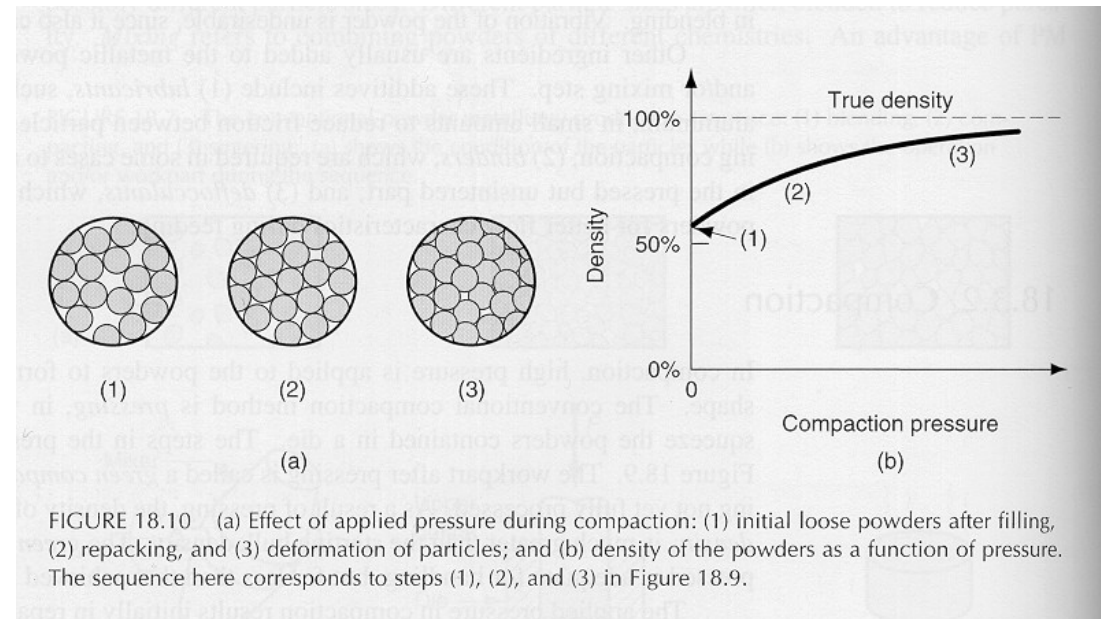
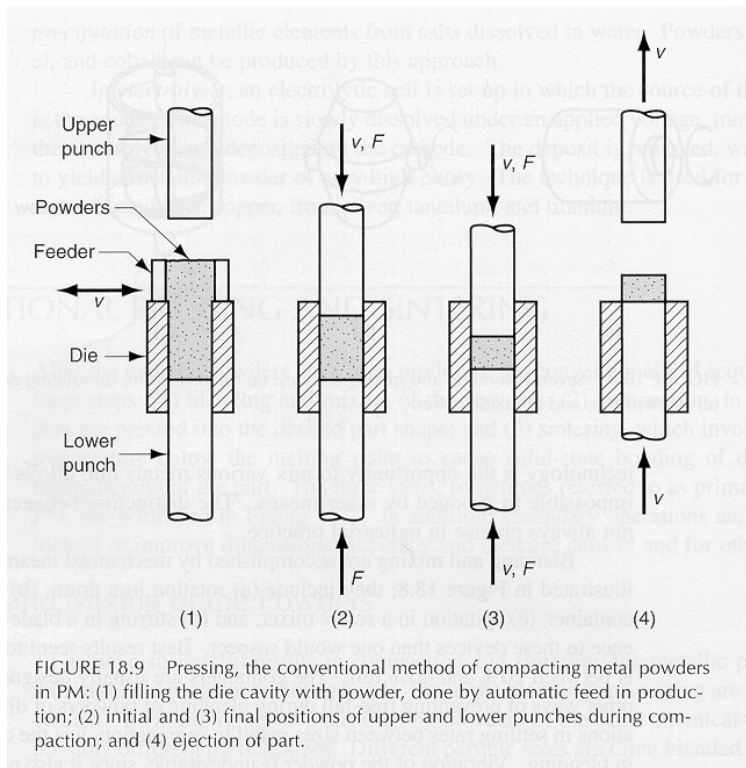
Metal powders can be explosive due to high surface area-to-volume ratio

Precautions include:

1. Grounding equipment
2. Preventing sparks and avoiding friction as a source of heat
3. Avoiding dust clouds, open flames, and chemical reactions

C. Compaction

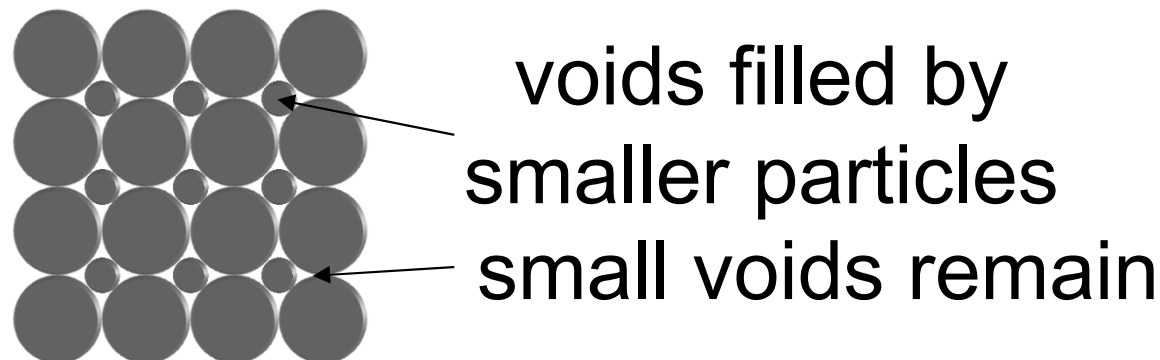
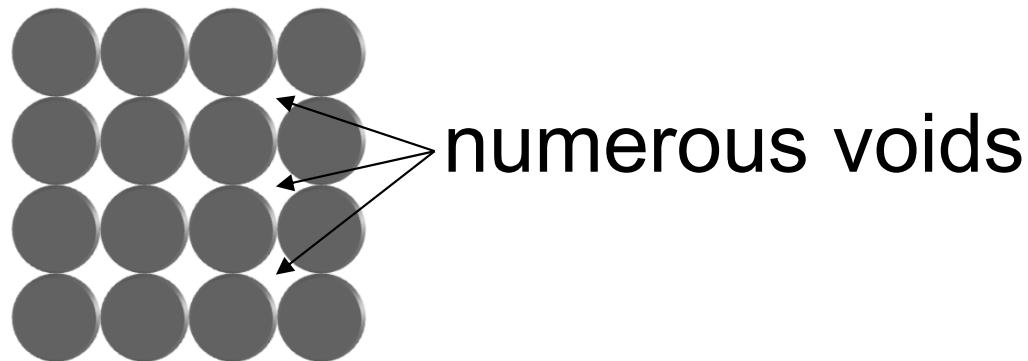
Compaction is the step where the blended powders are pressed into various shapes in dies.



Density

The density of the powder strongly influences the strength of the material obtained from compaction, which depends on the shape and porosity of individual micron-sized particulates.

- Mixing particles of different sizes allows decreased porosity and a higher packing ratio

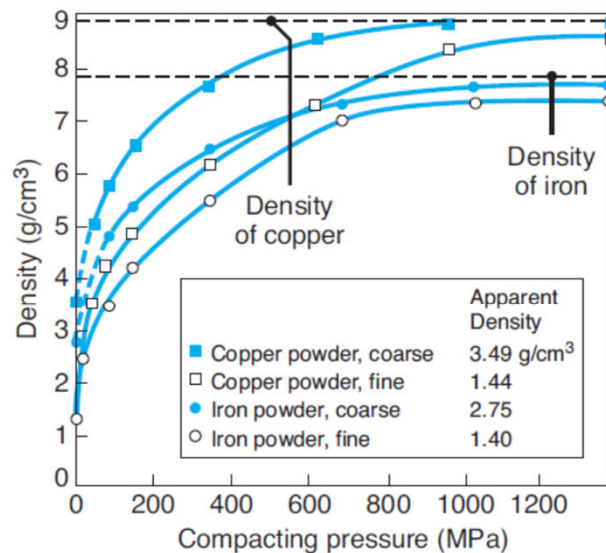




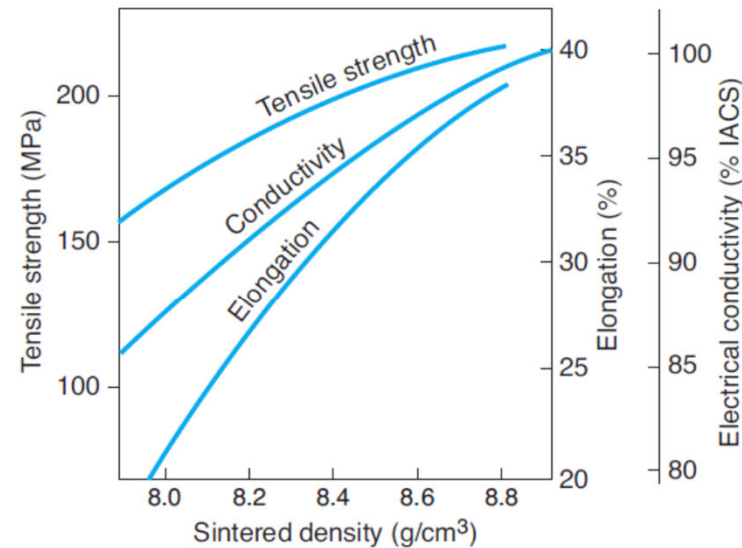
- Even if the metal particulates have the same diameter and are completely spherical, the actual packing density is typically on the order of 55–60%.
- This value may be improved by introducing nano-sized particles that will fill the voids among the larger particles.
- During subsequent high-temperature sintering, the larger particles will grow at the expense of the nanoparticles, leaving behind relatively small voids that are closed during the thermal treatment.



- Purposes of compaction are to obtain the required shape, density and particle-to-particle contact.
- Pressed powder is known as green compact (green density).
- Density depends on the pressure applied.
- Higher the density of the compacted part, the higher are its strength and elastic modulus.



(a)



(b)



- Compacting pressure required depends on the characteristics and shape of the particles, method of blending and lubricant.

Compacting Pressures for Various Powders

Metal	Pressure (MPa)
Aluminum	70–275
Brass	400–700
Bronze	200–275
Iron	350–800
Tantalum	70–140
Tungsten	70–140
Other materials	
Aluminum oxide	110–140
Carbon	140–165
Cemented carbides	140–400
Ferrites	110–165

It should be noted that for alloying metals such as copper and zinc, pressing is not necessary. The powders are simply placed in a mold and sintered, a process aptly referred to as loose-powder sintering.



Brief summary for compaction

- Blended powders are pressed together;
- The powder must flow easily into the die;
- Size distribution is an important fact:
 - They should not be all the same size;
 - Should be a mixture of large and small particles;
- The higher the density the higher the strength



Lubricant

A lubricant may also be mixed together for co-pressing with the composite powders, or to reduce friction between the powder mass and the surface of the die walls.

For Example:

- stearic acid (octadecanoic acid),
- stearin,
- zinc stearate,
- other waxy organic compounds (*e.g.*, palmates)

For this purpose, it is often sufficient to apply lubrication to the walls of the die, rather than introducing the organic compound to the metallic powders. If a significant amount of organic residue is left following compaction, it will be removed upon sintering, leaving behind large voids that will greatly detract from the finished material's overall strength.

Equipment

- Uses 100-300 ton press;
- Selection of the press depends on the part and the configuration of the part.



Fig: A 7.3 MN (825 ton) mechanical press for compacting metal powder.



Isostatic Pressing

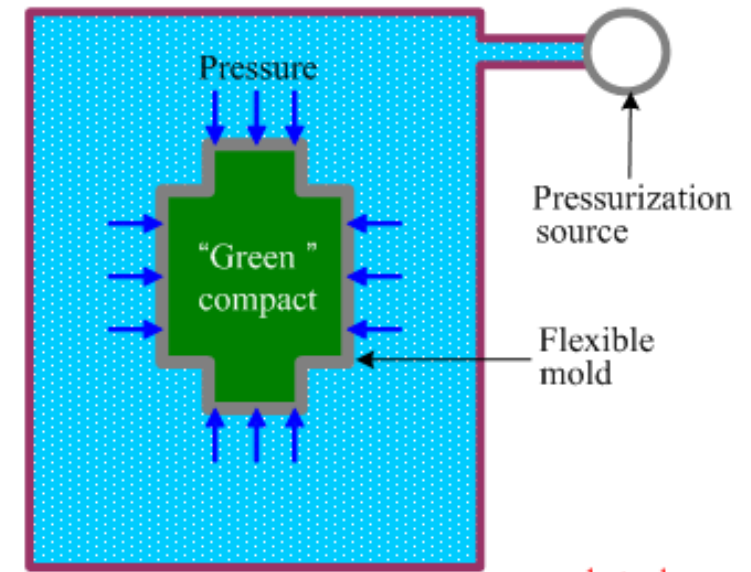
- Isostatic pressing is the powder compaction method involving applying pressure from multiple directions through a liquid or gaseous medium surrounding the compacted part.
- Cold isostatic pressing (CIP) is conducted at room temperature.
- Hot isostatic pressing (HIP) involves isostatic pressing conducted at increased temperature.

- Cold isostatic Pressing (CIP)

- A flexible (commonly [poly-urethane](#)) mold immersed in a pressurized liquid medium (commonly water) is used in the cold isostatic pressing method.

- Uses pressures up to 150 KSI.

- Typical application is automotive cylinder liners.

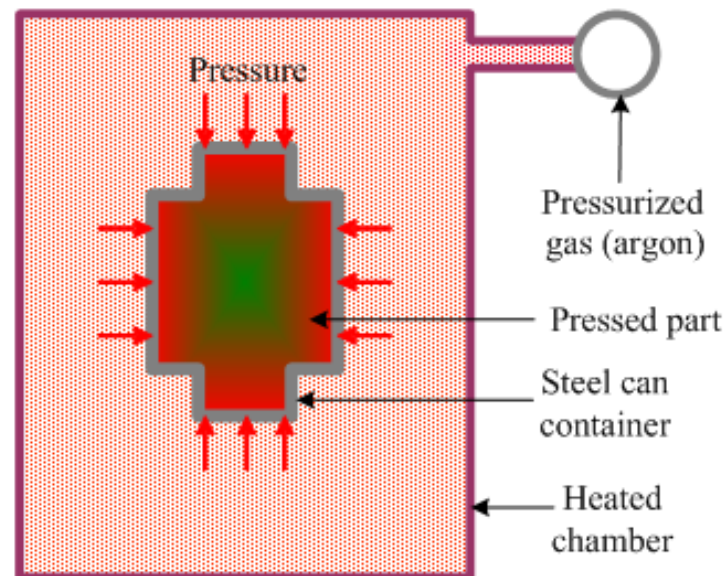


- The CIP method has the following advantages as compared to the [die cold pressing](#) method:

- better uniformity of compaction;

- more complex forms (for example long thin-walled tubes) may be compacted;

- Hot Isostatic pressing
 - Container is made of high-melting-point sheet metal
 - Uses an inert gas as the pressurizing medium
 - Common conditions for HIP are 15KSI at 2000F
 - HIP combines pressing and [sintering](#).
 - Mainly used for super alloy casting (aircraft and aerospace industries).





Advantages of HIP are:

- Produces fully dense compacts of uniform grain structure and density
- Handling larger parts

Limitations of HIP:

- Wider dimensional tolerances
- Higher equipment cost and production
- Small production quantities



Metal Injection Molding (MIM)

- MIM uses very fine metal powders blended with a polymer
- The molded greens are then placed in a furnace to burn off the plastics
- Advantages of injection molding
 - Produces complex shapes
 - Mechanical properties are nearly equal to those of wrought products

Forging

Products from powder forging (PF) are fully dense, good surface finish, good dimensional tolerances, and a uniform and fine grain size.

Rolling

- Also called roll compaction.
- Metal powder is fed into the roll gap in a two-high rolling mill and compacted into a continuous strip.

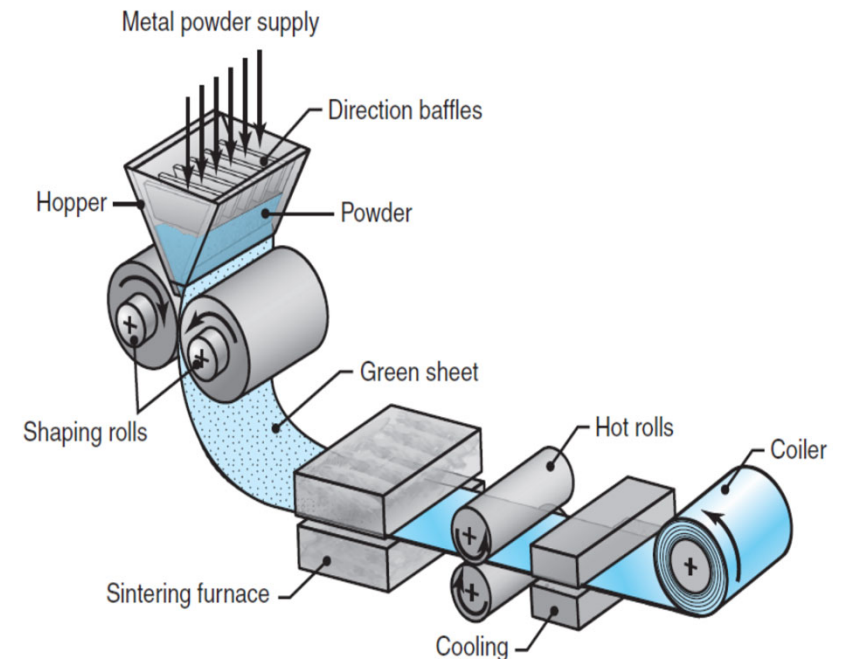




Table I. Various Methods for Forming Powders into Compacts

A. Forming of powders by application of pressure

1. Unidirectional pressing
 - a. Single action pressing
 - b. Double action pressing
 1. Floating die body
 2. Mechanically moved die body
2. Isostatic pressing
3. Powder rolling
4. Stepwise pressing
5. Powder extrusion
 - a. Powder direct
 - b. Powder canned
6. Powder swaging
7. Explosive compacting
8. Powder forging

B. Forming of powders without application of pressure

1. Loose powder sintering in a mold
 2. Slip casting
 3. Vibratory compacting
-

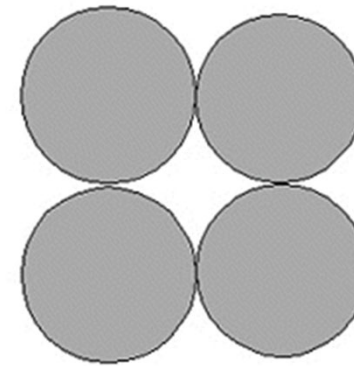


D. Sintering

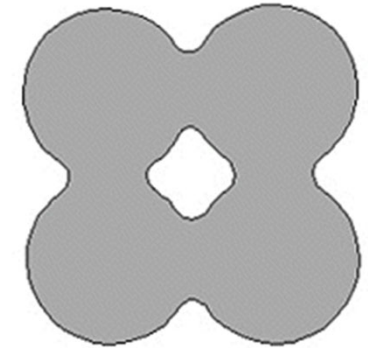
- Sintering - Green compacts are heated in a furnace to a temperature below melting point but high enough to allow bonding (fusion) of the individual particles.
- Improves the strength of the material.
- Proper furnace control is important for optimum properties.
 - Affecting mechanical properties are temperature, time, and processing history.
 - Porosity cannot be avoided completely due to voids remaining after compaction and gases evolve during sintering.

Continuous-sintering furnaces have 3 chambers:

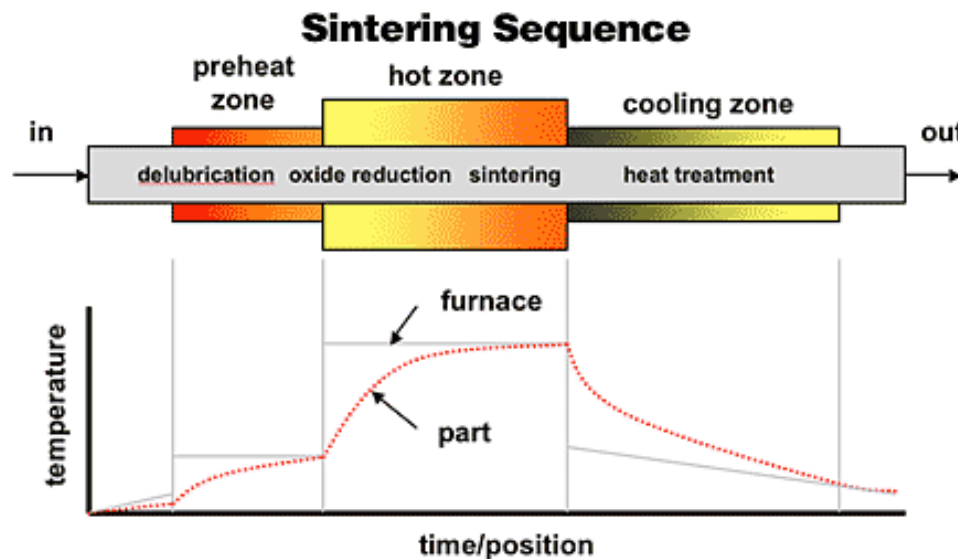
1. Preheat chamber
2. High-temperature chamber →
3. Cooling chamber



Representation of the initially unsintered powder particles.



Representation of Sintered Powder Particles



The diffusional processes that combine distinct powdered grains into one cohesive material.

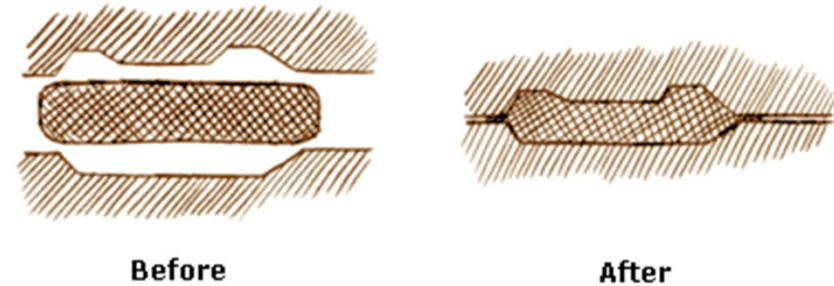
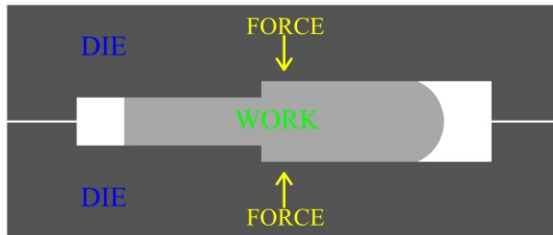


E. Finishing operations

A number of secondary and finishing operations can be applied after sintering, some of them are:

- Sizing
Cold pressing to improve dimensional accuracy.
- Coining
Cold pressing to press details into surface.
- Impregnation
Oil fills the pores of the part.
- Infiltration
Pores are filled with a molten metal.
- Heat treating, ;plating, painting.

SIZING OPERATION



COINING

Sizing:

- Sizing is done to refine dimensional accuracy.
- Pressures used are no more than the initial compacting pressure.
- Sized parts will be straighter, dimensional tolerances will be closer and surface finish will be improved.
- Little or no increase in density is achieved.

Coining:

- Pressures used are more than the initial compacting pressure.
- Increased part density in addition to improving dimensional accuracy.
- Increased mechanical properties.

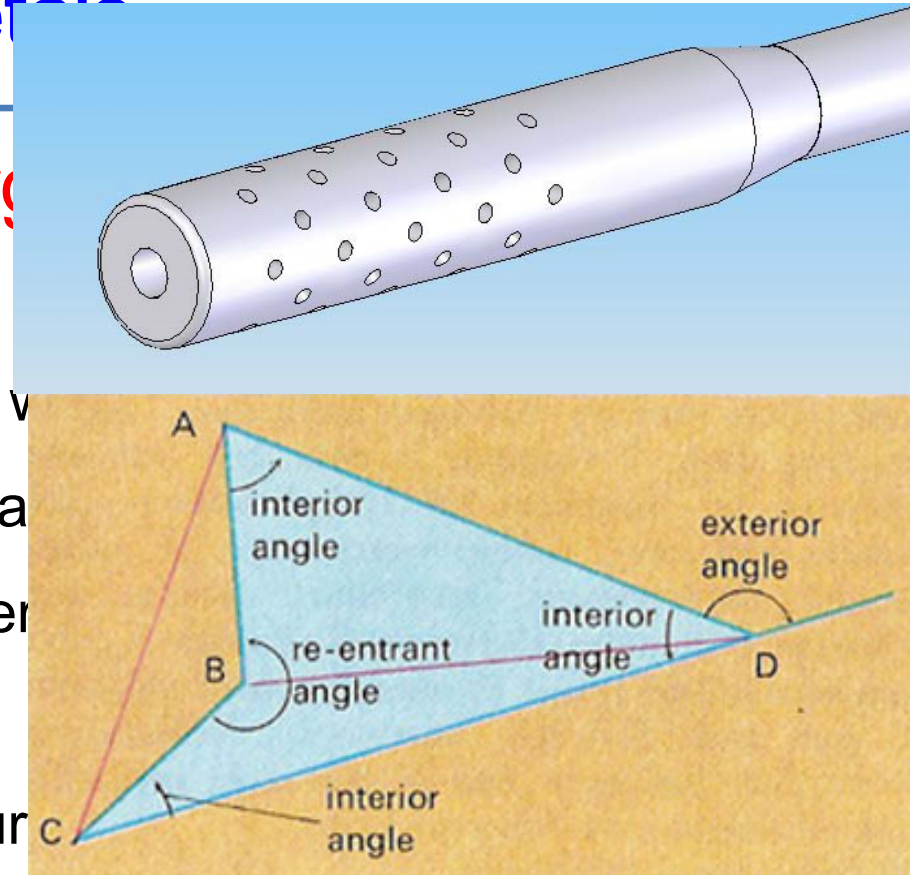


Advantages of powder metallurgy:

1. Lack of machining eliminates scrap losses.
2. Facile alloying of metals.
3. *In situ* heat treatment is useful for increasing the wear resistance of the finished material.
4. Facile control over porosity and density of the green and sintered material.
5. Fabrication of complex/unique shapes which would be impractical/impossible with other metalworking processes.
6. Rapid solidification process extends solubility limits, often resulting in novel phases.

Limitations of powder metallurgy

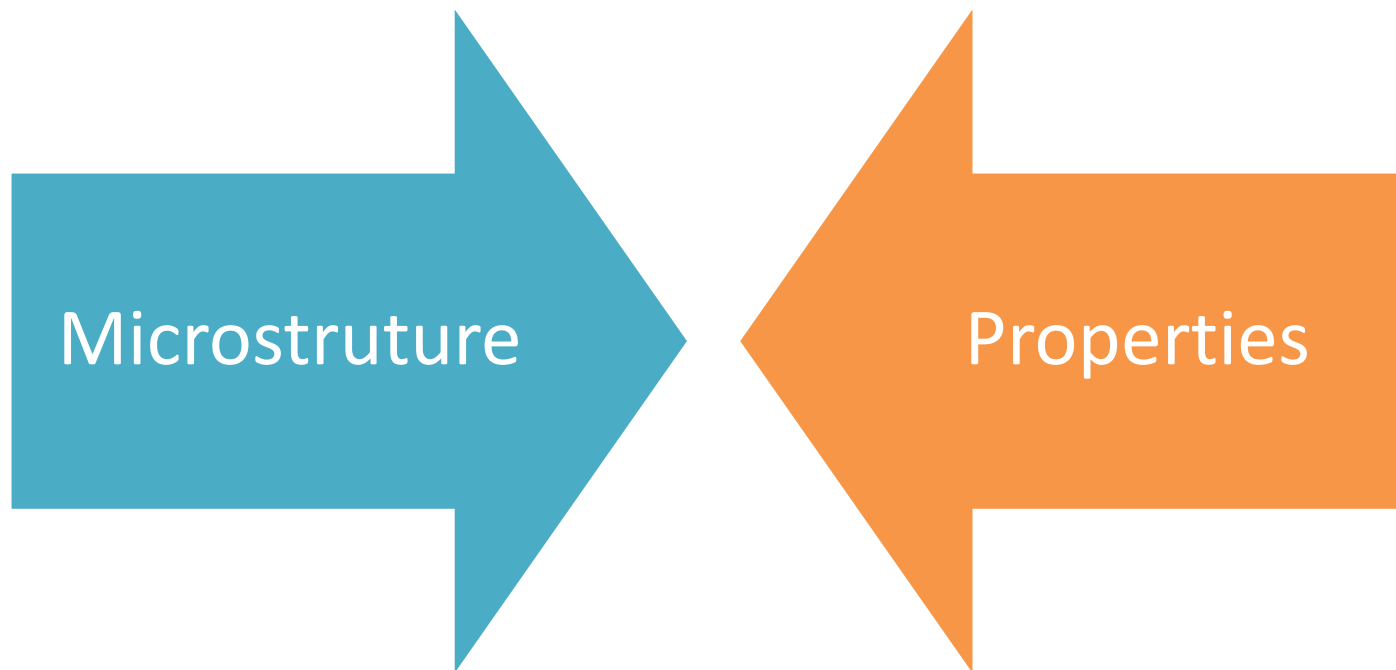
- Costs of powder production
- Limitations on the shapes and features with the process cannot produce re-entrant angles and radial holes in vertically pressed cylinders
- Pyrophoric risks
 - The extremely large exposed surface area to the bulk results in rapid oxidation upon exposure to air;
 - there is enhanced internal friction among the individual micron- or nano-sized individual particulates comprising the powder.
- Potential workforce health problems from atmospheric contamination of the workplace





3.2 Metallic structures and properties

The properties of any multiphase alloy depend upon the individual characteristics of the phases and how these phases are distributed in the microstructure.



- Changes in the microstructure
- Phase transitions
- Atomic packing

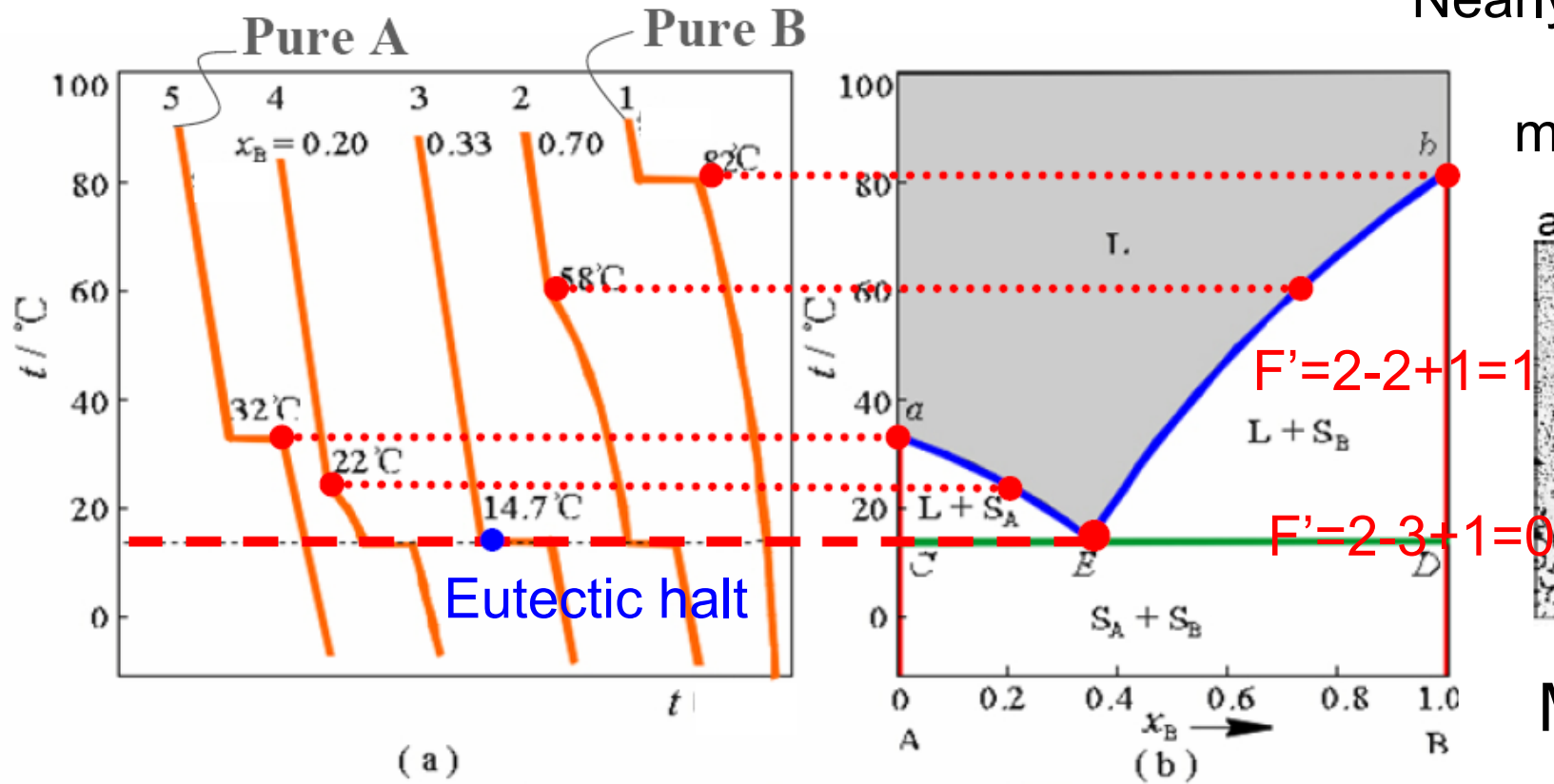


3.2.1 Phase behavior of iron-carbon alloys

- Type I alloys are completely miscible with one another in both liquid and solid states.
 - The different atoms that are arranged within a single lattice.
 - Solidification over a temperature range due to differing diffusion rates of the metals toward their preferred crystal arrangement
- Type II alloys are only miscible in the molten state, and will separate from one another upon cooling.
 - the alloying of metals or metals/nonmetals that are too dissimilar in their reactivities.

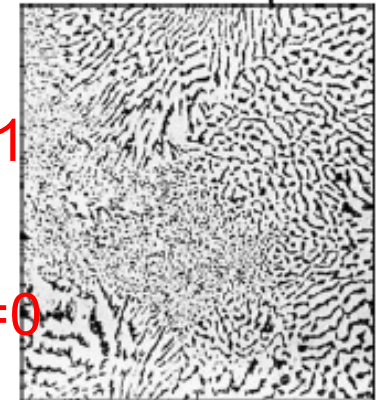


Binary eutectic alloy system

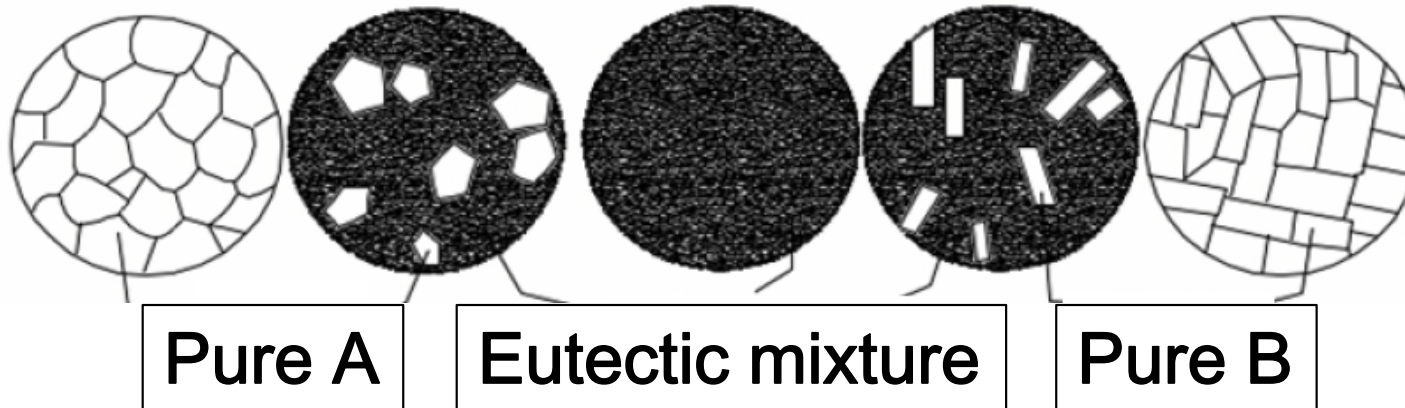


Nearly homogeneous mixture of microcrystals

at eutectic point

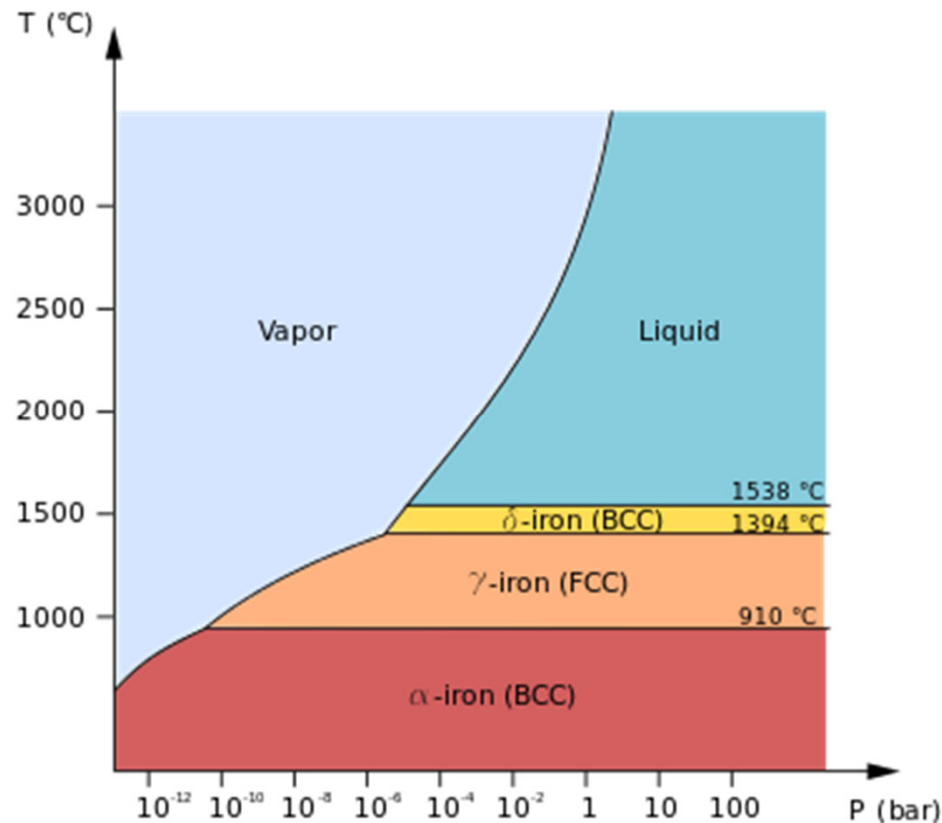


Microscopy
X-ray



Allotropes of pure iron

Pure iron exists as a variety of allotropes depending on the external temperature or pressure.



α-Fe (ferrite)

A BCC crystal structure

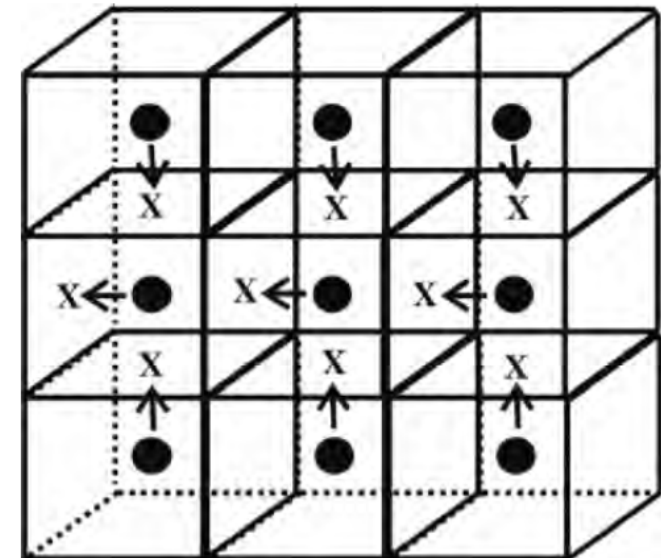
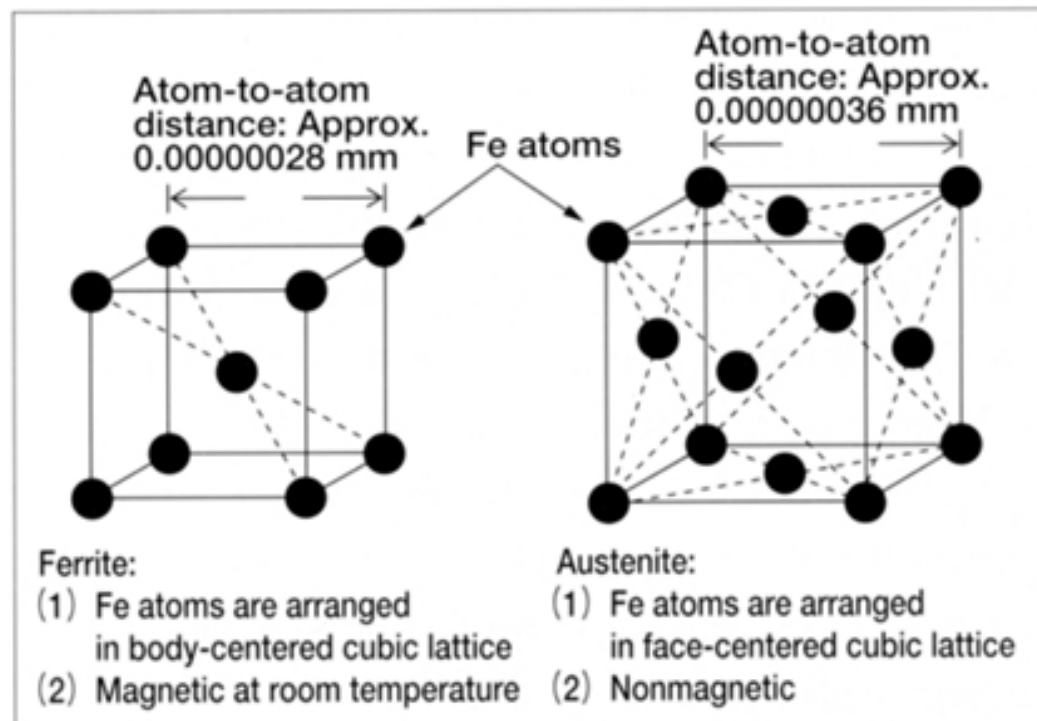
γ-Fe (austenite)

A FCC crystal structure

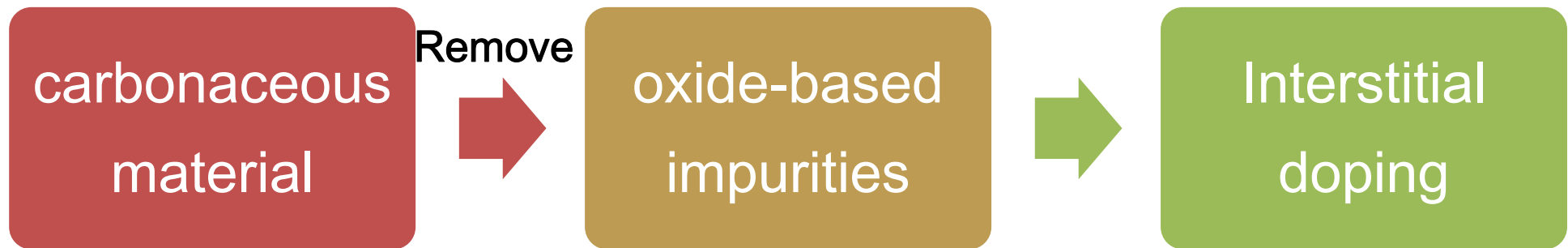
δ-Fe

A BCC crystal structure

- As the temperature is increased, thermally induced atomic motion increases, and it becomes more energetically favorable for atoms in the center of the standard bcc crystal lattice to migrate into face-centered positions of neighboring unit cells.
- As the magnitude of lattice vibrations continue to increase toward the melting point, the bcc structure is favored. This is due to the more open bcc structure being able to accommodate a larger range of vibrational motion than a relatively dense fcc array.



Iron-carbon alloys



- For instance, the carbon atoms in bcc ferrite are located only on empty face-centered positions.
- However, very few of these positions are occupied throughout the lattice, as the maximum solubility of carbon in α -Fe is only between 0.01 and 0.02 wt%.



The following are some features of adding Carbon to Iron:

- Steel is a crystalline substance containing less than 1.4% Carbon.
 - For Steel the Carbon must be present as Iron Carbide. Carbide of Iron is known as "CEMENTITE" It is a chemical compound of Iron and Carbon (Fe_3C).
 - Free Carbon (**graphite**) is present in Cast Iron.

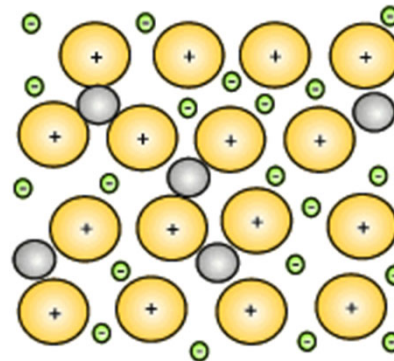
Classification. Three types of ferrous alloys:

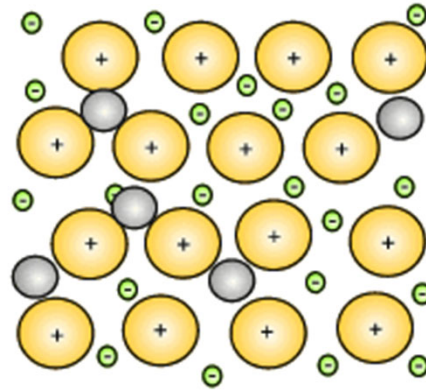
- ✓ Iron: less than 0.008 wt % C in α -ferrite at room T
- ✓ Steels: 0.15 – 1.4 wt % C (usually < 1 wt %)
 α -Fe (ferrite) + Fe_3C at room T
- ✓ Cast iron: 2.14 - 6.7 wt % (usually < 4.5 wt %)



- Carbon gives strength and hardness but at the expense of ductility.

the addition of carbon in the lattice acts as an “electron sink,” that is able to accept some of the delocalized electron-density from the metallic lattice. This results in a stronger interaction among all atoms in the lattice that adds to physical hardness, but detracts from the overall electrical conductivity, relative to pure iron.





Metal atoms have a lesser range of movement due to the “glue” formed by interstitial carbon atoms. As a result, external forces such as temperature and pressure will not as readily cause atomic movement and surface/bulk deformation or fracturing. This is the reason why pure iron is not particularly hard or physically durable, but steels are significantly improved in these properties.



- Since the dopant species are entrained within individual unit cells, the volume of each unit cell will increase concomitantly with the concentration of carbon.

Table 3.2. Unit Cell Dimensions of Iron Allotropes and Fe–C Alloys^a

Fe–C composition (crystal structure)	Unit cell parameters (Å)
α -Fe (ferrite, BCC)	$a = 2.8665$
γ -Fe (austenite, FCC)	$a = 3.555 + 0.044x^b$
δ -Fe (BCC)	$a = 2.9323$
Martensite (tetragonal)	$a = 2.867 - 0.013x^b$ $c = 2.867 + 0.116x^b$
Cementite (orthorhombic)	$a = 4.525$ $b = 5.088$ $c = 6.740$

^aValues taken from Cullity, B. D. *Elements of X-Ray Diffraction*, 2nd ed., Addison Wesley: Reading, MA, 1978.

^b x = wt% C in interstitial sites of the iron lattice.



- A decrease in density.
 - Since the change in volume is usually more significant than the increase in number of unit cell atoms, interstitial solids usually exhibit a decrease in density, relative to the pure allotrope.
 - For instance, the density of pure iron (7,874 kg m⁻³) shows a significant decrease upon interstitial placement of carbon in cast irons (ca. 7,400 kg m⁻³).

$$\rho = \frac{1.6604 \sum (n_l A_l + n_i A_i)}{V}$$

n_l, n_i : the number of regular lattice and dopant atoms per unit cell

A_l, A_i : the atomic weights of the regular lattice and dopant atoms.



Four phases in Fe-C alloys

□ α -ferrite - solid solution of C in BCC Fe

- Stable form of iron at room temperature.
- The maximum solubility of C is 0.022 wt%
- Transforms to FCC γ -austenite at 912 °C

□ γ -austenite - solid solution of C in FCC Fe

- The maximum solubility of C is 2.14 wt %.
- Transforms to BCC δ -ferrite at 1395 °C
- Is not stable below the eutectoid temperature (723 °C) unless cooled rapidly



□ δ -ferrite-solid solution of C in BCC Fe

- The same structure as α -ferrite
- Stable only at high T, above 1394 °C
- Melts at 1538 °C

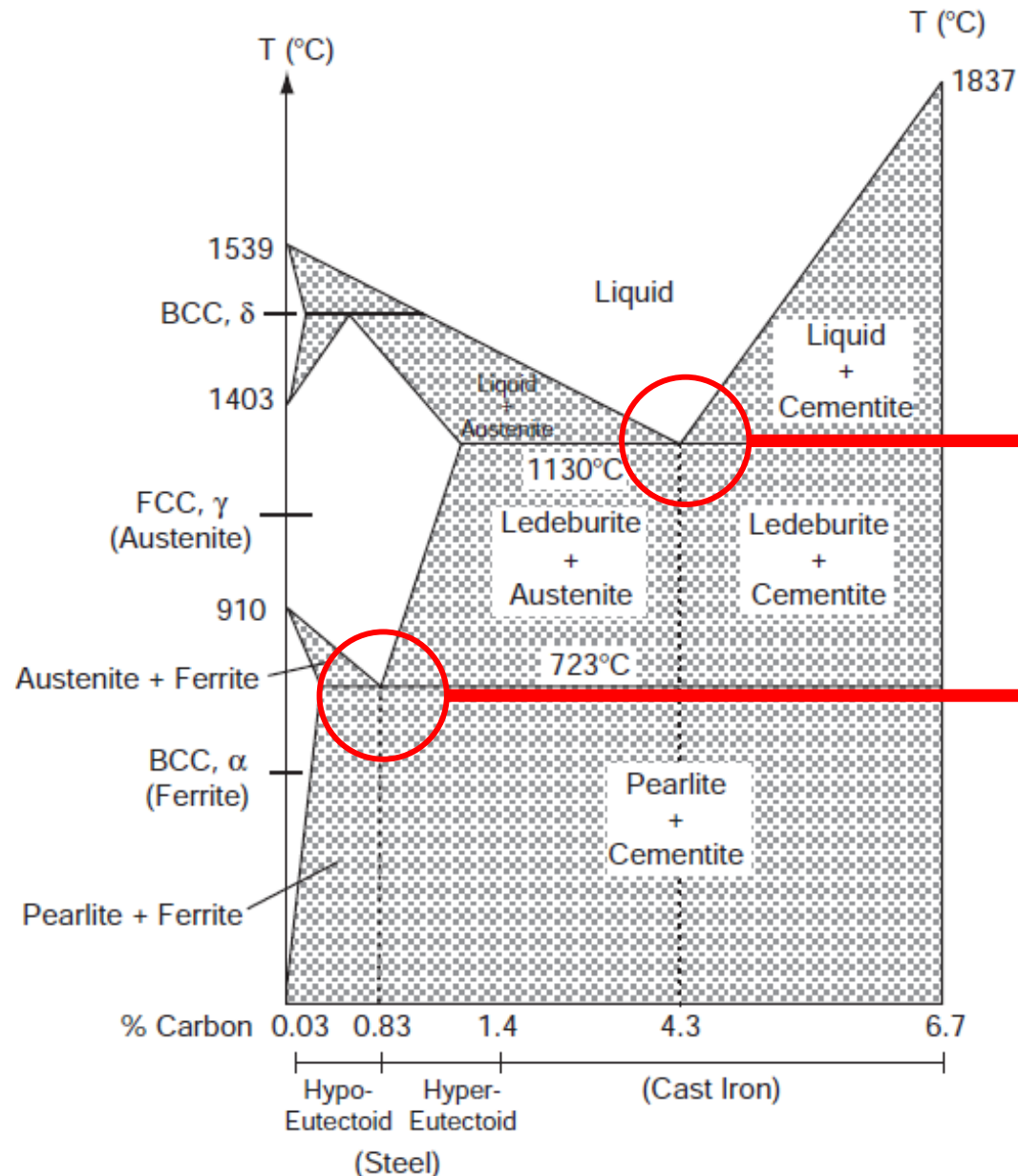
□ Fe_3C (iron carbide or cementite)

- This intermetallic compound is metastable, it remains as a compound indefinitely at room T, but decomposes (very slowly, within several years) into α -Fe and C (graphite) at 650 - 700 °C.



Cementite

- Hard and brittle
- It should be noted that cementite is actually a metastable phase, with graphite representing the most stable form of carbon at equilibrium. However, it is difficult to obtain stable graphite nuclei in steels (with $< 2\%C$) since the Fe/Fe₃C metastable equilibrium is preferentially formed.
- Since the bonding character of neighboring atoms is broken across the grain boundary, the diffusion of impurities occurs more readily in these areas.
- It has been determined experimentally that the strength of steel increases with carbon content up to the eutectoid composition, and then begins to drop as cementite nuclei are formed in the material.



Eutectic point

4.30 wt% C, 1130 °C
 $L \leftrightarrow \gamma + \text{Fe}_3\text{C}$

Eutectoid point

The material above this point is solid rather than a liquid.

- Hypoeutectoid
- Hypereutectoid

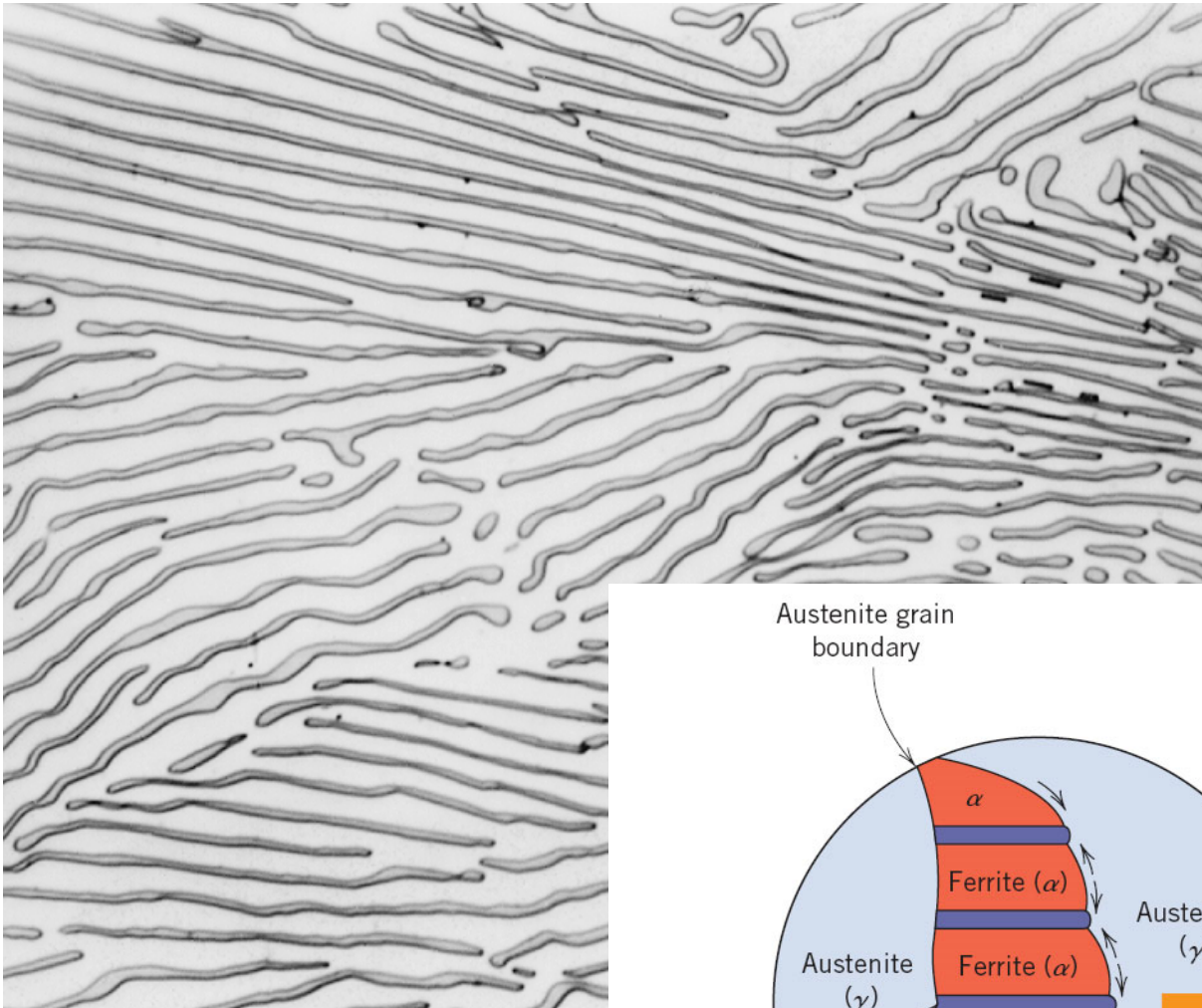
0.83 wt% C, 723 °C
 $\gamma \leftrightarrow \alpha + \text{Fe}_3\text{C}$

Figure 3.10. The equilibrium phase diagram for the iron-carbon system.



Microstructure of eutectoid steel

- When alloy of eutectoid composition (0.83 wt % C) is cooled slowly it forms pearlite, a lamellar or layered structure of two phases: α -ferrite and cementite (Fe_3C)
- The layers of alternating phases in pearlite are formed for the reason as: redistribution C atoms between ferrite (0.022 wt%) and cementite (6.7 wt%) by atomic diffusion.
- Mechanically, pearlite has properties intermediate to soft, ductile ferrite and hard, brittle cementite.



In the micrograph, the dark areas are Fe_3C layers, the light phase is α -ferrite.

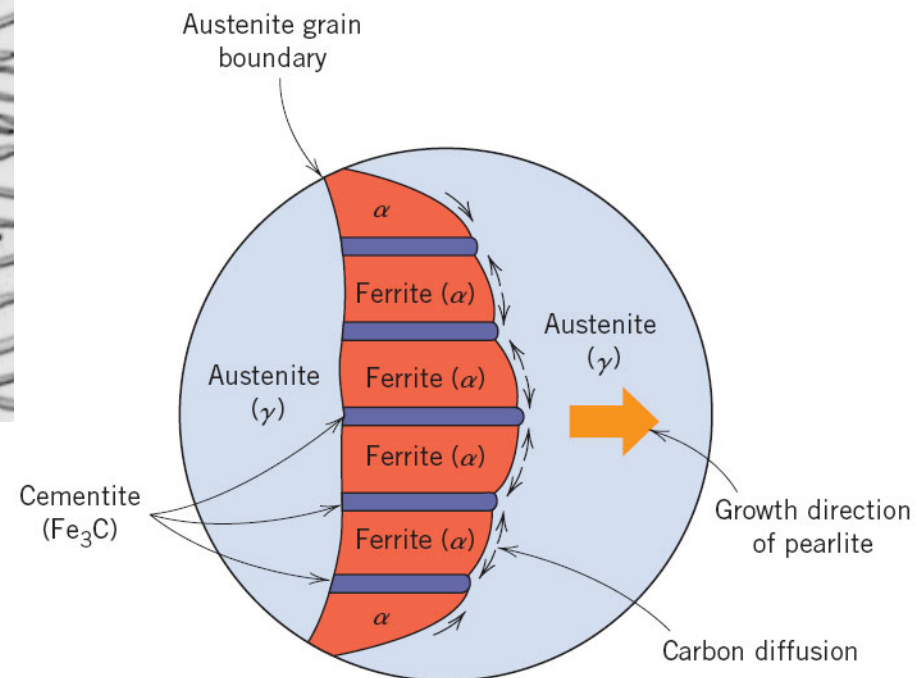
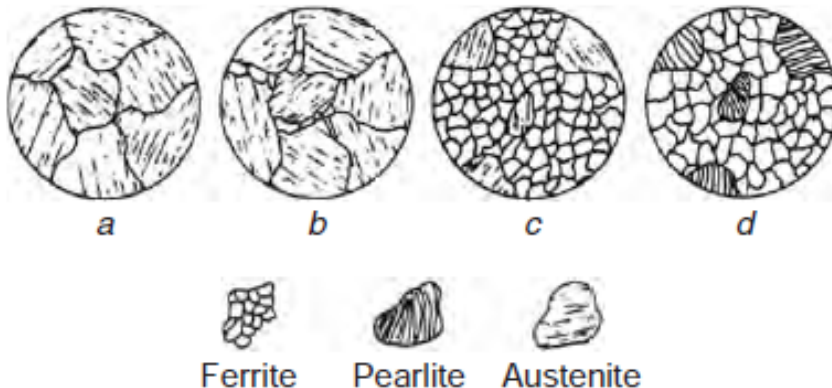
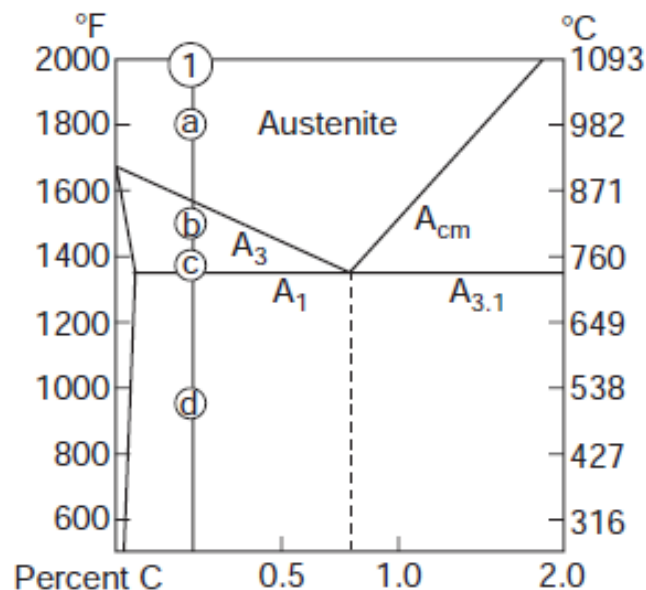


Figure 9.28 Schematic representation of the formation of pearlite from austenite; direction of carbon diffusion indicated by arrows.

• Hypoeutectoid steel

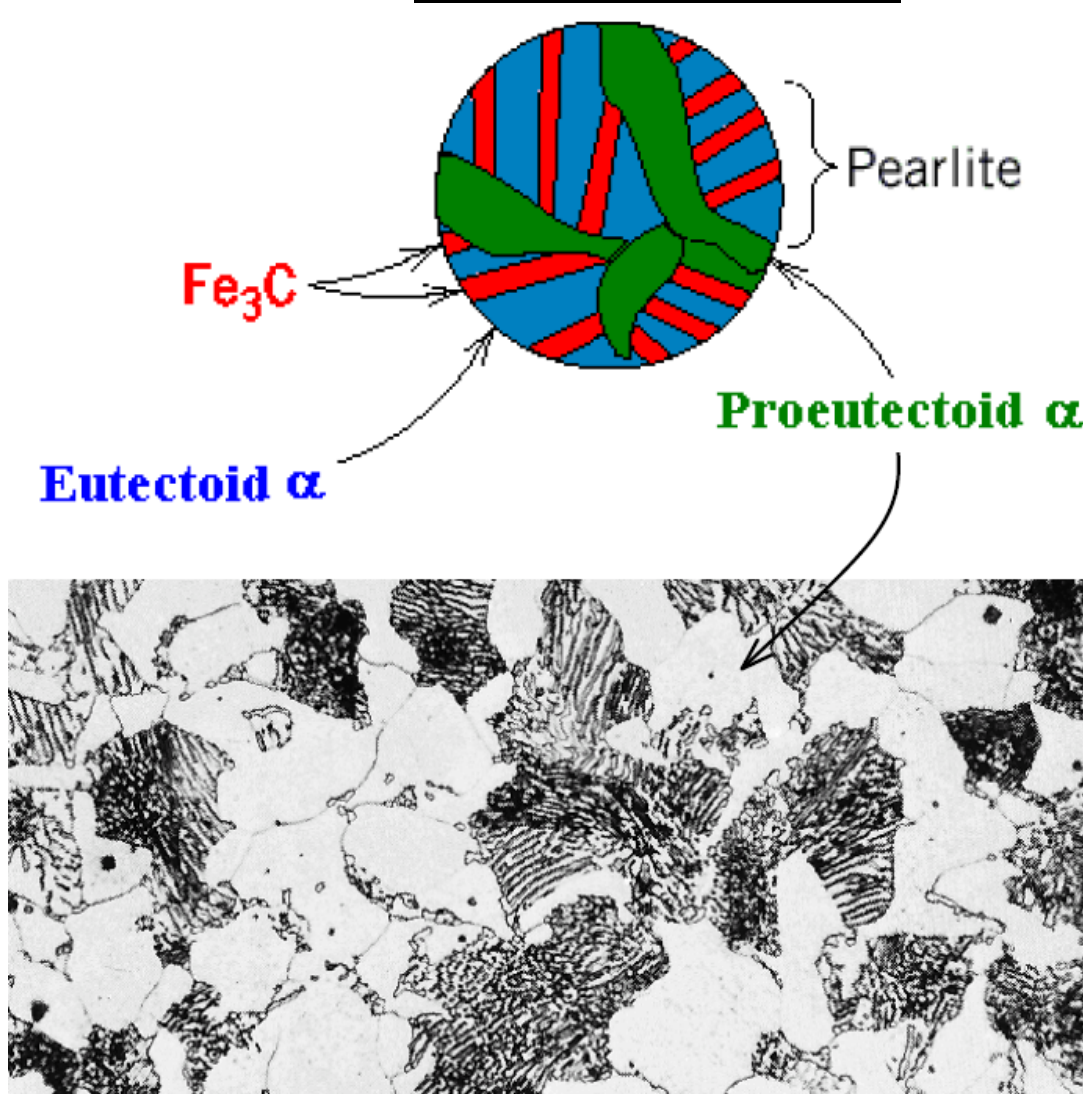
Compositions to the left of eutectoid ($< 0.83 \text{ wt } \% \text{ C}$) alloys.



- Austenite steel is slowly cooled.
- Ferrite begins to form along the austenite grain boundaries.
- Further cooling results in a ferrite-rich phase, with some remaining austenite crystals.
- At the eutectoid point of 723°C , the residual austenite is converted to pearlite, yielding a state that contains both ferrite and pearlite crystals upon further cooling.

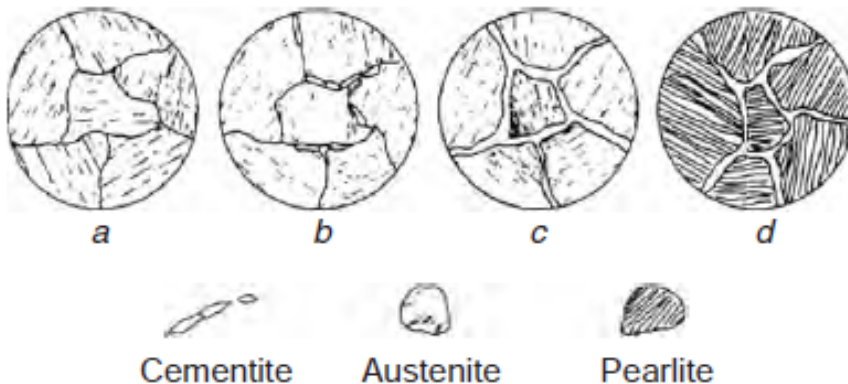
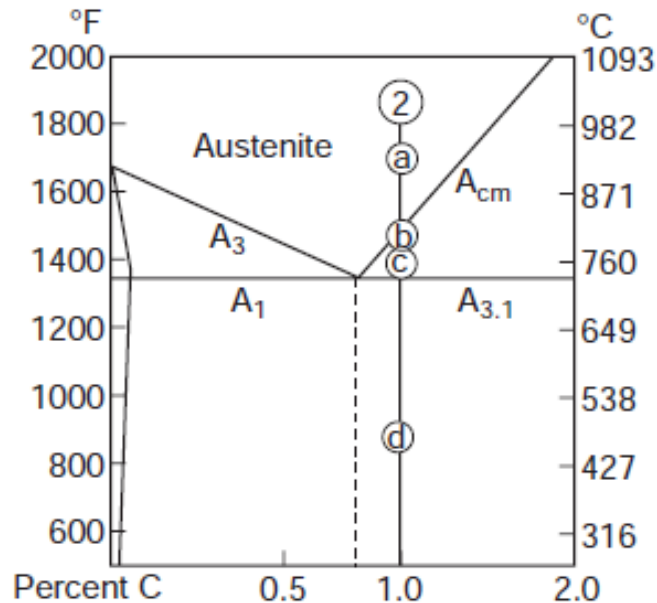


Hypoeutectoid alloys contain proeutectoid ferrite (formed above the eutectoid temperature) plus the eutectoid perlite that contain eutectoid ferrite and cementite.



• Hypereutectoid steel

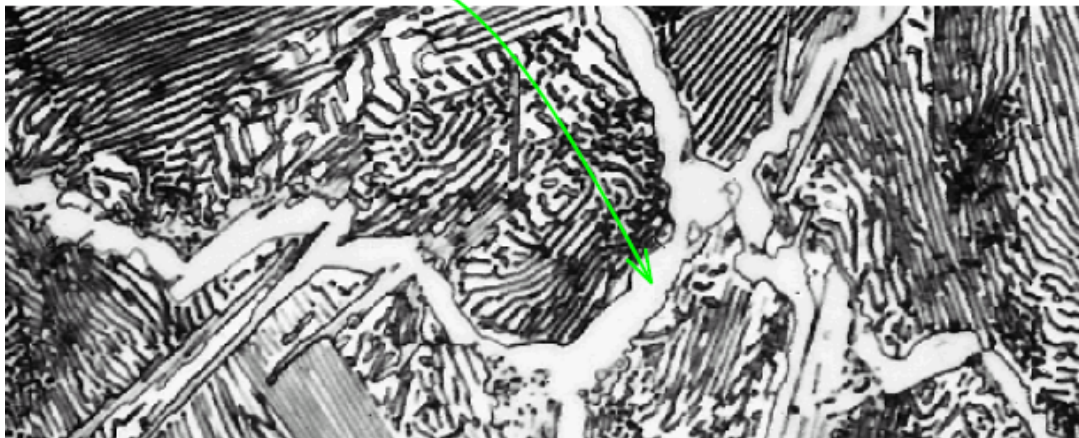
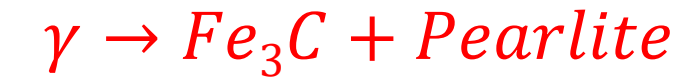
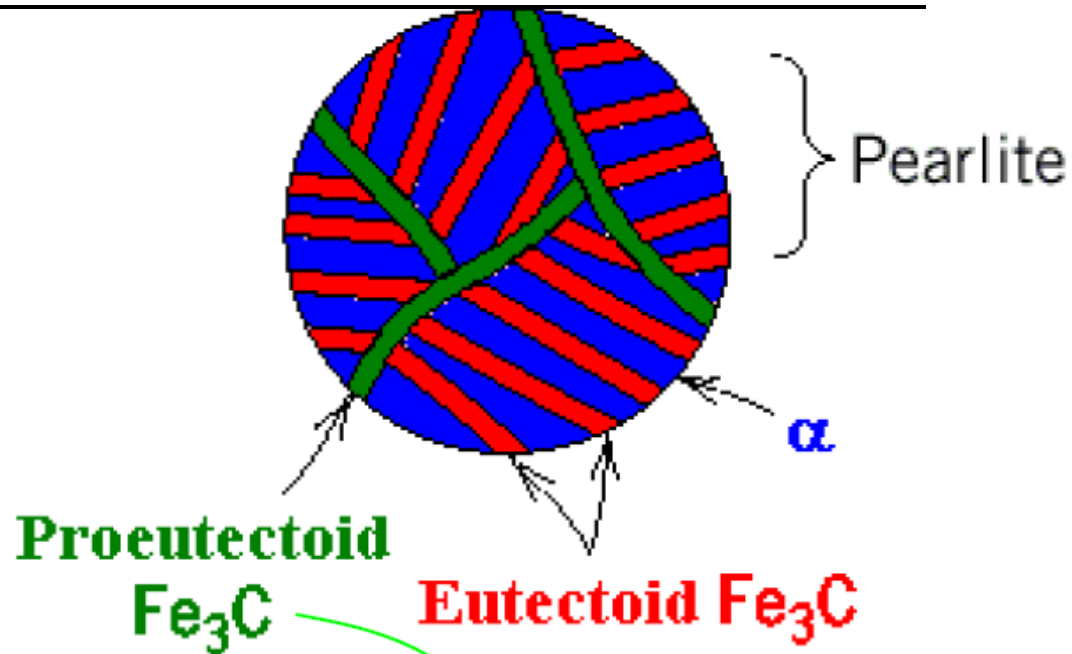
Compositions to the right of eutectoid ($0.83 \text{ wt } \% \text{ C} < x < 2.0 \text{ wt } \% \text{ C}$) alloys.



- Hypereutectoid steel contains significantly greater carbon concentrations;
- Cooling results in the precipitation of the excess carbon in the form of hard/brittle cementite, Fe_3C , nuclei that form along austenite grain boundaries.
- At the eutectoid point of 723°C , the residual austenite is converted to pearlite, yielding a state that contains both pearlite and cementite crystals upon further cooling.



Hypereutectoid alloys contain proeutectoid cementite (formed above the eutectoid temperature) plus perlite that contain eutectoid ferrite and cementite.





- **Cast Iron**

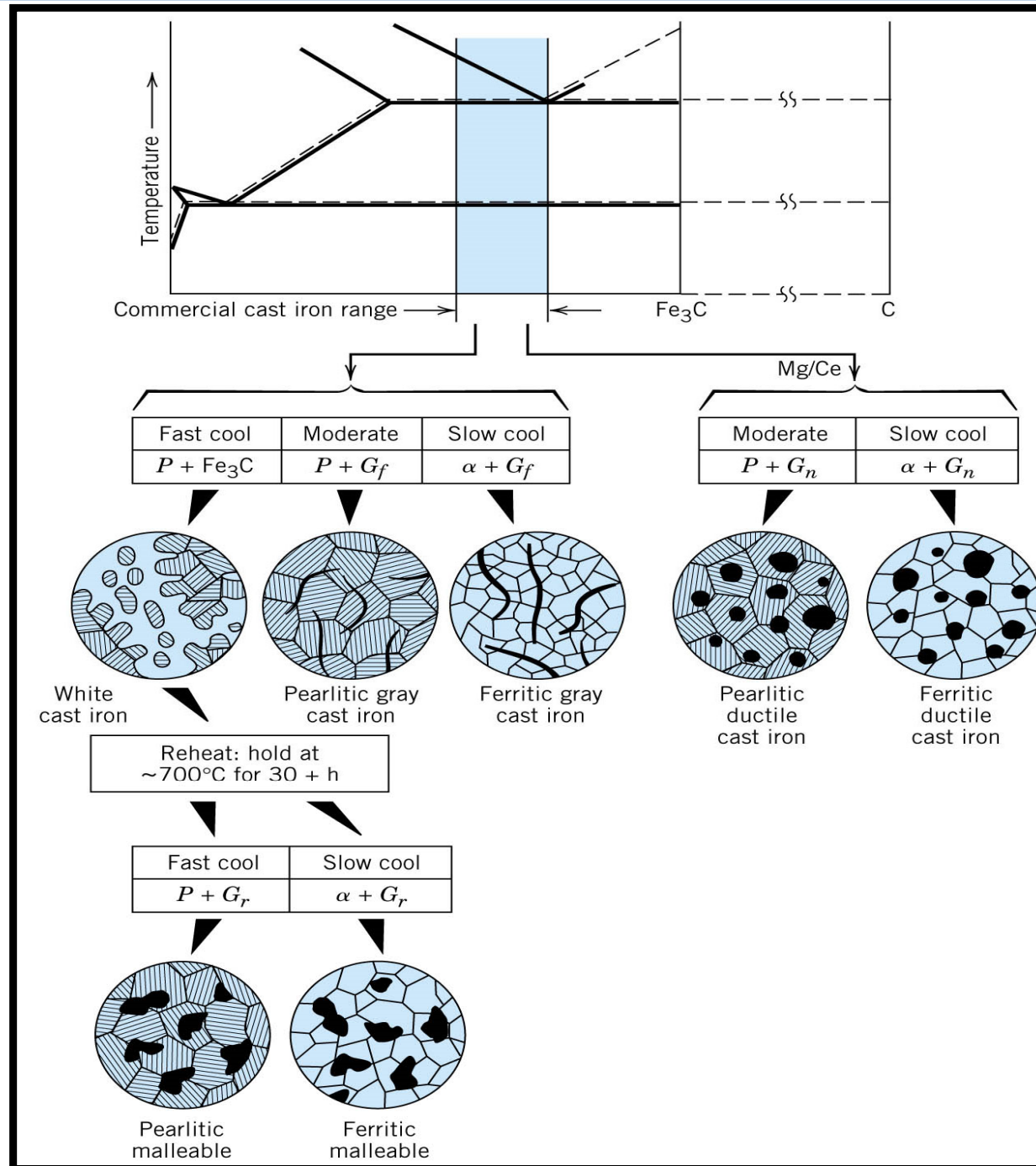
- Supersaturated iron lattices yield a material known as *cast iron*, a ternary Fe–C–Si alloy containing much higher carbon than steel, typically around 3–5 wt% C.
- As the liquid is cooled at the eutectic temperature, solidification of ledeburite (tiny austenite crystals embedded in a matrix of cementite).
- At carbon concentrations less than the eutectic (< 4.3 wt% C), ledeburite and austenite will form a solid solution. By contrast, increasing carbon concentrations will result in ledeburite/cementite solutions.



- As the name implies, cast irons are cast from their molten states into molds to yield the desired shapes. Due to the oversaturation of carbon present in these solids, cast iron is not suitable for structural applications. However, cast iron is extremely inexpensive to produce, making this material one of the most heavily used materials in industry for the manufacture of tools, valves, and automotive parts.

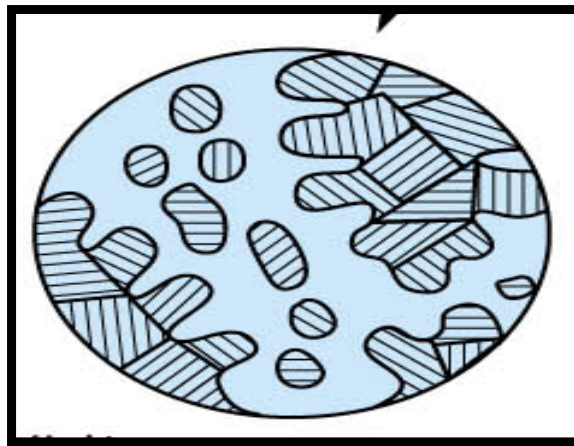


- If Carbon in Cast Iron is in the form of Carbide we have "White Cast Iron"
- If the Carbon is free in the form of Graphite we have "Grey Cast Iron".



White Cast Iron

- Due to large amounts of iron carbide presence, the structure of white iron is very hard, wear resistance and brittle.



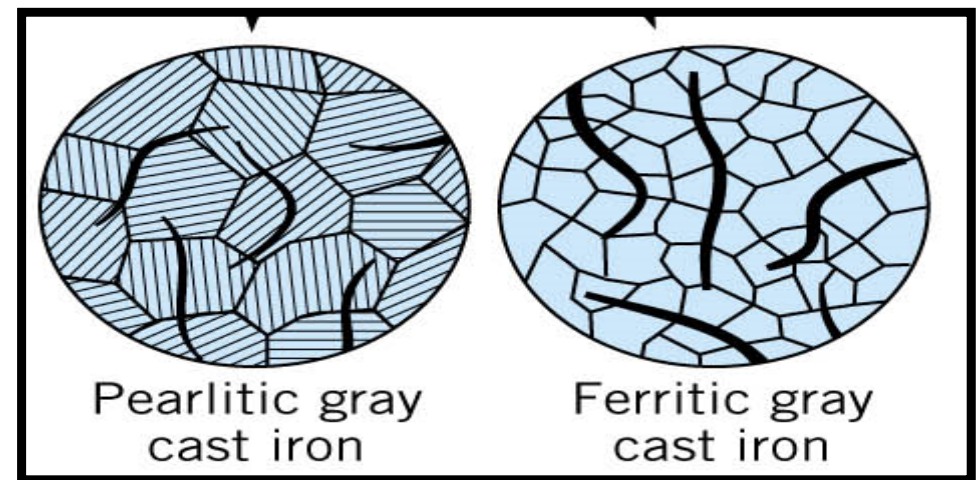
- It is obtained either by cooling gray iron rapidly or by adjusting the composition by keeping the carbon and silicon content low.
- Products from white iron include [railway brake shoes](#).



railway brake shoes

Gray Cast Iron

- Composition of 2.5% to 4% carbon and 1% to 3% silicon.
- Graphite exists largely in the form of flakes.
- Properties of gray iron :
 - Low (negligible) ductility
 - Weak in tension
 - Strong in compression
 - Good vibration damping



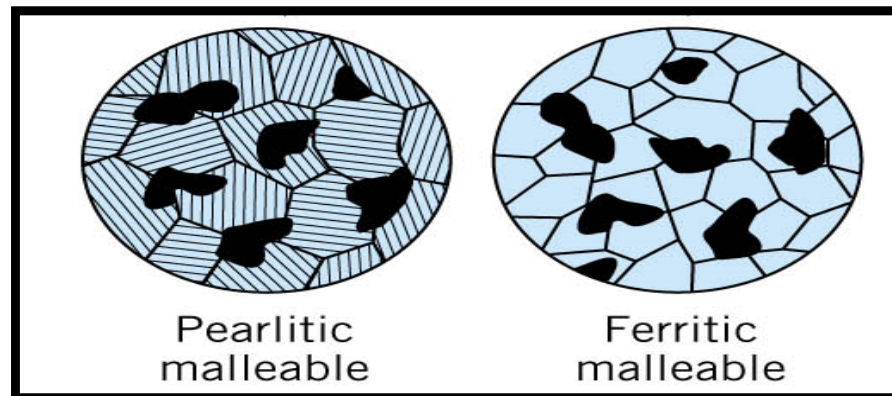
- Products from gray iron include automotive **engine blocks** and **heads**, **motor housings**, and **machine tool bases**.

Malleable Iron

Obtained by annealing white iron in an atmosphere of carbon monoxide and carbon dioxide, between 800°C~900°C, for several hours.

• 2 types of malleable iron :

- **Pearlite malleable (white malleable)** – upon fast cooling of white iron
- **Ferrite malleable (black malleable)** – upon slow cooling of white iron

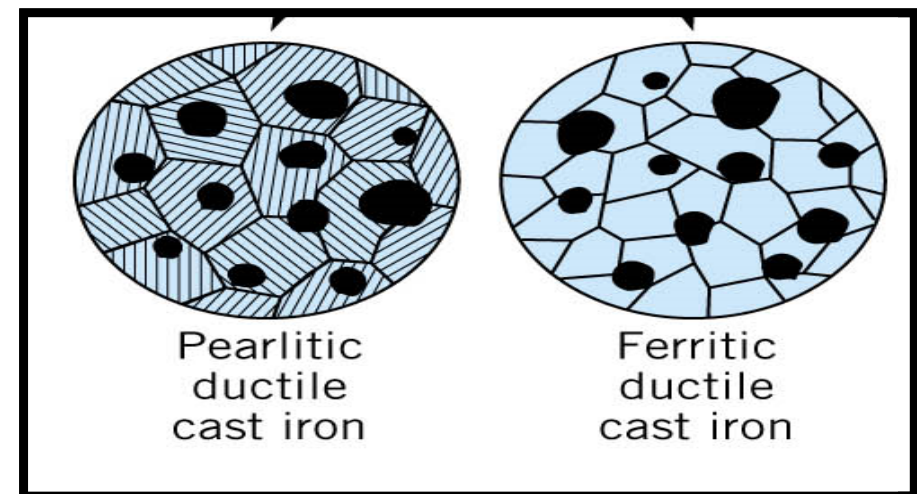


- The structure has good ductility, strength and shock resistance.
- Typical products include **pipe fittings** and **flanges**, **railroad equipment parts**.

Nodular Cast Iron (球磨铸铁 Ductile Iron)

- This is an iron with the composition of gray iron in which the molten metal is chemically(added with magnesium) treated before pouring to cause the formation of graphite spheroids rather than flakes.

- shock resistant , stronger and more ductile iron.

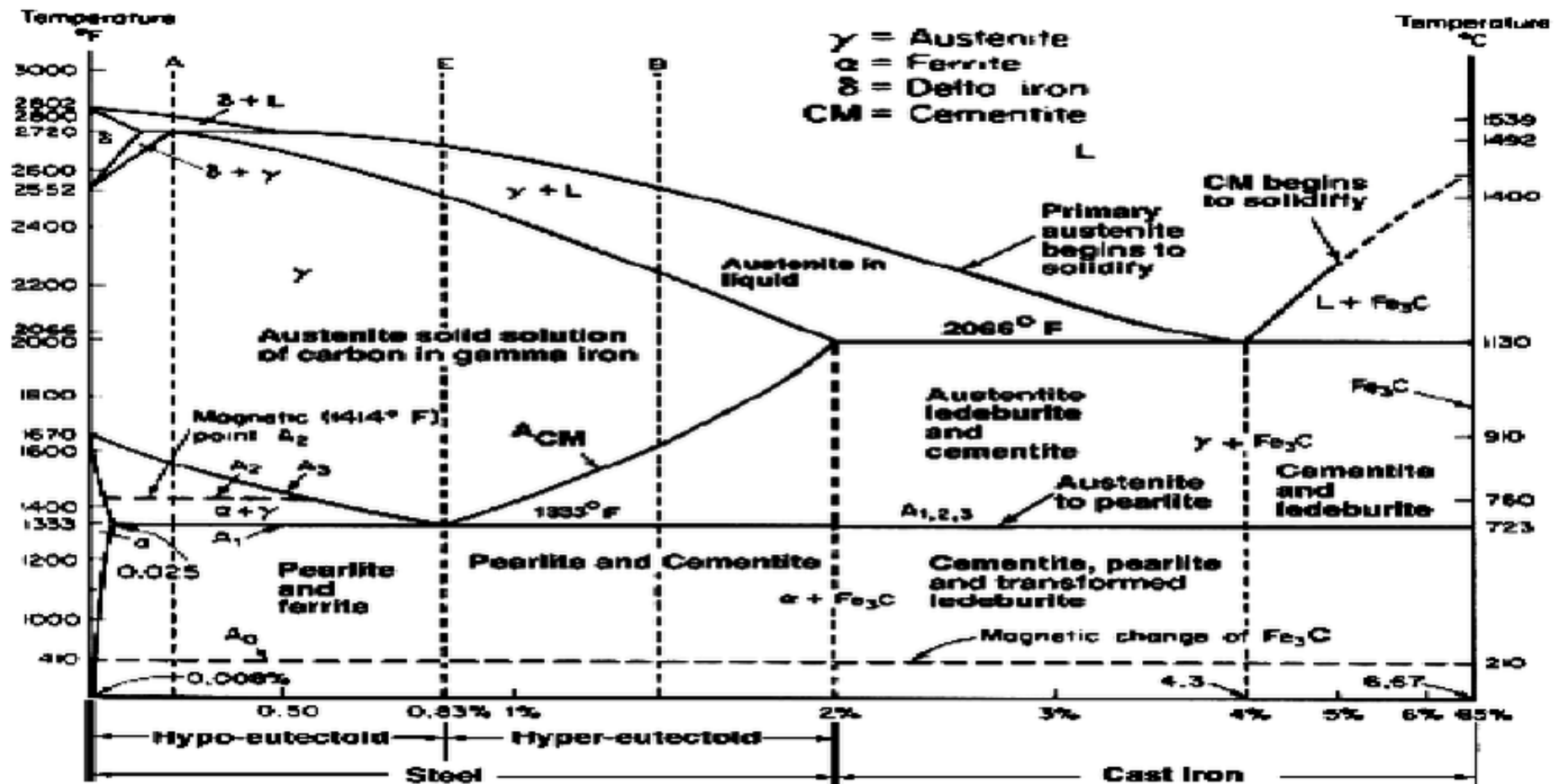


- Applications include **machinery components** requiring **high strength** and **good wear resistance**.



A number of dopants may be added to facilitate the preferential formation of graphite rather than cementite, which serve as a nucleation site:

- The major nucleation mechanism in cast iron doping is called *inoculation*: the formation of sulfide species upon the addition of strong sulfide formers such as calcium, barium, cerium, or strontium. These sulfides possess lattice parameters very similar to the graphite crystal structure, serving as substrates for nucleation/epitaxial growth of graphite.
- Other common graphitizer dopants are Si (typically added as metal ferrosilicon compounds), Ni, and Cu;
- By contrast, Cr, Mo, V, and W are antigraphitizers, promoting the formation of carbides ($\text{Cr}_7\text{C}_3/\text{Cr}_{23}\text{C}_6$, $\text{Mo}_2\text{C}/\text{Fe}_3\text{Mo}_3\text{C}$, $\text{VC}-\text{V}_4\text{C}_3$, $\text{W}_2\text{C}/\text{Fe}_3\text{W}_3\text{C}$, respectively).



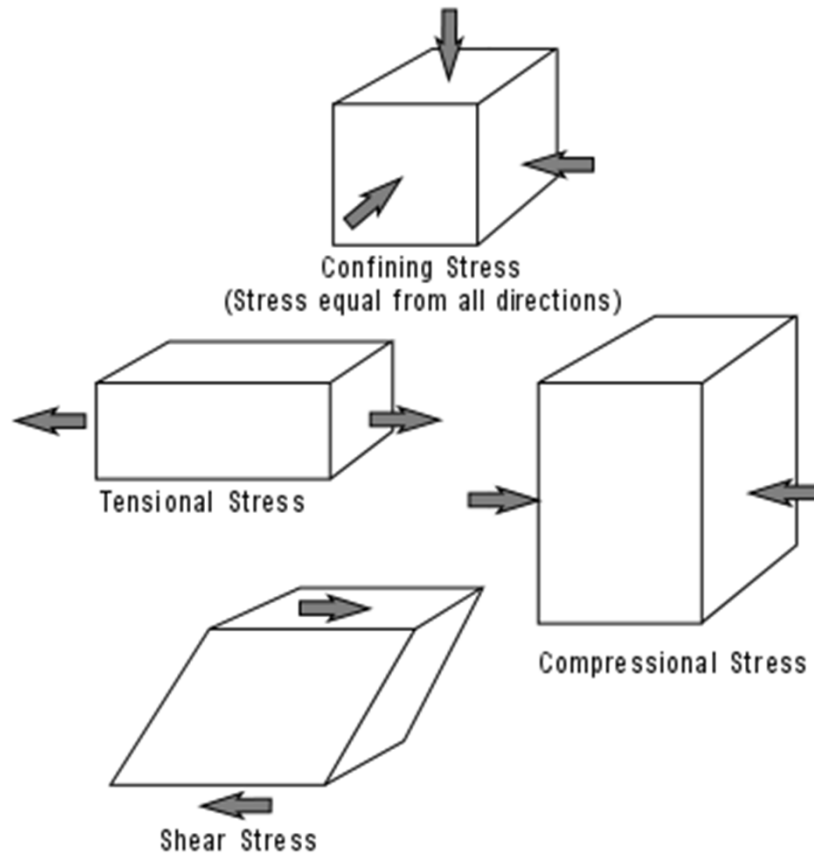


3.2.2. Hardening Mechanisms of Steels



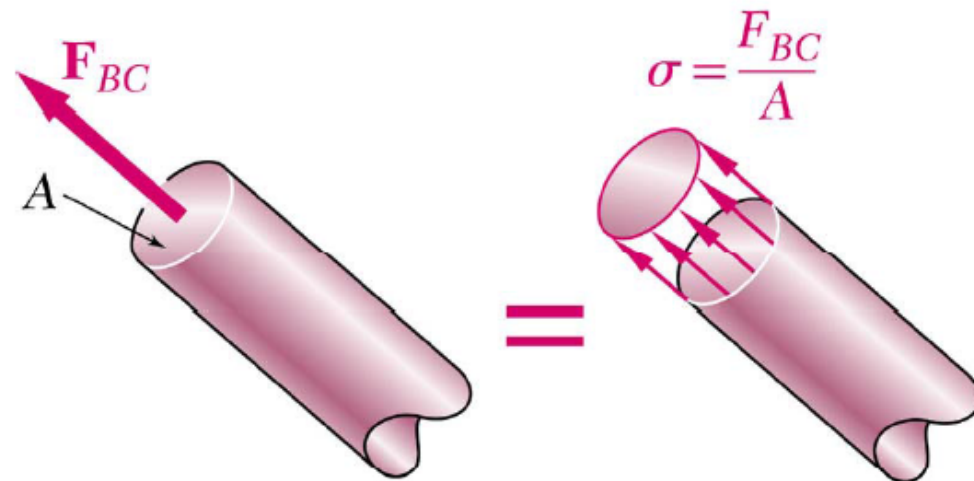
Hardening strategies:

- Strain hardening,
- grain size hardening,
- Precipitation alloying,
- Solute hardening,
- Tempering,
- Surface hardening.



Stress = Force per unit area

$$\sigma = \frac{F}{A}$$



Strain characterizes a deformation

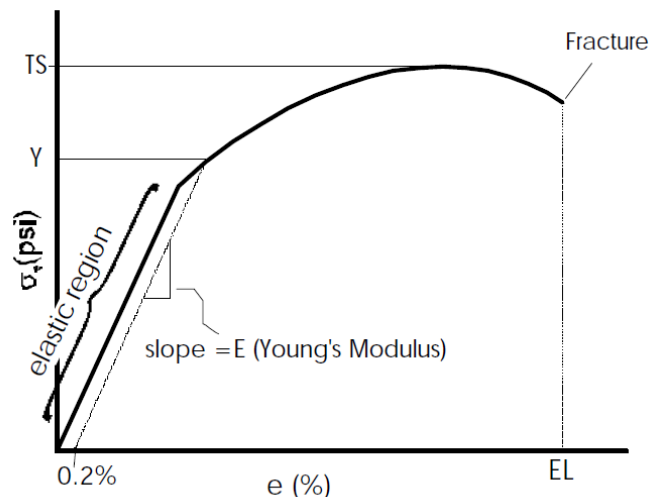
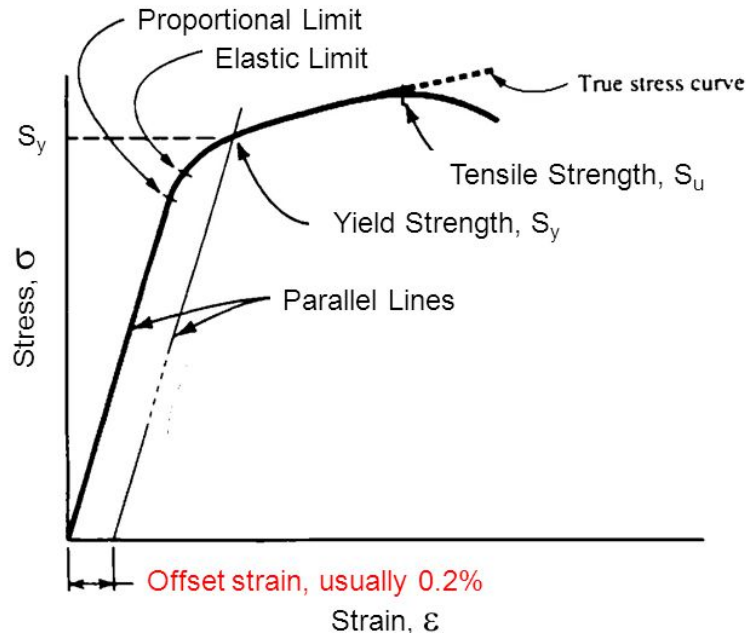


Tensile test

The typical method used to assess the strength of a metal is a tensile test, where the metal is clamped to upper and lower jaws, and pulled until it fractures.



stress vs. strain curves



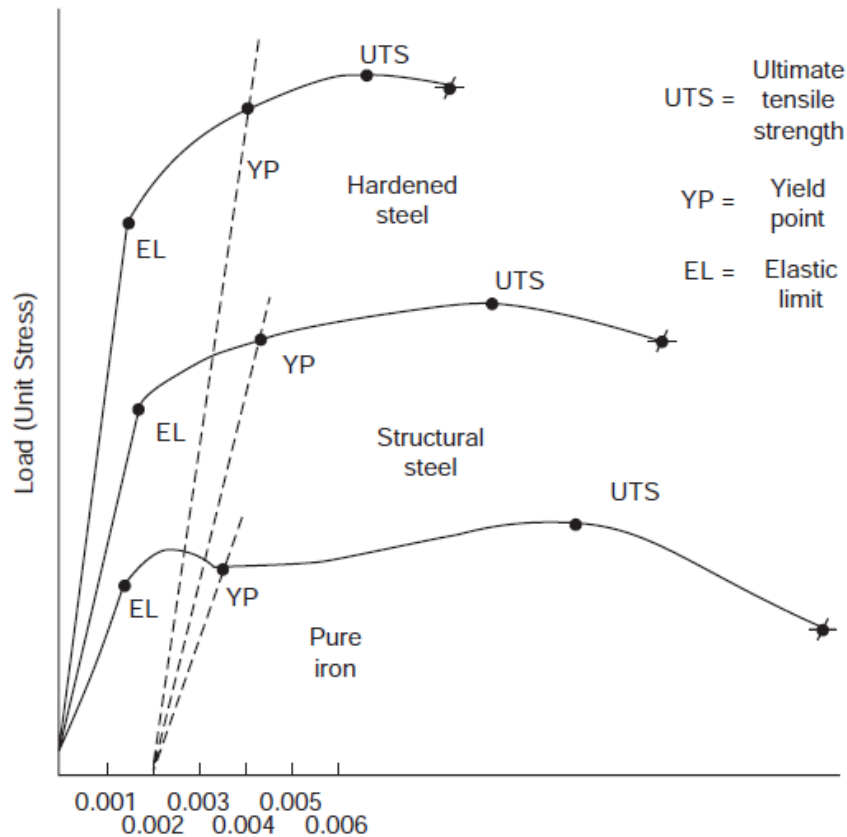
The elastic limit (EL) is the greatest stress that a material may withstand and still revert back to its original shape when the stress is removed.

The yield strength the stress which will produce a small amount of permanent deformation, generally equal to a strain of 0.002 or 0.2%.

Young's Modulus, denoted as E , is the slope of the engineering stress strain curve in the elastic region.

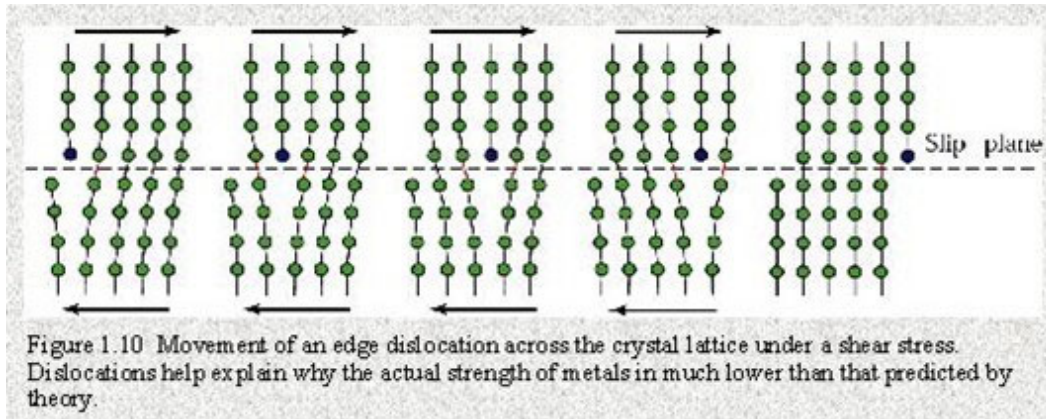


The typical stress vs. strain curves for iron, steel, and hardened steel



- The Ultimate Tensile Strength (UTS) is the highest stress which the material can undergo before the onset of necking.
- Necking is localized deformation which is the first step toward fracture. When necking occurs, the specimen begins to deform locally, hence the observed drop in load and engineering stress toward the end of the curve.

Strategy I: Strain hardening (work hardening)



Dislocation
defects

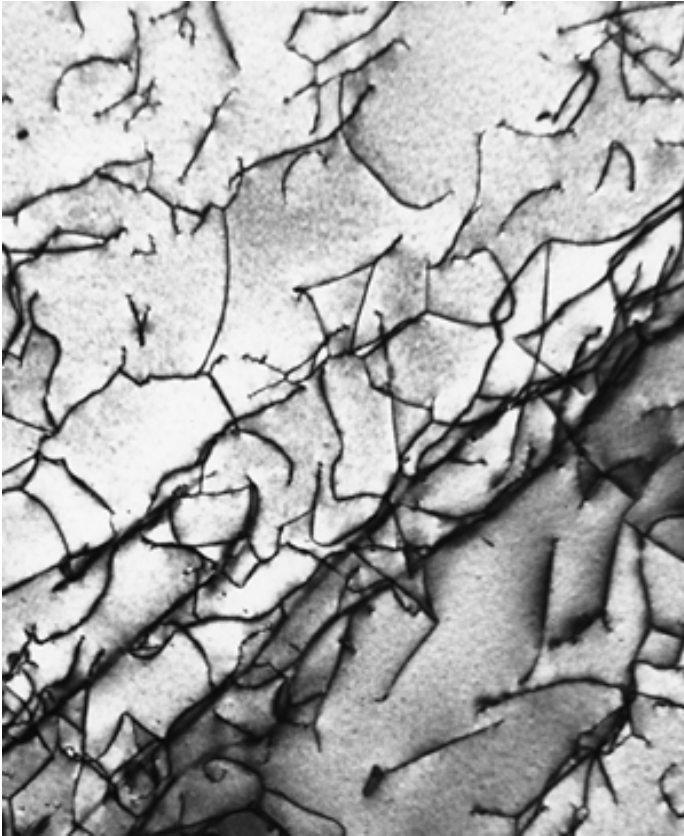
Deformation

The ability of a metal to plastically deform depends on the ability of dislocations to move.

Virtually the hardening techniques rely on this simple principle: Restricting or Hindering dislocation motion renders a material harder and stronger.



Work hardening (bending/hammering the cold material)

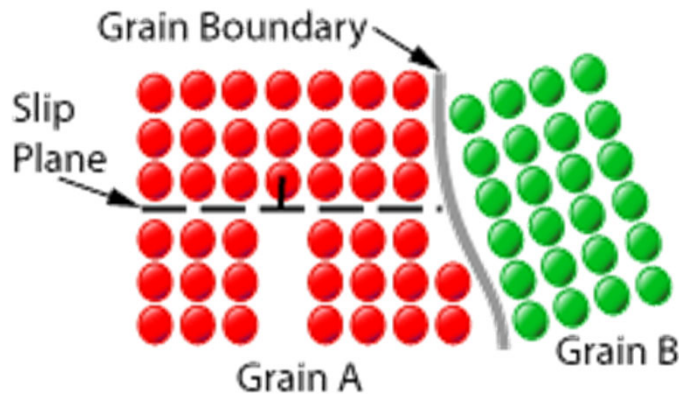


- A process referred to as work hardening (bending/hammering the cold material) is used to break up crystallites, introducing dislocations in the material.
- The more dislocations exist, the more they will interact with one another, becoming pinned or entangled with one another.
- This will impede the movement of dislocations and strengthening of the material.
- This is done at low temperature so the metal atoms cannot rearrange themselves.⁷⁴



- Cold Working
 - ✓ This is done at low temperature ($T < 0.5T_m$, where T_m =melting point), so the metal atoms cannot rearrange themselves, which would negate the effect.
 - ✓ Consequently, low-melting metals such as Sn and Pb may not be cold-worked at room temperature.
- Hot Working
 - ✓ In contrast, hot working is performed at temperatures above the recrystallization temperature of the metal.
 - ✓ Although hot working requires less energy and induces larger deformations than cold working, most metals exhibit surface oxidation that may deleteriously affect its overall properties and applications.

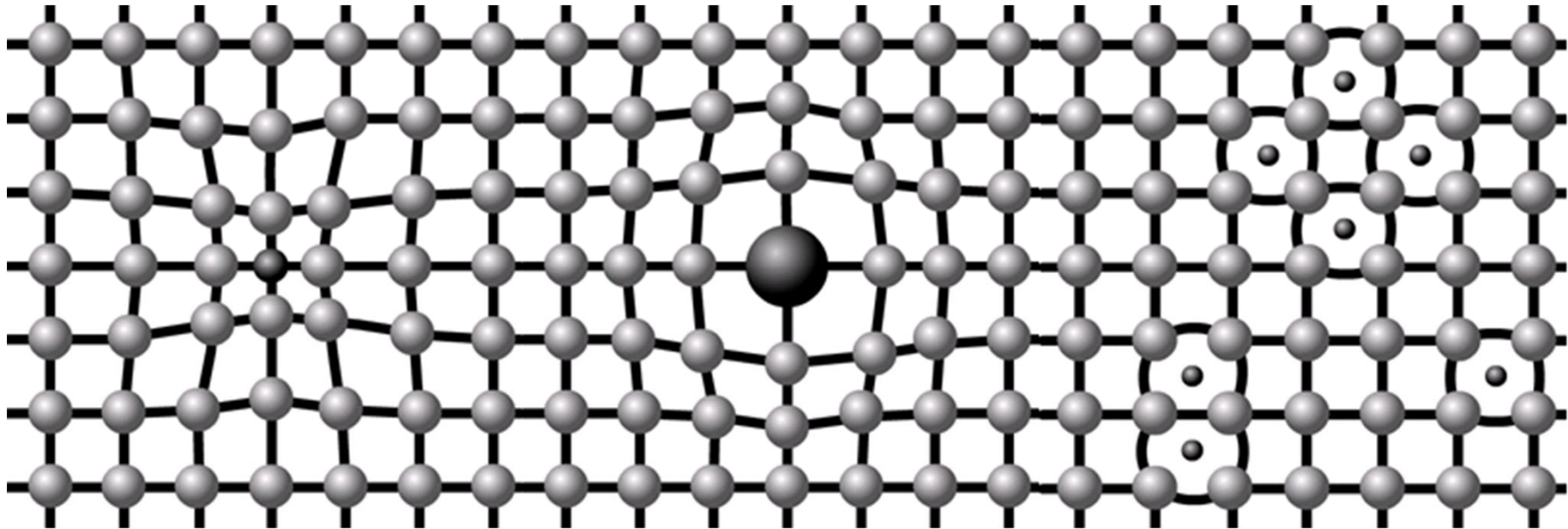
Strategy II: grain size hardening



Because adjacent grains have different orientations, the boundary between grains acts as a barrier to dislocation movement and the resulting slip.

- The smaller the grains, the shorter the distance atoms can move along a particular slip plane.
- Therefore, smaller grains improve the strength and hardness of a material.
- The size and number of grains within a material is controlled by the rate of solidification from the liquid phase.
- Increase Rate of solidification from the liquid phase to reduce grain sizes.

Strategy III: solid-solution alloying



- Substitutional solid solution
- Interstitial solid solution

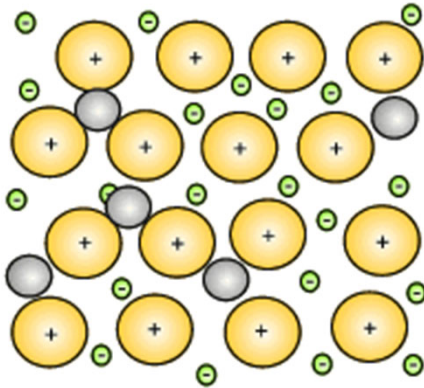


Strategy IV: Precipitation hardening (heat treatment)

The strength and hardness of some metal alloys may be improved by the formation of extremely small, uniformly dispersed particles (precipitates) of a second phase within the original phase matrix.

Alloys that can be precipitation hardened or age hardened:

- Copper-beryllium Cu-Be
- Copper-tin Cu-Sn
- Magnesium-aluminum Mg-Al
- Aluminum-copper Al-Cu
- High-strength aluminum alloy



The stresses that these defects create in the crystal lattice are ‘pinning points’ that restrict the motion of dislocations and so strengthen the material.

For example, if the alloying agent is carbon, hard crystallites of iron carbide may form, changing the microstructure to impede slip. By comparison, austenite usually does not contain iron carbide, and is quite susceptible to slip.

This process, referred to as precipitation hardening induces an increase in hardness through the formation of homogeneous dispersions that reduce slippage between grains.

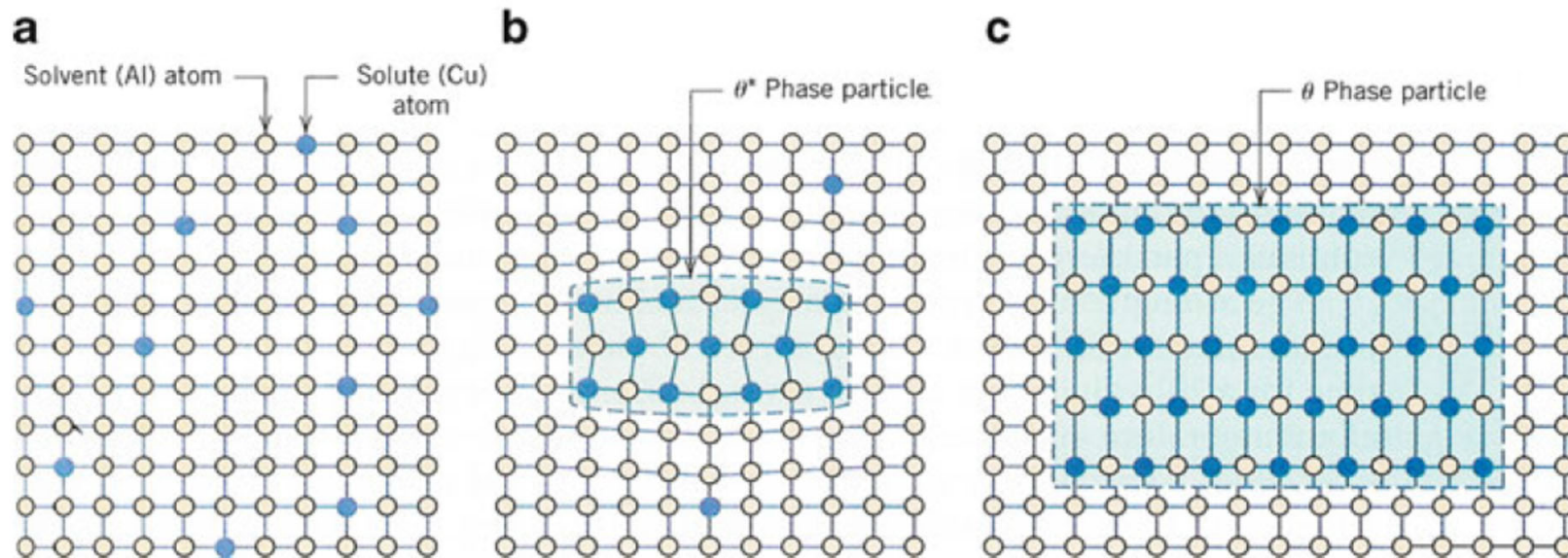
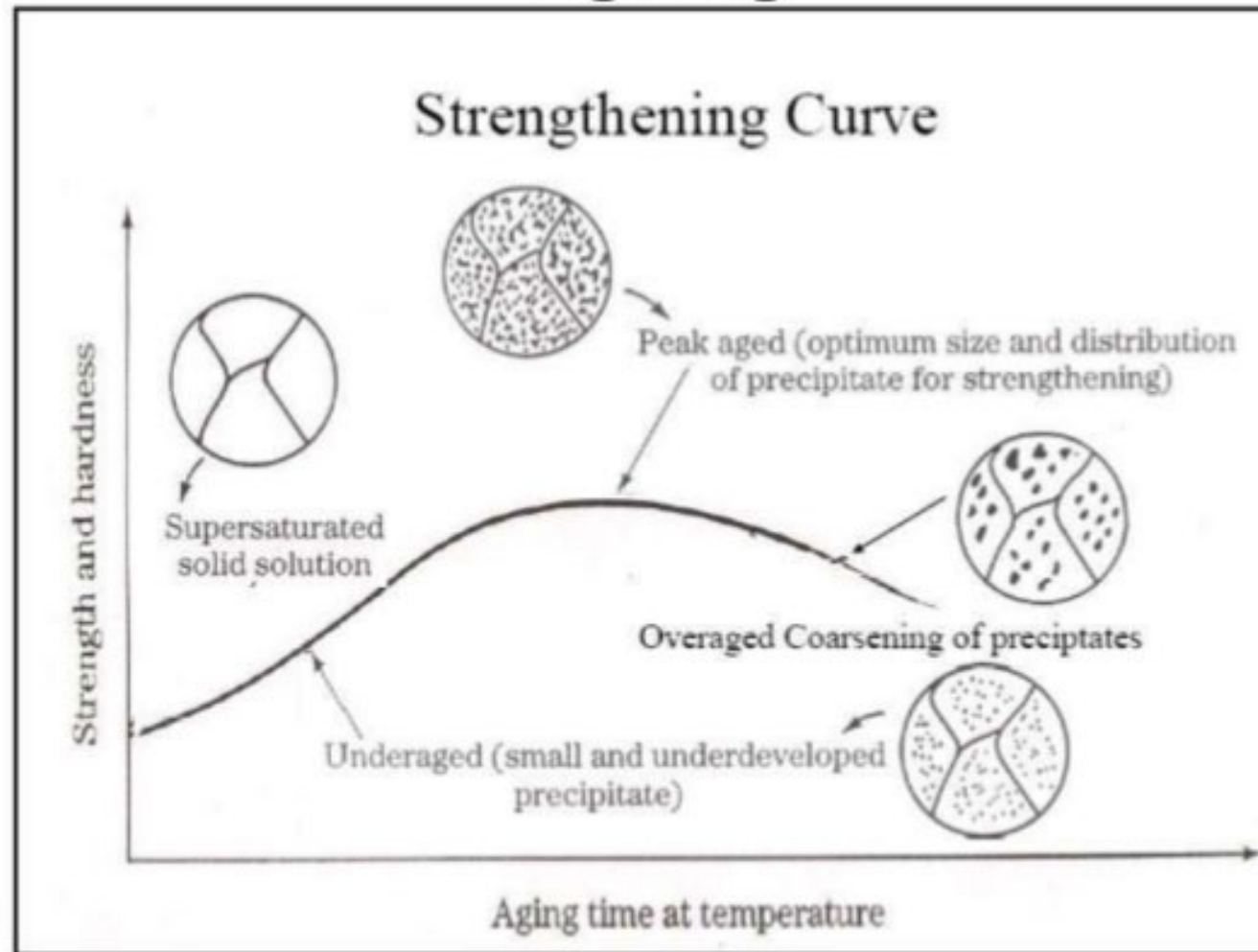


Figure 3.21. Top: The Al-rich side of the Al-Cu phase diagram. Bottom: schematic of the formation of the equilibrium precipitate σ phase: (a) a supersaturated α solid solution, (b) a transition, θ'' , precipitate phase, and (c) the precipitate θ phase within the α -matrix phase. Reproduced with permission from Callister, W. D. *Materials Science and Engineering: An Introduction*, 7th ed., Wiley: New York, 2007. Copyright 2007 John Wiley & Sons, Inc.



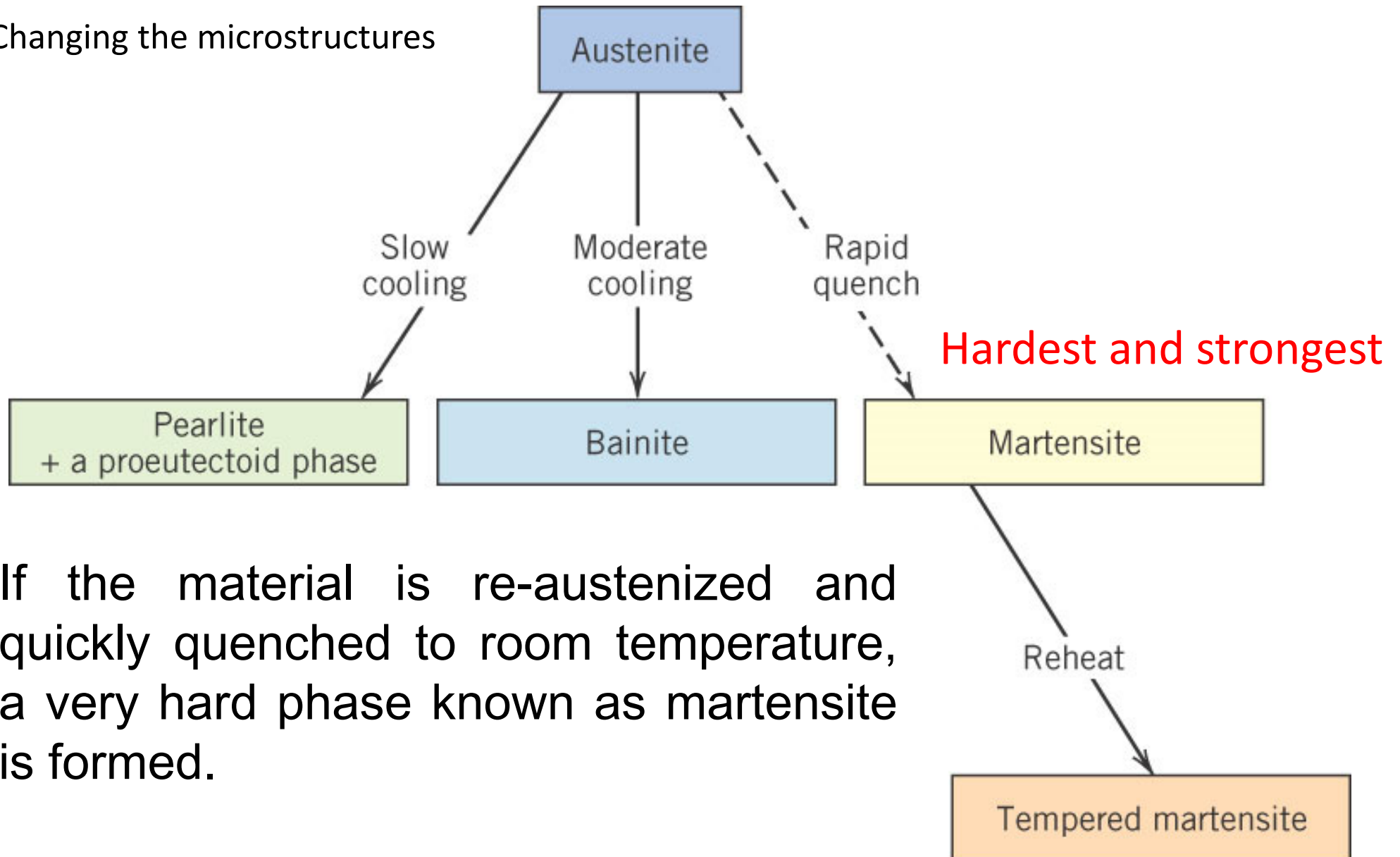
Ageing





Transformation hardening

Changing the microstructures



If the material is re-austenized and quickly quenched to room temperature, a very hard phase known as martensite is formed.

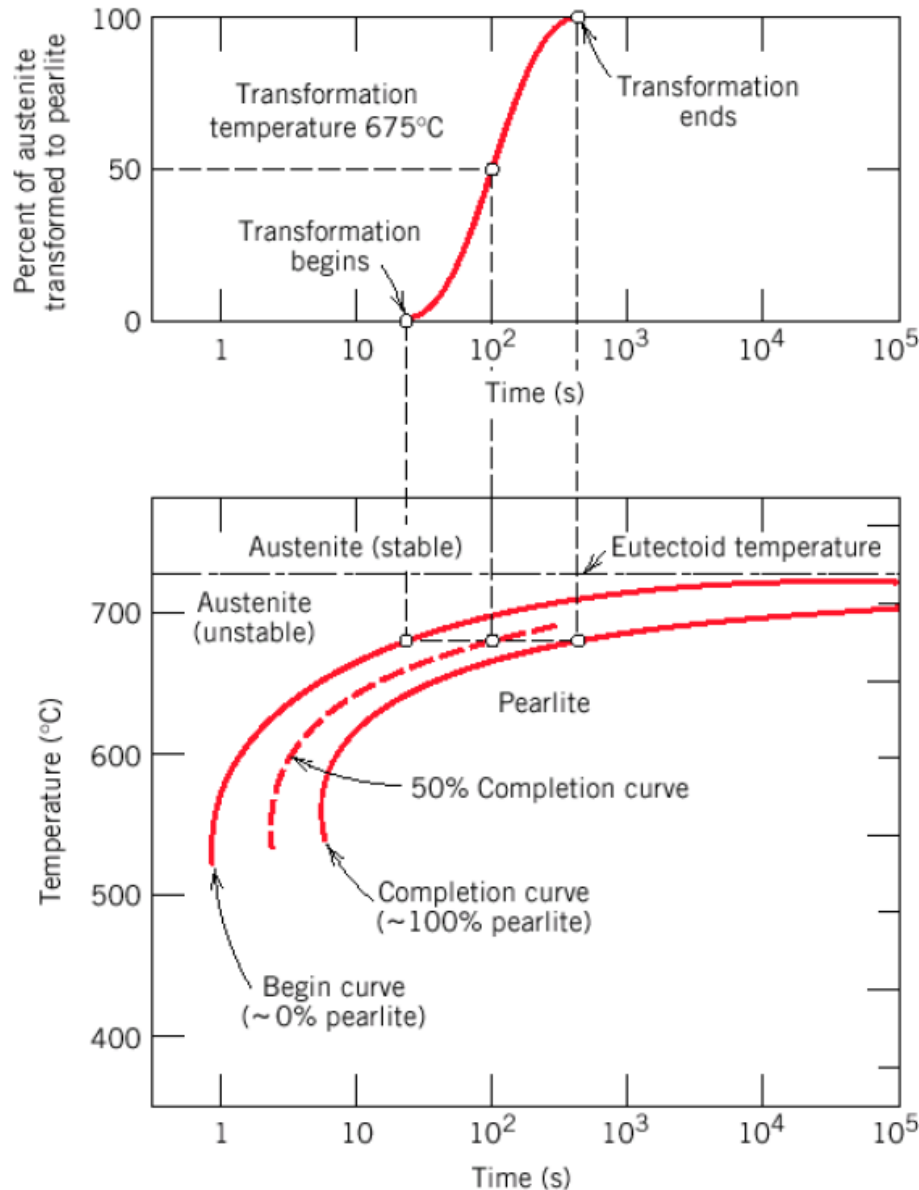


The temperature range is not the only crucial variable affecting the properties of the material, but rather the rate of heating and cooling.

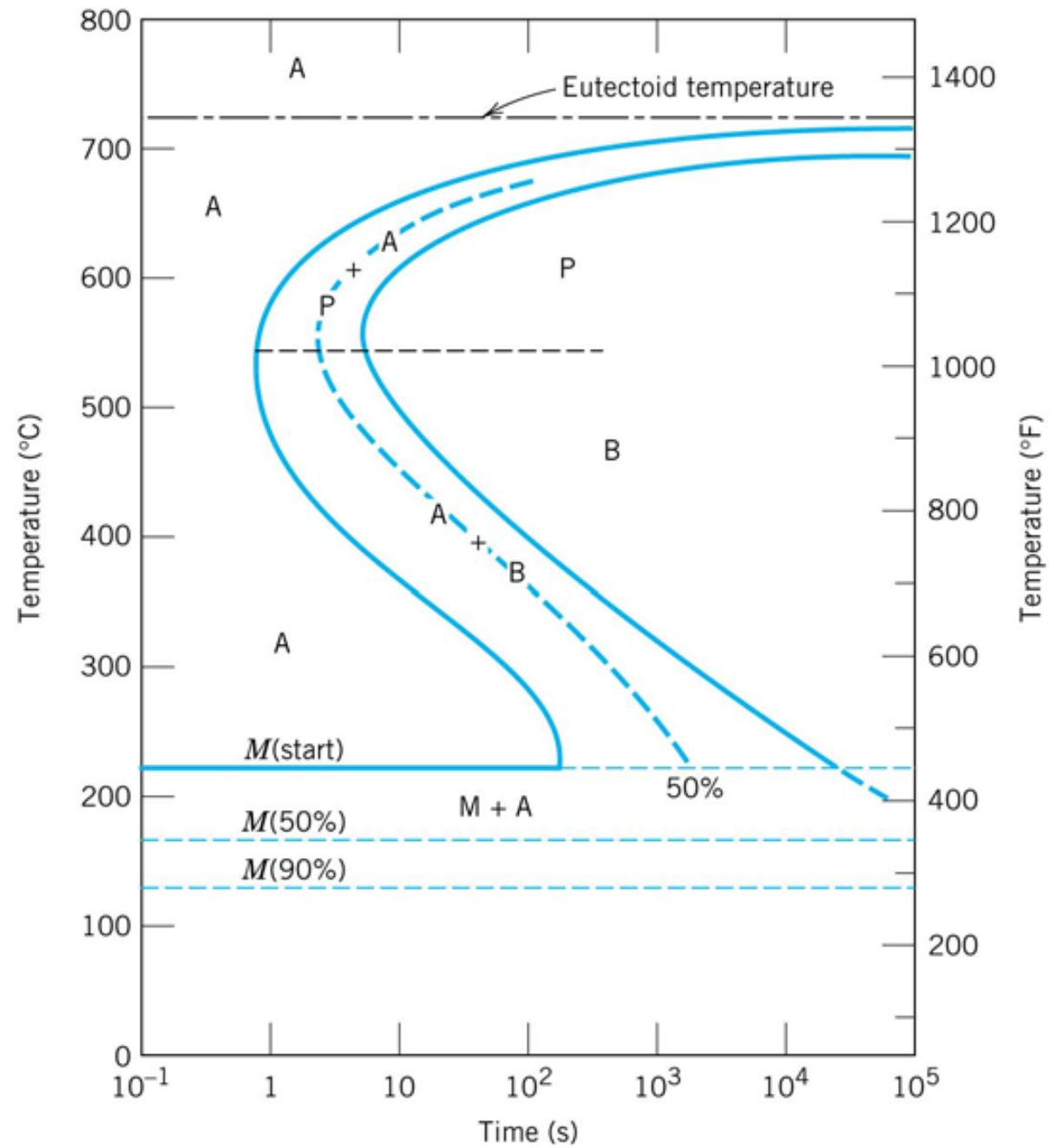
- When austenitic steel is very slowly cooled to room temperature, the resulting solid will be soft and malleable.
- When the same steel is rapidly quenched in cold water to temperatures less than 250°C , the normal phase transformations to ferrite or pearlite (depending on %C present) are suppressed. Rather, the γ -Fe phase is converted to martensite, the hardest and strongest of all possible Fe–C microstructures.

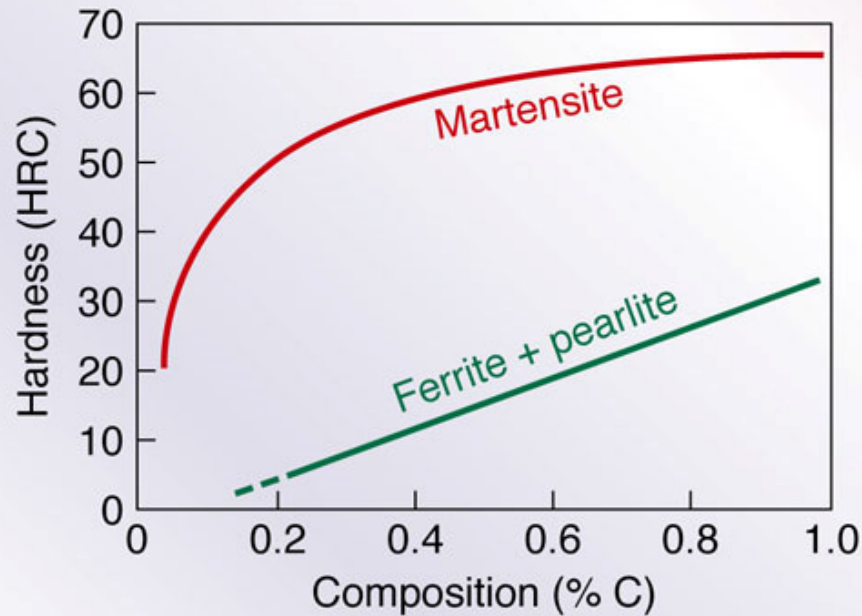


Isothermal Transformation (or TTT) Diagrams (Temperature, Time, and % Transformation)

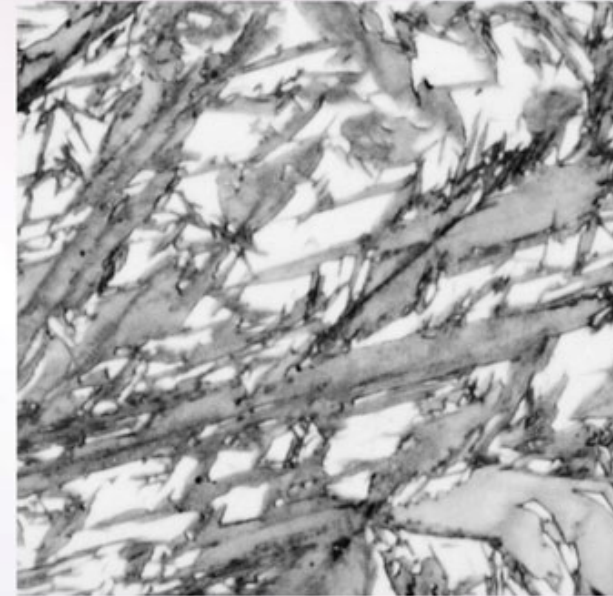


- TTT diagram is plotted for different length of times and temperature
- These curves gives us the information of transformation for different phases
- With the help of TTT diagram we can control the phase transformation by changing the conditions





(a)



(b)

(a) Hardness of martensite as a function of carbon content. (b) Micrograph of martensite containing 0.8% carbon. The gray plate-like regions are martensite; they have the same composition as the original austenite (white regions). Magnification: 1000x.

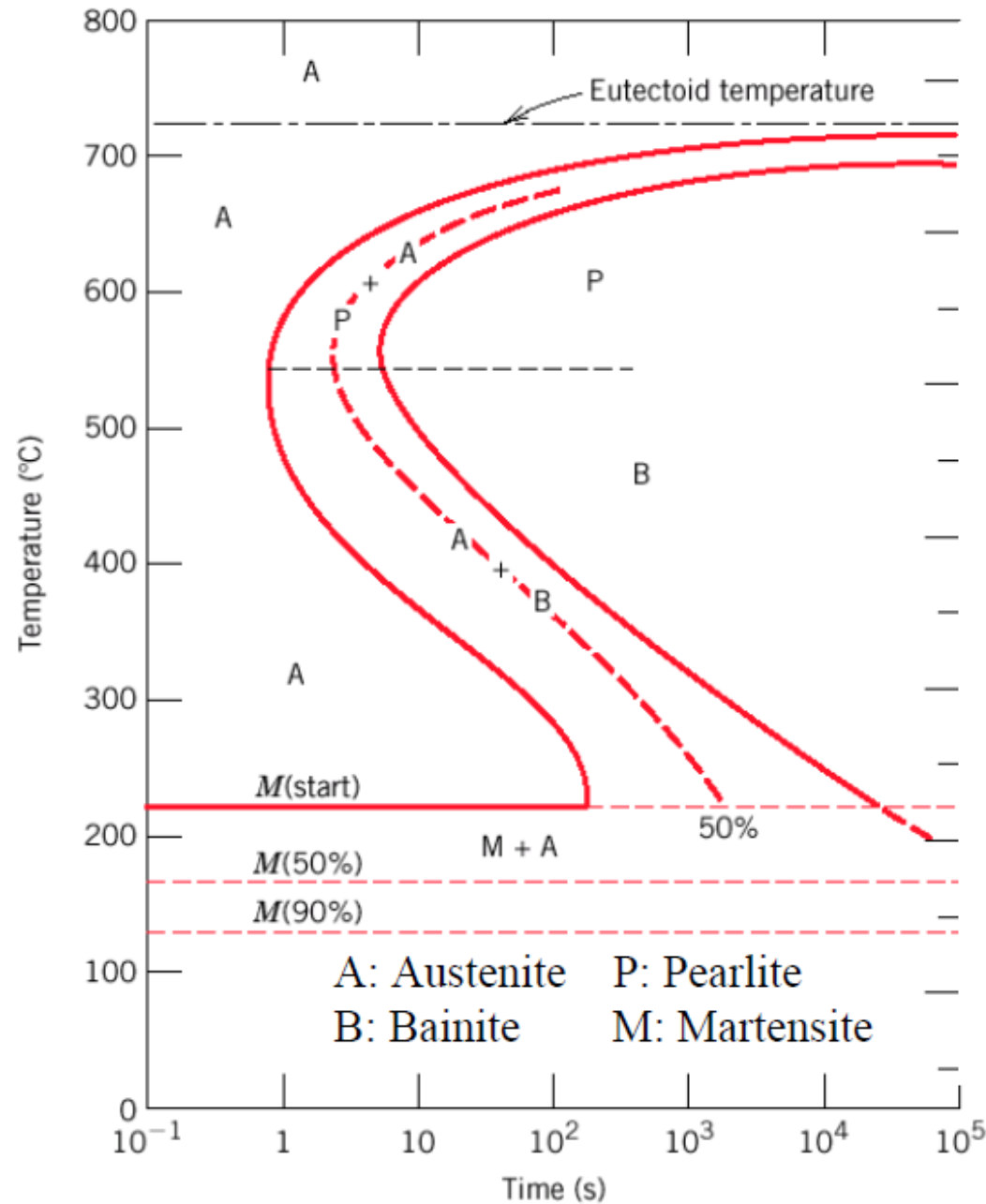


Martensite

- Martensite forms when austenite is rapidly cooled (quenched) to room T.
- It forms nearly instantaneously when the required low temperature is reached. The austenite-martensite does not involve diffusion → no thermal activation is needed, this is called an **athermal transformation**.
- Martensite is metastable - can persist indefinitely at room temperature, but will transform to equilibrium phases on annealing at an elevated temperature.
- Martensite can coexist with other phases and/or microstructures in Fe-C system.
- Since martensite is metastable non-equilibrium phase, it does not appear in phase Fe-C phase diagram.



TTT Diagram



Austenite-to-martensite is diffusionless and very fast. The amount of martensite formed depends on temperature only.

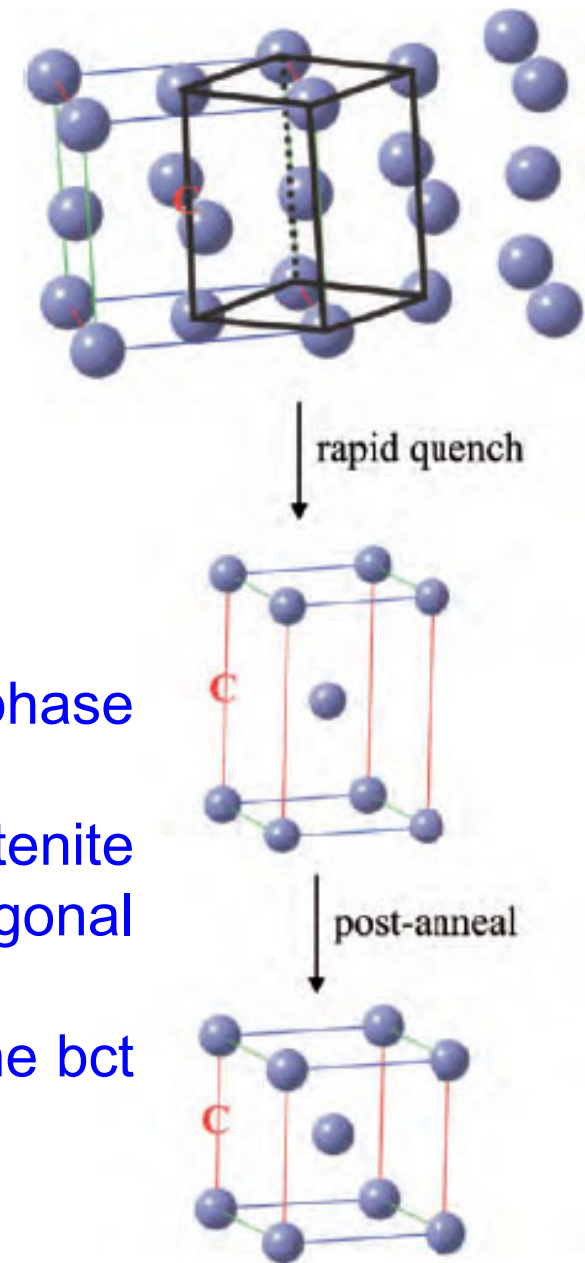


Alloying metals that are present in austenite will decrease the temperature required to start martensite growth.

$$M_s(^{\circ}\text{C}) = 539 - 423(\%C) - 30.4(\%Mn) - 17.7(\%Ni) \\ - 12.1(\%Cr) - 7.5(\%Mo)$$

The equation quantifies the effects of dopant concentrations on the temperature required for the onset of martensite formation, M_s . The greatest effects are seen for the austenite-forming elements of C, Mn, and Ni where even small concentrations result in a sharp decrease in M_s .

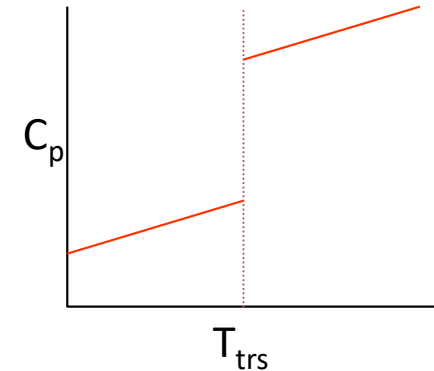
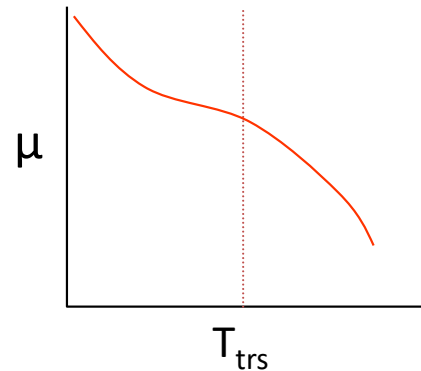
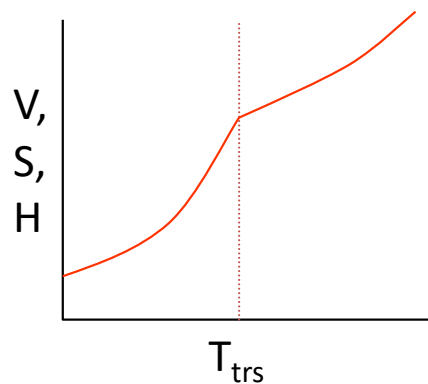
- Each atom displaces a small (sub-atomic) distance to transform FCC γ -Fe (austenite) to martensite which has a Body Centered Tetragonal (BCT) unit cell (like BCC, but one unit cell axis is longer than the other two).
- Crystallographic representation of the phase transformation from austenite to martensite.
- Two neighboring fcc unit cells of austenite associate, resulting in a body-centered tetragonal (bct) unit cell.
- A post-anneal known as tempering converts the bct structure to α -Fe.





2. Second-order phase transition

The first derivative of the chemical potential with respect to temperature is continuous but its second derivative is discontinuous..



At equilibrium,

$$\mu(\alpha) = \mu(\beta), \Delta G = 0,$$

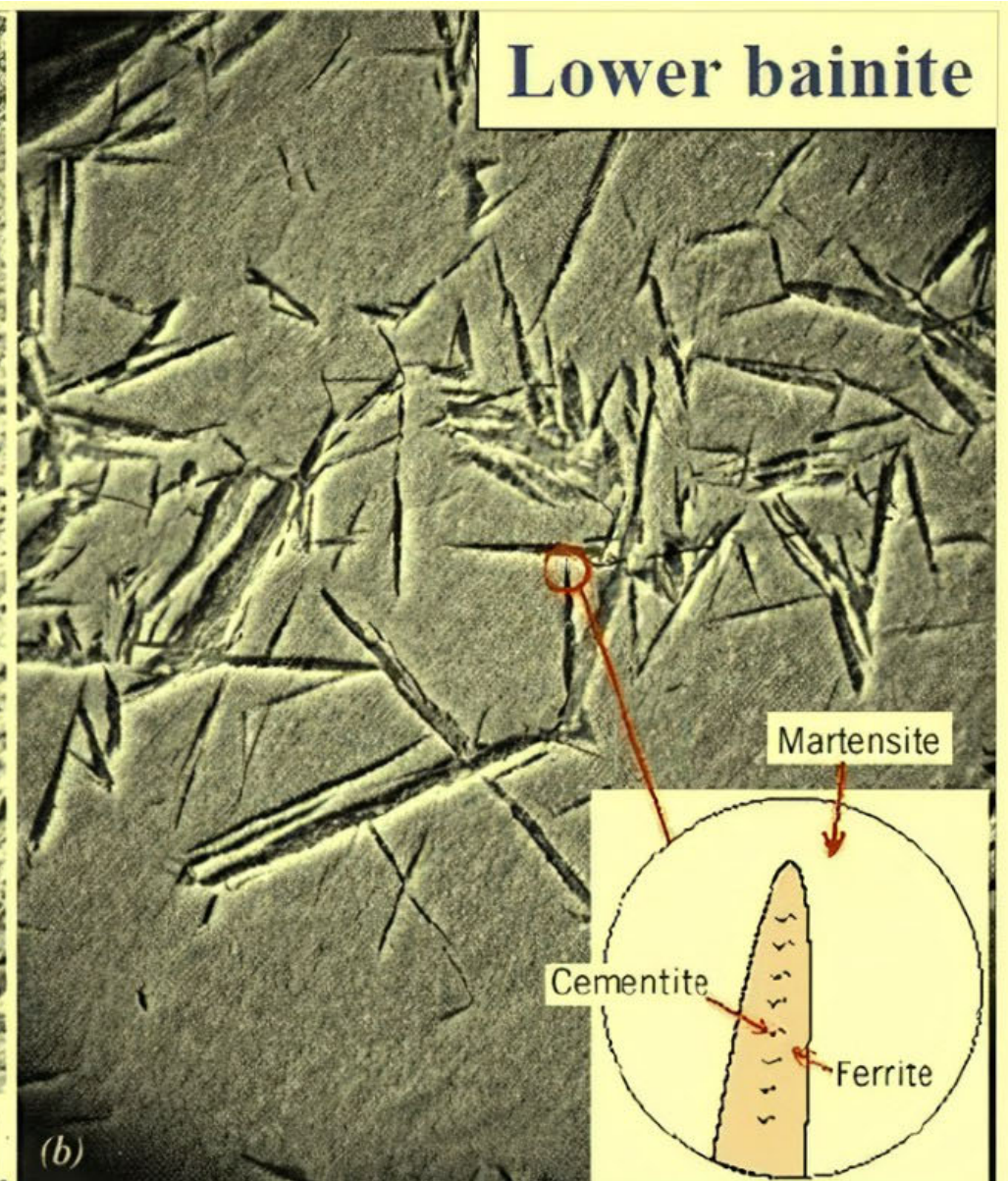
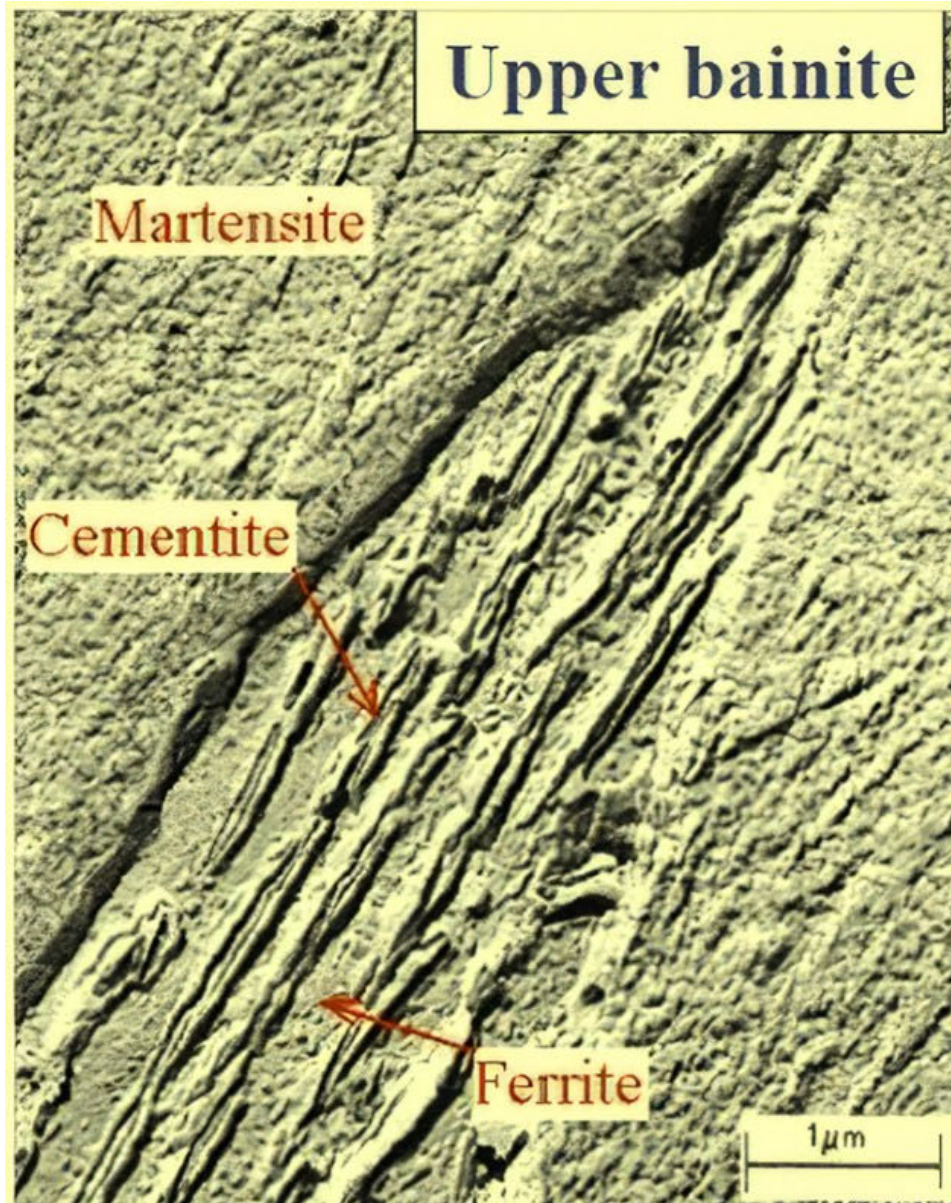
$$\text{but, } \Delta H = 0, \Delta S = 0, \Delta V = 0.$$

C_p is discontinuous but a finite value

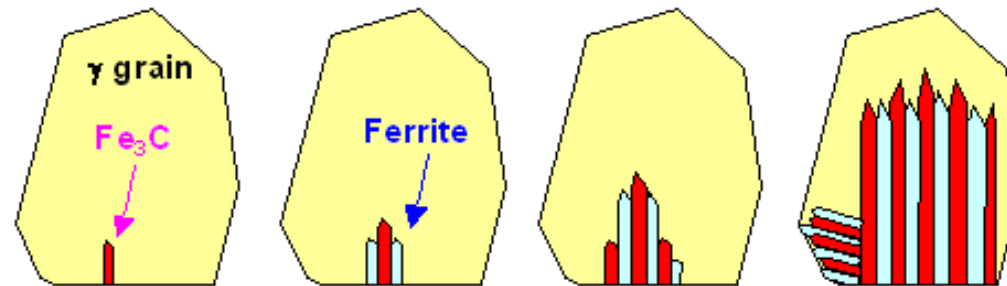
Bainite

- Bainite consists of acicular (needle-like) ferrite with very small cementite particles dispersed throughout.
- The carbon content is typically greater than 0.1%.
- Bainite transforms to **iron and cementite** with sufficient time and temperature (considered semi-stable below 150°C).

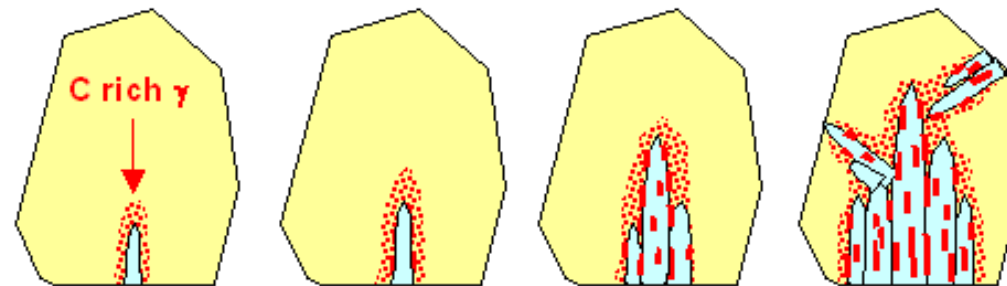




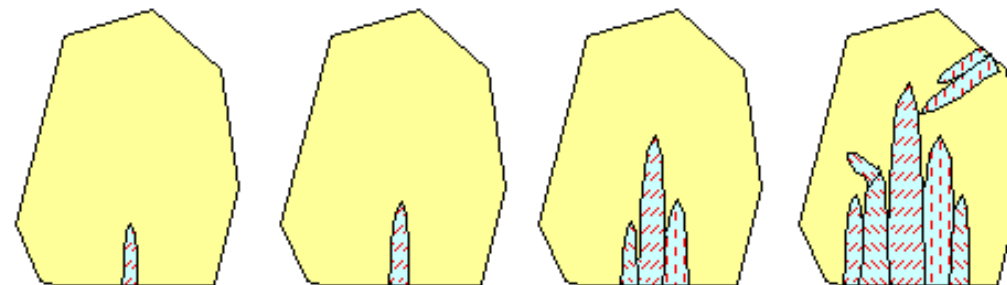
Pearlite Formation



Upper Bainite Formation



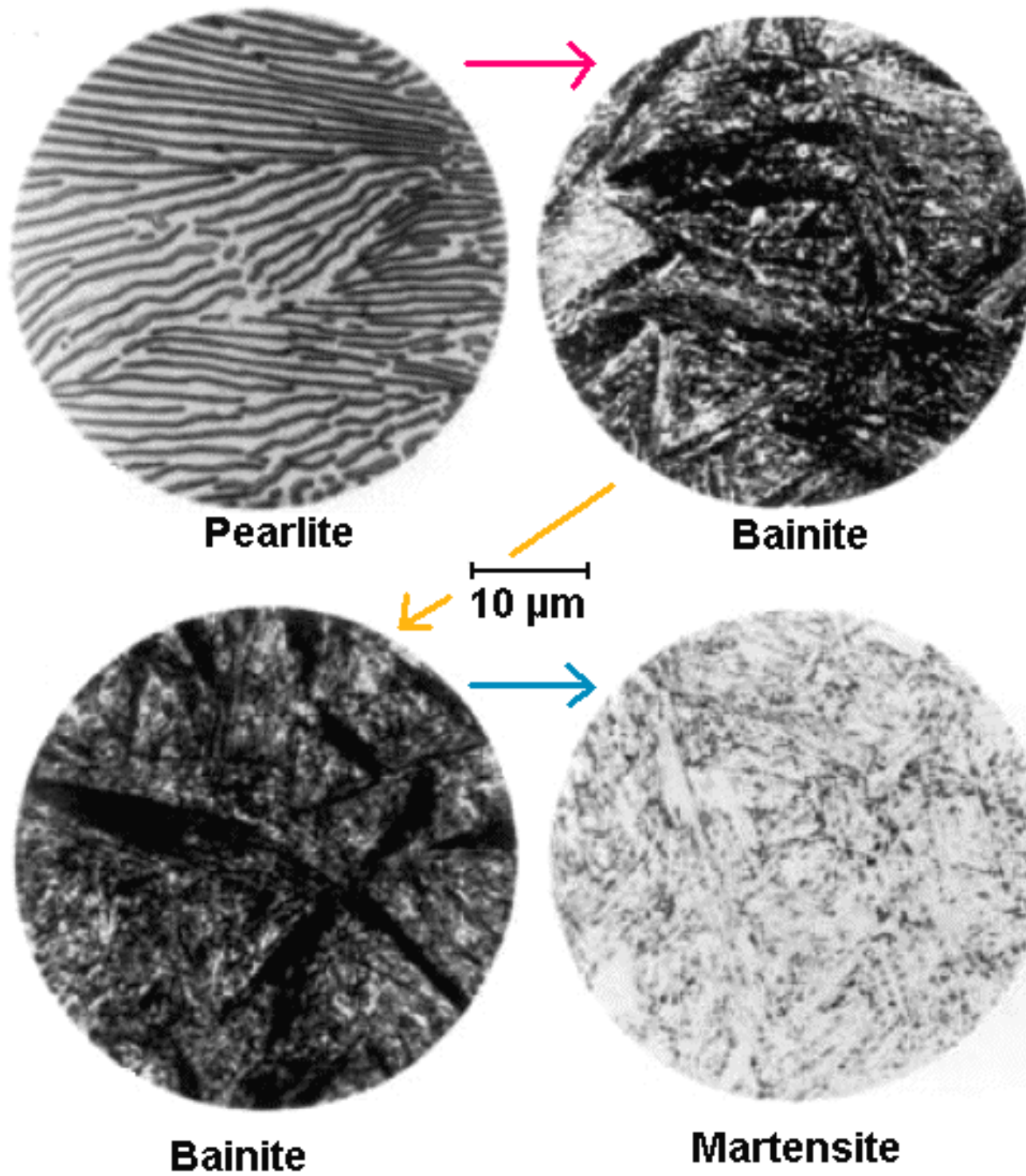
Lower Bainite Formation



Nucleation

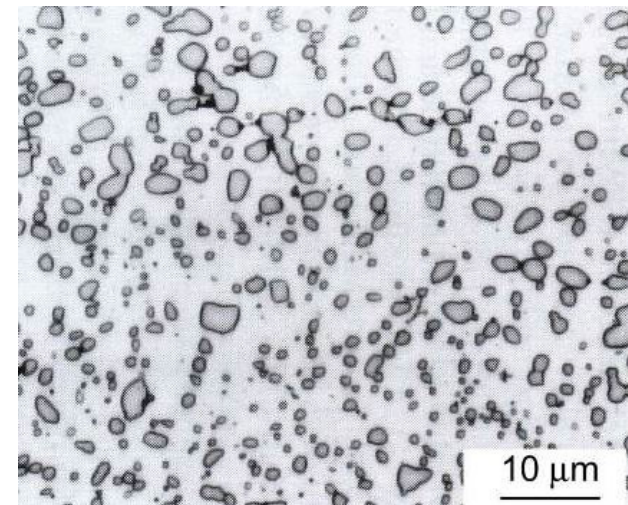
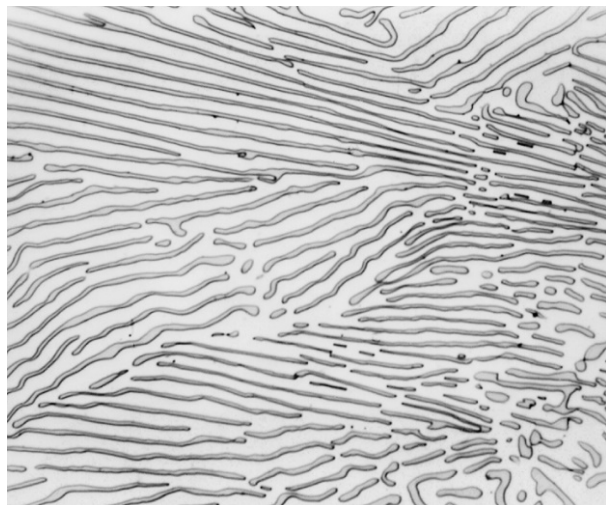
Growth

Appearance



Spheroidite

- Annealing of pearlitic or bainitic microstructures at elevated temperatures just below eutectoid (e.g. 24 h at 700 C) leads to the formation of new microstructure –**spheroidite** - spheres of cementite in a ferrite matrix.
- Composition or relative amounts of ferrite and cementite are not changing in this transformation, only **shape of the cementite inclusions is changing**.
- Transformation proceeds by C diffusion – needs high T.
- Driving force for the transformation - reduction in total ferrite - cementite boundary area.





Spheroidite

The prolonged high temperature environment affects the microstructure through interrupting Fe–Fe and Fe–C bonds. This allows for cementite (Fe_3C) regions to agglomerate into spheres, yielding a new morphological structure.

- Higher toughness
- Lower hardness



Tempering

One serious drawback of the above austenite/rapid quench method for martensite formation is the possibility of distorting and cracking the metal due to the rapid cooling event. During the quenching process, thermal stresses arise from the varying cooling rates experienced by outer and interior areas of the steel. In addition, there is a volume change when austenite is transformed to martensite.

- Two methods that have been used to reduce quenching stresses are martempering and austempering

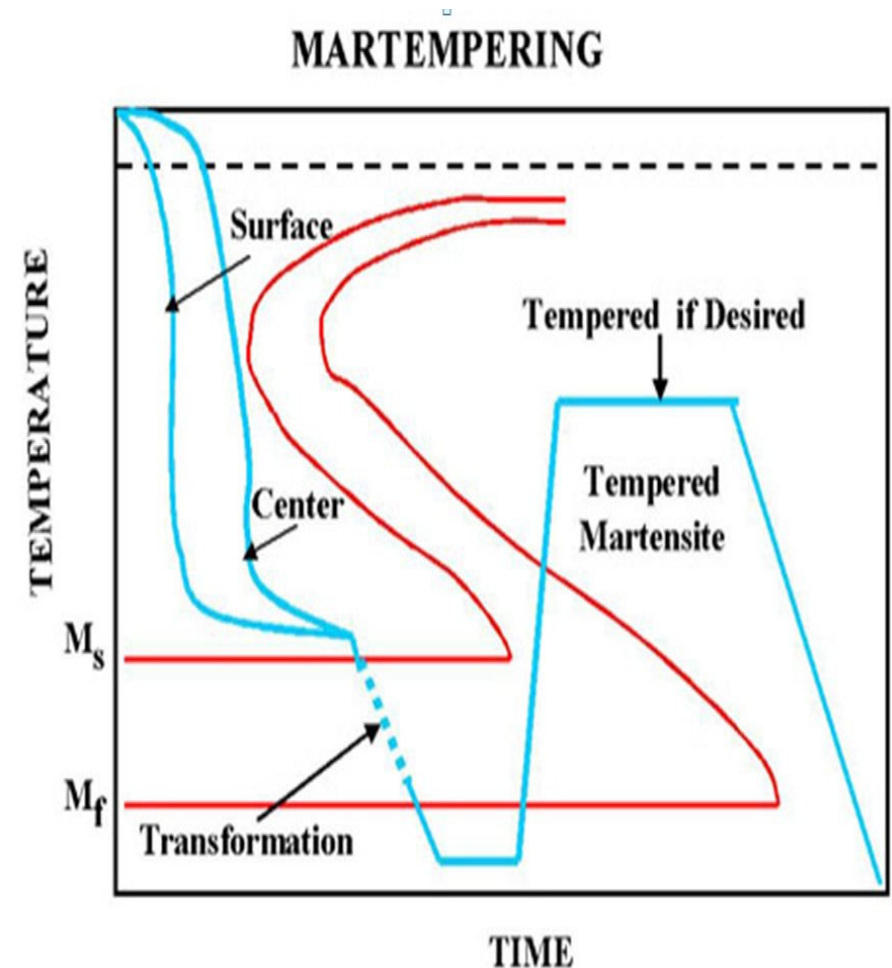


Tempered Martensite

Martensite is so brittle that it needs to be modified for practical applications. This is done by heating it to 250-650 °C for some time (tempering) which produces **tempered martensite**, an extremely fine-grained and well dispersed cementite grains in a ferrite matrix.

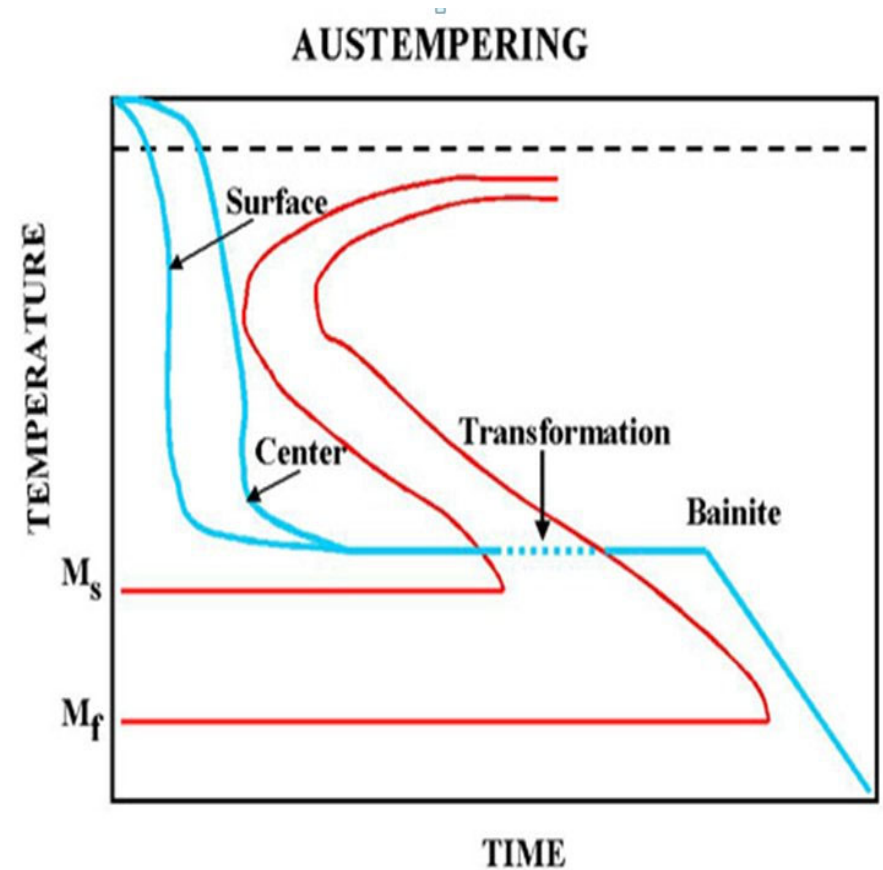
- Tempered martensite is less hard/strong as compared to regular martensite but has enhanced ductility (ferrite phase is ductile).
- Mechanical properties depend upon cementite particle size: fewer, larger particles means less boundary area and softer, more ductile material.
- Particle size increases with higher tempering temperature and/or longer time (more C diffusion) - therefore softer, more ductile material.

Martempering allows the transformation of austenite to martensite to take place at the same time throughout the structure of the metal part. By using interrupted quench, the cooling is stopped at a point above the martensite transformation region to allow sufficient time for the center to cool to the same temperature as the surface. Then cooling is continued through the martensite region, followed by the usual tempering process.

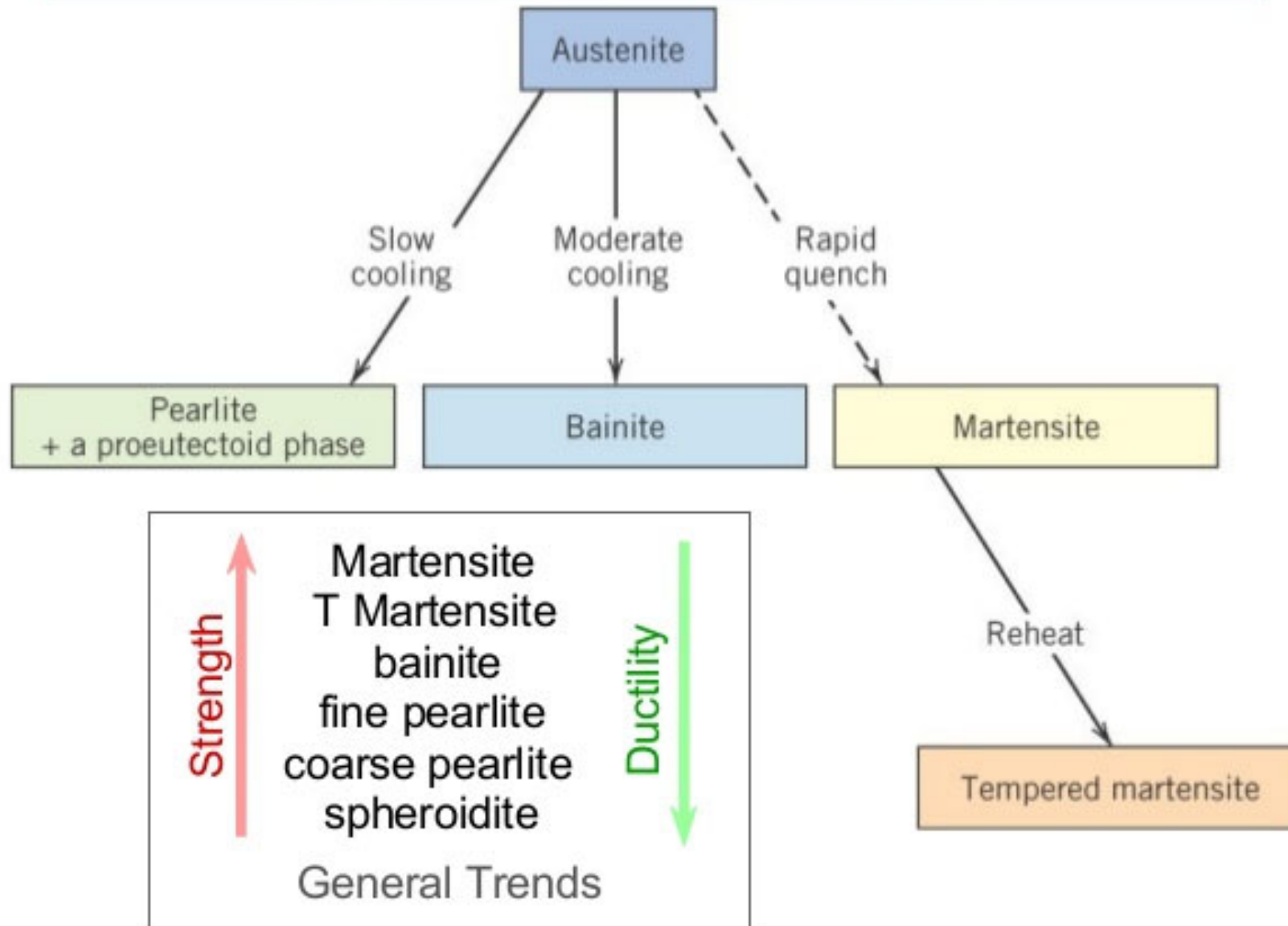




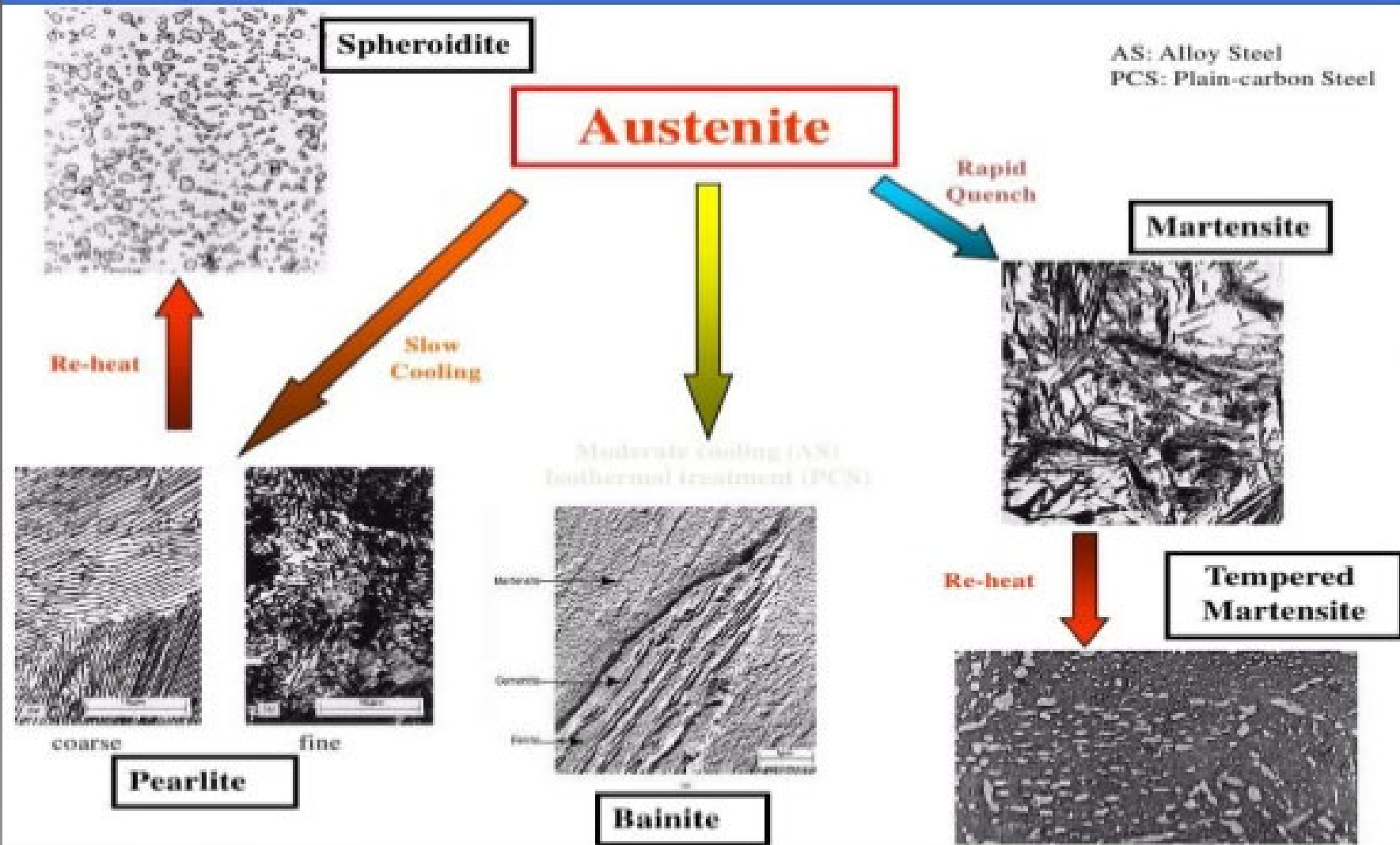
Austempering: the austenized steel is quenched at a rate faster than that required for pearlite formation, but above the temperature required for martensite growth. Hence, rather than transforming to martensite, the center and surface are converted to bainite – a strong material that shares the hardness of martensite with the toughness of pearlite. Bainitic steels are frequently used for the fabrication of gardening tools such as shovels.



Possible Transformations



Different transformed products of Austenite





Strategy V: Surface Hardening

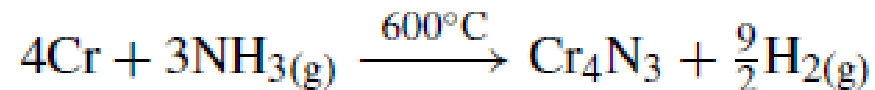
The above changes in the microstructure upon annealing do not only apply to the bulk material, but also for the surface.

- If an iron material is placed at high temperature in the presence of carbon vapor, a procedure known as **carburization** occurs, where carbon atoms diffuse into the surface of the steel, increasing the surface hardness.
- There must be careful control of the annealing atmosphere; if the steel is brought into contact with an oxidizing atmosphere, decarburization of the surface will occur through preferential formation of CO_2 .



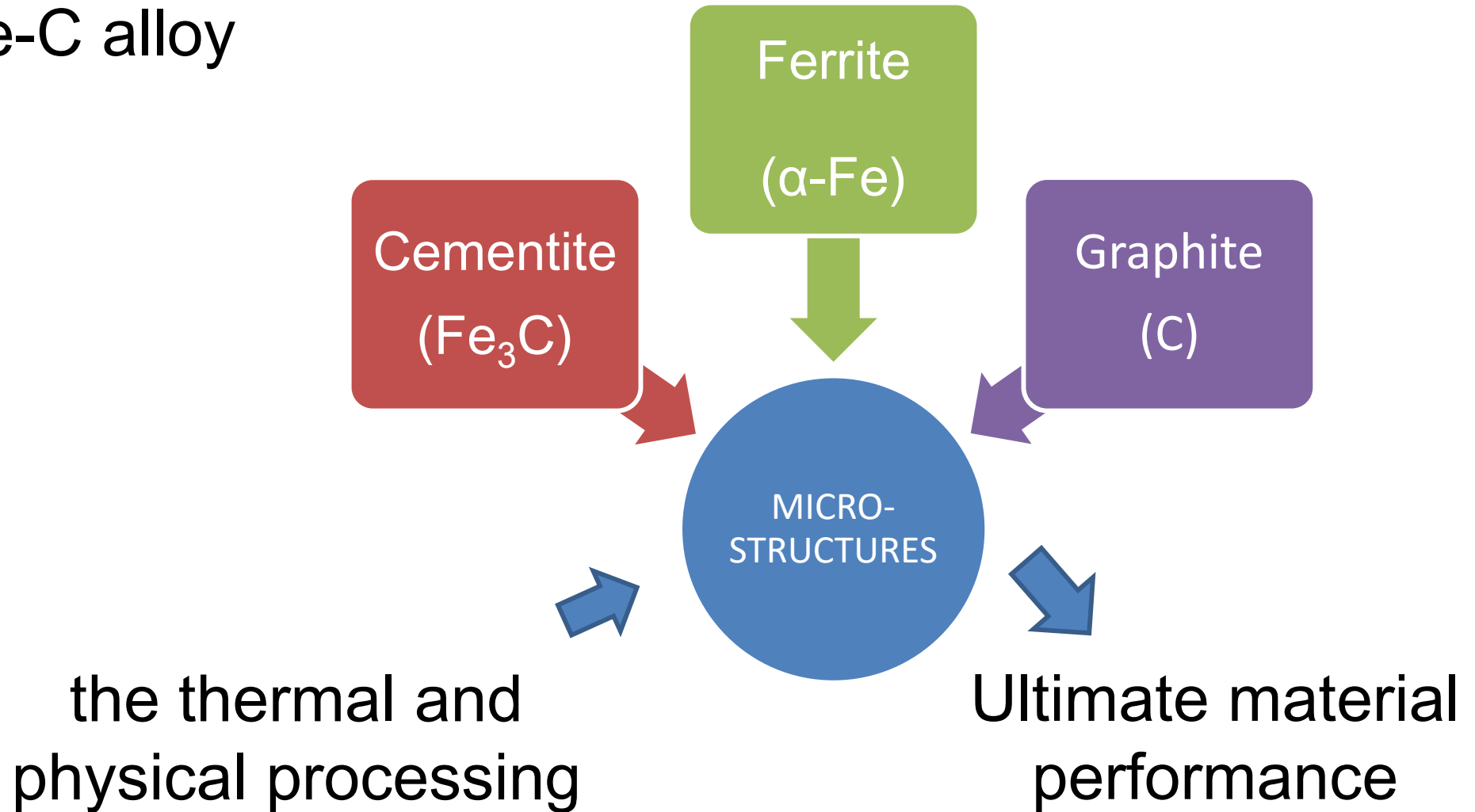
- Other surface hardening techniques introduce nitrogen to steels containing metals such as Al, Cr, and V that form stable nitrides.

Surface hardening techniques add a variety of attractive properties to steel components such as increasing wear and stress resistances, decreasing the odds of fracturing, and increasing corrosion resistance.





Fe-C alloy





3.2.3. Stainless Steels

A variety of other elements may also be present in steel that will alter its overall physical properties.

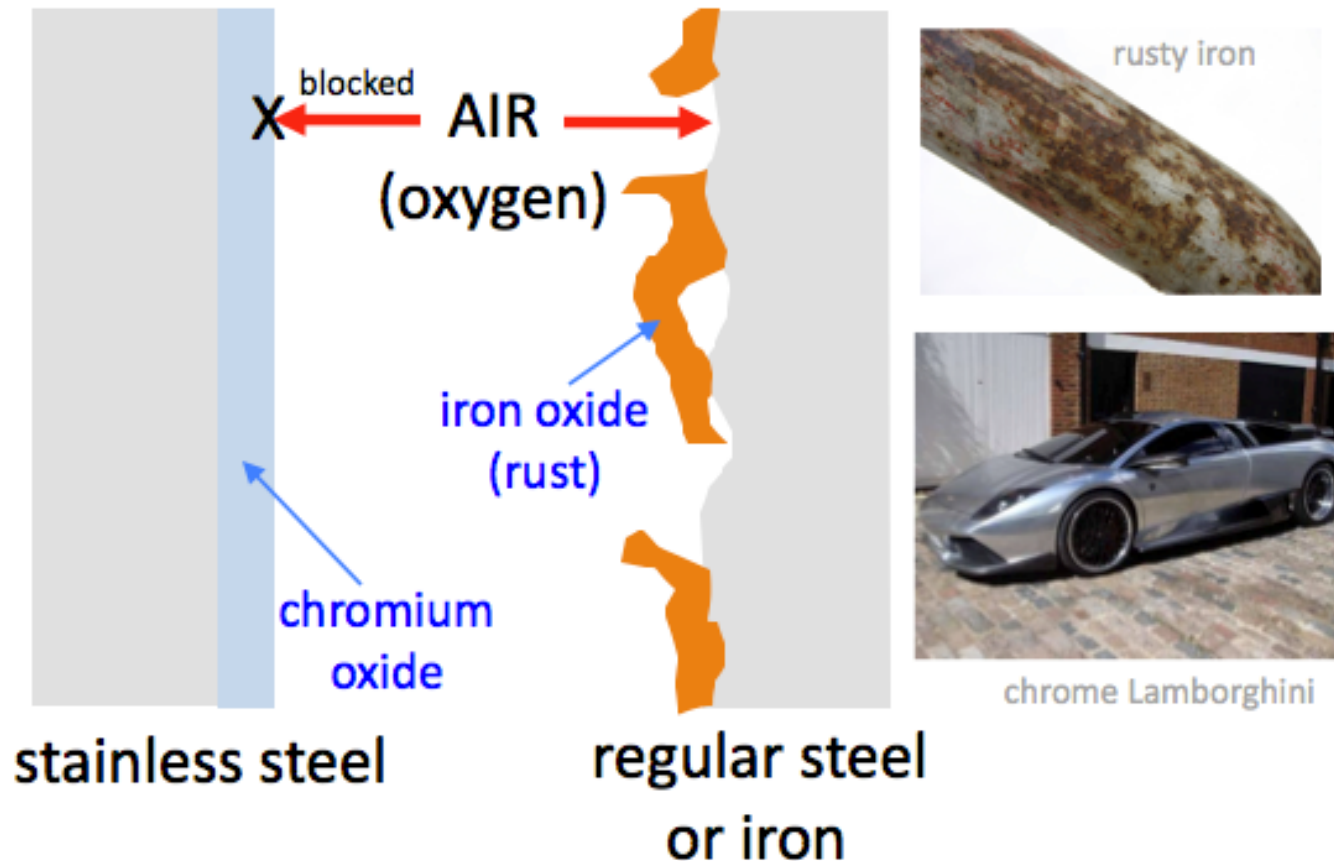
For example, all steels contain Mn.

- Assisting in hardening mechanisms,
- Removing sulfur and oxygen atoms from the matrix.
 - This prevents FeS formation and removes bubbles in the molten state of steels, both of which would greatly contribute to brittleness of the final product.



Stainless Steels

Chromium oxide blocks corrosion



Stainless Steels

Steel containing <0.30 wt%C and chromium concentrations >10.5 wt%

The addition of Cr results in the formation of a native layer of Cr_2O_3 , providing corrosion resistance. As the concentration of Cr is increased, the material is concomitantly less predisposed to rust.



$$\Delta G^\circ = -nFE^\circ$$

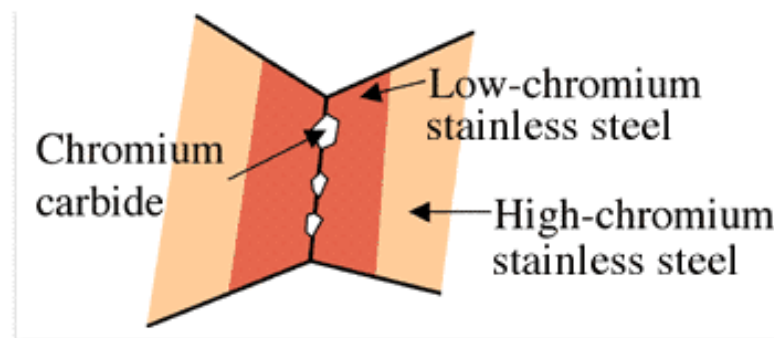


The difference between stainless and plated steels:

- Stainless steels will self-repair itself when scratched. Since the chromium is homogeneously dispersed throughout stainless steel, a scratch will serve to expose additional Cr sites forming additional layers of the protective oxide.
- The application of a protective coating over steel will only be an effective barrier as long as it remains intact. When this coating is penetrated by a scratch/crack, the bare steel is exposed to the surrounding environment allowing the possibility for corrosion.
- Aluminum and silicon are also added to steel that also form native oxides that are effective in preventing surface corrosion of the underlying metal.

Sensitization (Weld decay)

- When stainless steels are overheated (ca. 400-800°C) for a prolonged period.
- During the heating, the chromium in the stainless steel reacts with carbon in the steel and forms particles of chromium carbide at the grain boundaries. The regions near the grain boundaries become depleted in chromium.





- The sensitization process can be reversed by reheating the steel to temperatures in excess of $1,000^{\circ}\text{C}$ for a period long enough to re-dissolve the chromium carbide particles and form a homogeneous solid solution.
- Rapid cooling must then be introduced to suppress the reformation of carbide.

A stainless steel to be used at high temperatures:

- ✓ Low C compositions
- ✓ Doping with carbide-forming metals such as V, Ti, or Ta that are more easily oxidized than Cr.
- ✓ More chromium must be added as the concentration of carbon is increased to ensure effective corrosion resistance.



Classification

There are currently over 200 commercially available types of stainless steels.

Table 3.3. General Types and Properties of Stainless Steels

Type	Concentration	Properties/applications
Martensitic	11–20 wt% Cr 0.15–0.75 wt% C	High hardness, magnetic/cutlery, blades, surgical instruments, valves, springs
Austenitic	16–26 wt% Cr 35 wt% Ni 20 wt% Mn	High and low temperature resistance, ductility, superior corrosion resistance/kitchen sinks, ovens, reaction vessels, food processors, gutters
Ferritic	10.5–30 wt% Cr <1 wt% C, N, Ni	Magnetic, inexpensive/automotive exhaust and fuel lines, cooking utensils, bank vaults, washing machines, dishwashers
Duplex (austenitic– ferritic)	18–26 wt% Cr 4–7 wt% Ni 2–3 wt% Mo	Weldable, high tensile strength, Cl^- ion resistance (acidic environments)/desalination plants, food pickling plants, petrochemical plants, pulp and paper industries



Austenitic stainless steels

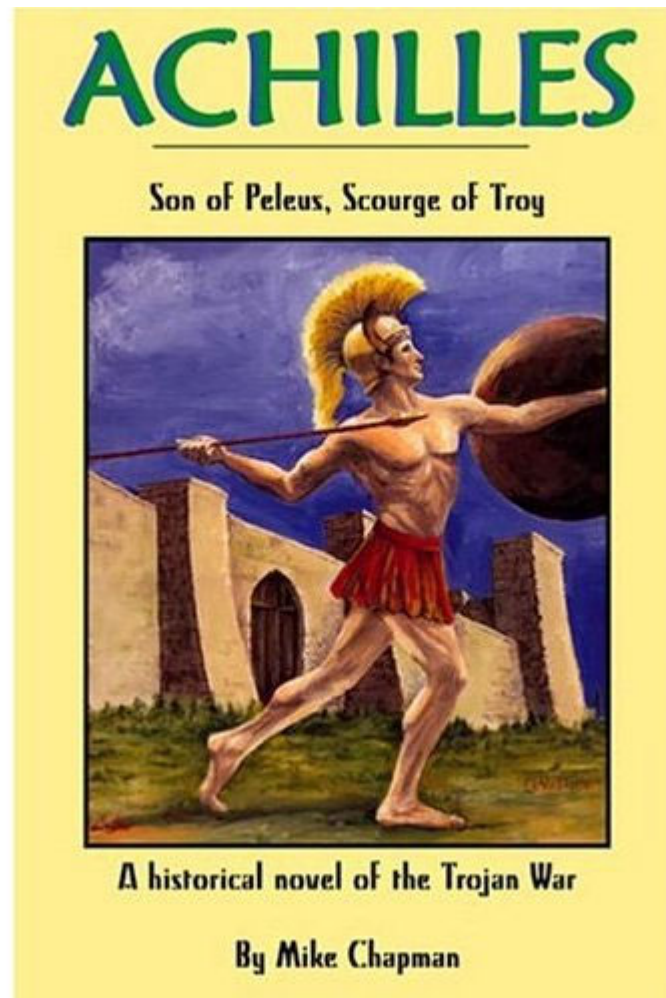
The industrial applications for austenitic stainless steels far outweigh the other types due to their facile work hardenability and high corrosion resistance.

As we have seen, the fcc austenite phase is not stable at temperatures below 723°C ; however, austenite-stabilizers such as Ni, Mn, Cu, C, or N may be added to extend the stability of this phase down to room temperature.

Due to high concentrations of easily oxidizable elements such as Cr, Ni, and Mn, the corrosion resistance is the greatest for austenitic stainless steels.



However, their **Achilles' heel** is their reaction with chloride ions. Cl^- will preferentially react with the metal centers, forming MCl_x rather than a protective coating of M_xO_y .





Ferrite stainless steels

- Since ferritic stainless steels contain more carbon than other classes, they are relatively harder to weld and shape than other varieties, which have historically limited their applications.
- However, since the 1960s, processes such as “argon oxygen decarburization” have resulted in steels with less carbon, allowing for smaller concentrations of chromium to be used. As a result, the price for ferritic stainless steel has dramatically dropped, and a number of applications now employ these materials– more than 2/3 of which include automotive exhaust systems.



Duplex stainless steels

Table 1. Difference in the properties of ferritic and austenitic stainless steels.

Properties	Ferritic	Austenitic
Toughness	Moderate	Very high
Ductility	Moderate	Very high
Weldability	Limited	Good
Thermal expansion	Moderate	High
Stress corrosion cracking resistance	Very high	Low
Magnetic properties	Ferro magnetic	Non-magnetic



Duplex stainless steels

- Duplex stainless steels feature the “best of both worlds” since they contain both ferritic and austenitic phases.
- The ferritic phase helps to circumvent the problems associated with stress-corrosion cracking in chloride environments, while the austenitic component helps to improve the generally low strength and ductility of purely ferritic steels.
- This biphasic steel is generated through careful heating/cooling of the Fe–Cr–Ni alloy. When these materials initially solidify, they are 100% ferritic in nature; subsequent cooling causes the precipitation of austenite within individual ferrite crystals.



Martensitic stainless steels

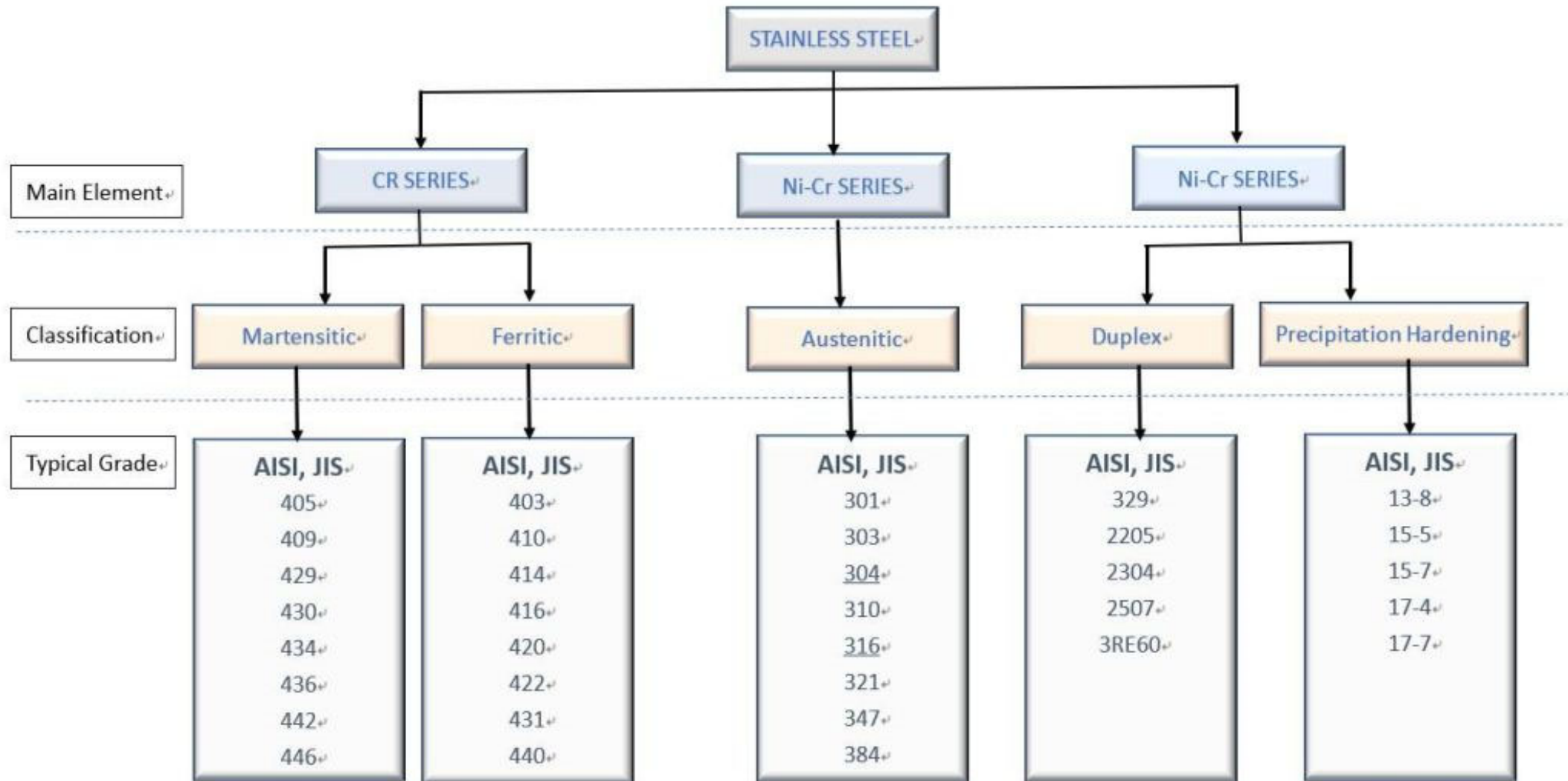
Strengthening of stainless steels is carried out through cold-working processes –rolling into sheets or drawing into wires/rods at temperatures around 25°C. This generates a strong material through formation of a distorted bcc lattice that is roughly analogous to martensite.

Martensitic steels have poor weldability and ductility. Corrosion resistance of these steels is moderate (slightly better than in ferritic steels).



Table 3.3. General Types and Properties of Stainless Steels

Type	Concentration	Properties/applications
Martensitic	11–20 wt% Cr 0.15–0.75 wt% C	High hardness, magnetic/cutlery, blades, surgical instruments, valves, springs
Austenitic	16–26 wt% Cr 35 wt% Ni 20 wt% Mn	High and low temperature resistance, ductility, superior corrosion resistance/kitchen sinks, ovens, reaction vessels, food processors, gutters
Ferritic	10.5–30 wt% Cr <1 wt% C, N, Ni	Magnetic, inexpensive/automotive exhaust and fuel lines, cooking utensils, bank vaults, washing machines, dishwashers
Duplex (austenitic– ferritic)	18–26 wt% Cr 4–7 wt% Ni 2–3 wt% Mo	Weldable, high tensile strength, Cl^- ion resistance (acidic environments)/desalination plants, food pickling plants, petrochemical plants, pulp and paper industries





3.2.4. Nonferrous Metals and alloys

A. The coinage metals

1 H																	2 He	
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne	
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
55 Cs	56 Ba	57 *La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra	89 +Ac	104 Rf	105 Ha	106 Sg	107 Ns	108 Hs	109 Mt	110 110	111 111	112 112	113 113						
			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu		
			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr		

Non-metals

Metals

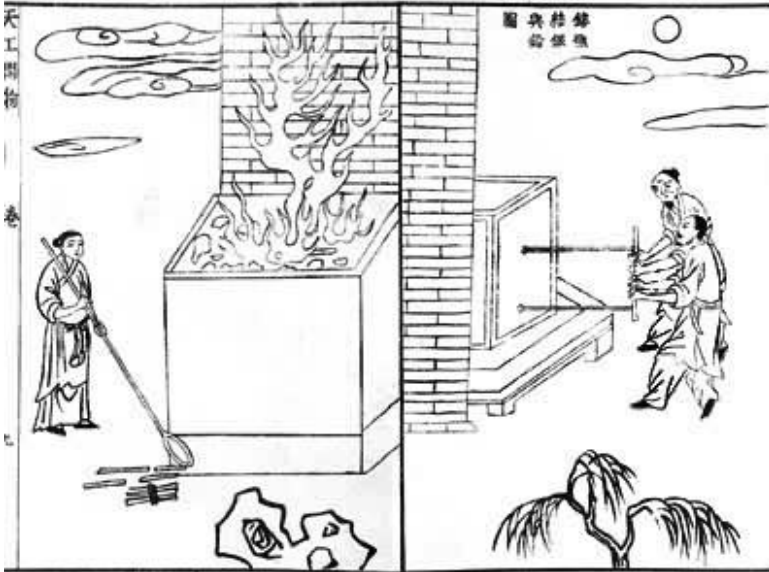
Metalloids



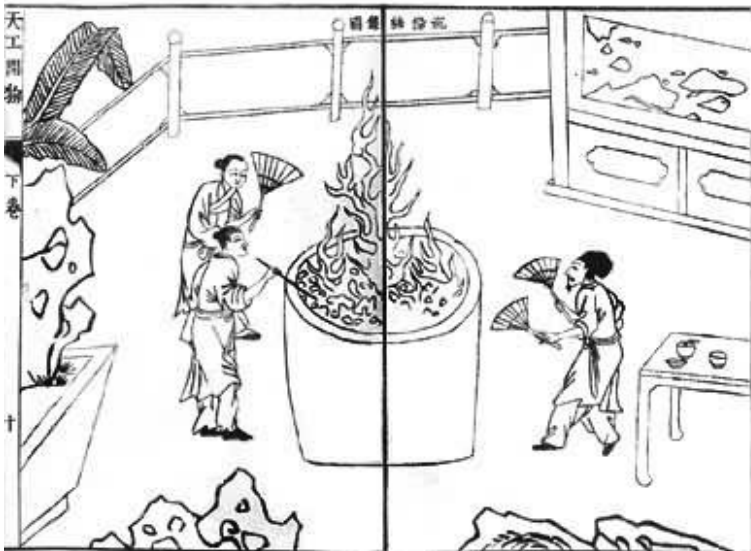


Cupellation

3000 B. C.

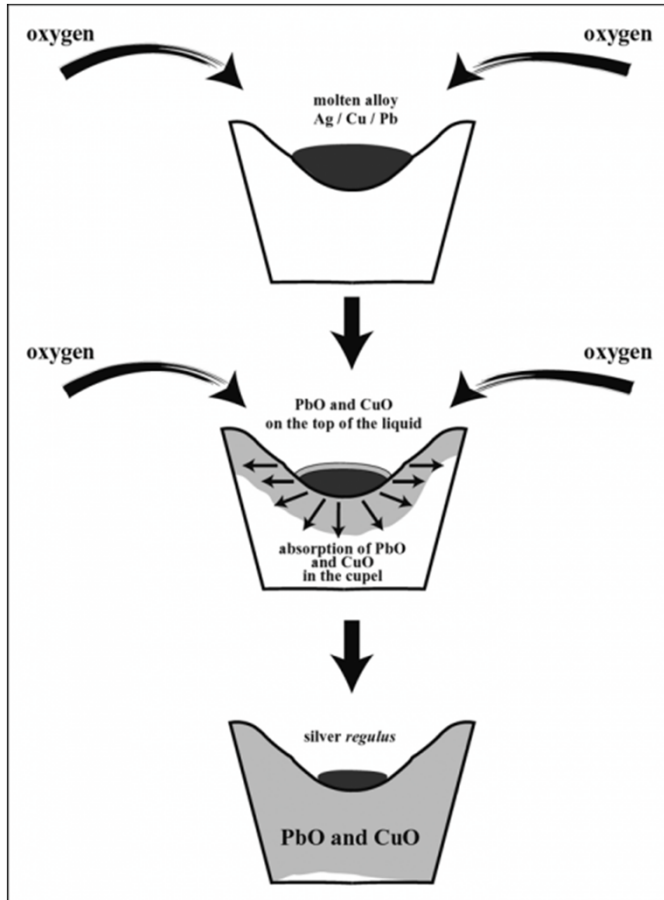


《天工开物》中的“熔礁结银与铅”图



《天工开物》中的“沉铅结银”图





- Lead was added to the ore and heated to a temperature of *ca.* 800 °C in air.
- Any gold or silver in the sample was dissolved in the liquid metal, separating it from the undissolved matter. The insoluble material primarily containing iron, copper, and zinc compounds was discarded.
- The remaining precious metals were brought to a higher temperature by blowing the fire with bellows. This raised the temperature to the point where lead oxide formed rapidly.
- This was usually performed in a hearth comprising clay or crushed bones; the PbO would be absorbed into the hearth, while the precious metal deposited at the surface.

Melting temperature of
Pb: 328°C



Properties

- High melting points, High density, and Inertness

$nd^{10}(n+1)s^1$ Coinage metals

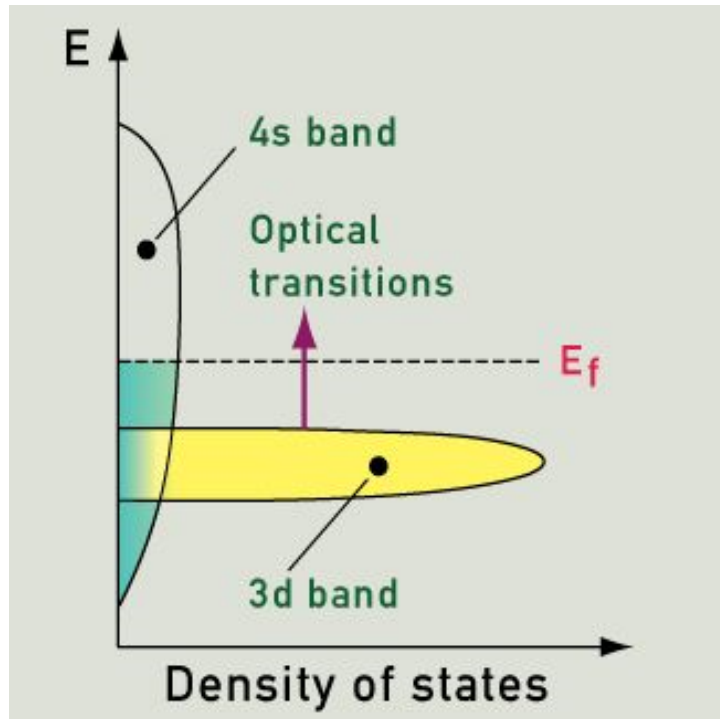
$np^6(n+1)s^1$ Alkali metals

A filled d shell is far less effective at shielding an outer s electron from nuclear attraction than the less diffuse p shell. As a result, the first ionization energy of the coinage metals is much higher than the alkali metals, and their bonding is significantly more covalent in nature.



- Reactivity
 - ✓ The un-reactivity of the coinage metals increases dramatically from Cu to Au;
 - ✓ Copper and silver readily react with sulfur and halogens; Acid rain produced from gaseous sulfur compounds (SO_x) react with copper surfaces, eventually resulting in the formation of a basic copper sulfate film, as evidenced by a green-blue-patina color.
 - ✓ Gold is completely unreactive to all reagents except very strong oxidizing acids such as aqua regia (3:1 HCl/HNO_3).
 - ✓ Another useful property of copper and silver are their antifungal behaviors.

- Colors



By definition, the highest-occupied energy level at 0K is referred to as the *Fermi level*

Electrons absorb energy from incident light, and are excited from lower energy levels to higher, vacant energy levels.

If an energy level (like the 3d band) holds many more electrons (than other energy levels) then the excitation of electrons from this highly occupied level to above the Fermi level will become quite important.



Copper and gold are the only metals that exhibit colors in their bulk state.

- ✓ For **copper**, the gap from the top of the 3d-band to the Fermi surface is *ca.* 544 nm. Hence, energy in the green/blue region of the spectrum may be absorbed, resulting in an observed red/orange color.
- ✓ For **gold**, the gap is *ca.* 400 nm, corresponding to absorption in the blue region of the spectrum and an observed golden color.
- ✓ By contrast, the analogous energy gap for **silver** is 311 nm, resulting in UV absorption and an observed white/silver color (*i.e.*, equal reflection of all visible wavelengths).



- Silver and copper have the highest electrical and thermal conductivities of all metals in the Periodic Table.
- Gold is the most electronegative metal, behaving as a halogen through formation of Au^- ions in the presence of strongly electropositive metals such as cesium.



Gold Alloys

“Carat” refers to the weight of precious stones (1 carat = 200 mg).

“Karat” is used to describe the purity of metals.

“Fineness” refers to the weight portion of the precious metal in the alloy.

For example,

24 karats represents 100% purity and a fineness of 1,000;

18 karats represents a purity of 75% and a fineness of 750.



Unalloyed gold is almost never used for applications due to its softness.

A number of alloying agents may be added to gold such as Ag, Cu, Zn, Ni, Pt, and Pd. As required for solid solutions, each of these dopants is of a similar size, and has an fcc crystal lattice that matches that of gold.

these dopants to gold:

- improving the strength
- imparting colors



Table 3.4. Commonly Used Alloys for Colored Golds

Composition (karat)	%Au	%Ag	%Cu	%Zn	%Ni	%Ti	%Pd	%Fe	%Si	%Co
<i>Yellow gold</i>										
23	99.0						0.9			
18	75.0	13.0	12.0							
18	75.0	15.0	10.0							
14	58.3	4.0	31.2	6.4	0.1			0.05	0.01	
14	58.3	24.8	26.8	0.14						
10	41.7	11.7	40.8	5.83					0.03	
10	41.7	5.5	43.8	9.0						
<i>White gold</i>										
18	75.0		2.23	5.47	17.8					
18	75.0	15.0					10.0			
14	58.3			28.3	4.8	8.6				
14	58.3	32.2					9.5			
10	41.7		29.2	12.1	15.1					2.0
10	41.7	47.4		0.9			10.0			
<i>Rose gold</i>										
18	75.0	5.0	20.0							
14	58.3	2.1	39.6							
10	41.7	2.8	55.5							



Copper Alloys

As a more stringent application of the Hume-Rothery sizes of alloying metals, if the difference in radii is less than 8%, the metals will be soluble throughout the full range of compositions.

Nickel and copper radii are 1.49 and 1.45Å, respectively.

There are over 20 different alloys that are used in industry based on the mutual solubility of copper and nickel in all proportions.



- Monel (68% Ni, 32% Cu) is used to handle corrosive materials such as F_2 or HF.
- US currency coinage such as quarters and nickels are alloys of Cu/Ni, containing 91.7% Cu and 75% Cu, respectively.
- Pennies are predominantly nickel, with a thin layer of copper deposited through electroplating.





第三套人民币1元硬币由铜镍合金组成。



第四、五套人民币1元硬币由钢芯镀镍组成。





第一套是长城币，第一套是长城币，1980年至1985年，只发行了4年，是中国流通硬币的开山鼻祖，它是有70%的铜和30%的锌，组成的合金材质，因发行量少，流通时间短，现在的收藏价，根据年份的不同，已经翻了几百倍到上万倍，当然这只是它的收藏价值，也并不是含有黄金的原因。



第二套是梅花5角，1991年到1994年，材质为进口铜材，同样为铜锌合金，1995年到2001年的材质为国产铜，一共发行的11年，从2016年开始银行只收不发，逐渐退出市场，收藏价现在到了几十到几百元，其中93年的因铜制好，发行量少，收藏价值最高，如果你手中还有梅花五毛硬币，可不要随便花了。



第三套是荷花币，发行时间是2002年到至今，它的材质也和以前有很大的不同，从70%的铜芯改成了钢芯镀铜，上面的中华人民共和国字样也改成了中国人民银行，我们手上常见的五毛硬币，都是这种荷花币，十多年间，发行量有上百亿枚。



The atomic radii of Cu, Sn($1.45\text{\AA}$), and Zn($1.42\text{\AA}$) are also nearly identical.

Cu/Sn alloy: bronze

Cu/Zn alloy: brass

bronze

The strength of a bronze increases with the tin content; however, its toughness and malleability decreases.

The maximum strength of bronze occurs at ca. 30% Sn, but at this concentration the alloy is much too brittle for most applications due to the formation of Cu_3Sn particles.



Brass

Although the radii of Cu and Zn satisfy the Hume-Rothery constraints for solid solutions, these metals do not share the same crystal lattice.

The coinage metals: fcc

Zinc: hcp

- For small concentrations of Zn, the fcc α -brass structure is formed.
- As the Zn concentration reaches 50%, the bcc β -brass (CuZn) phase predominates.



- Additional zinc may continue to dissolve in this phase until a concentration of *ca.* 75% Zn is reached, which results in the final HCP phase referred to as ϵ -brass, CuZn_3 .
- Beyond this concentration, additional zinc results in the HCP η -brass phase which is no longer considered an alloy, but pure Zn.

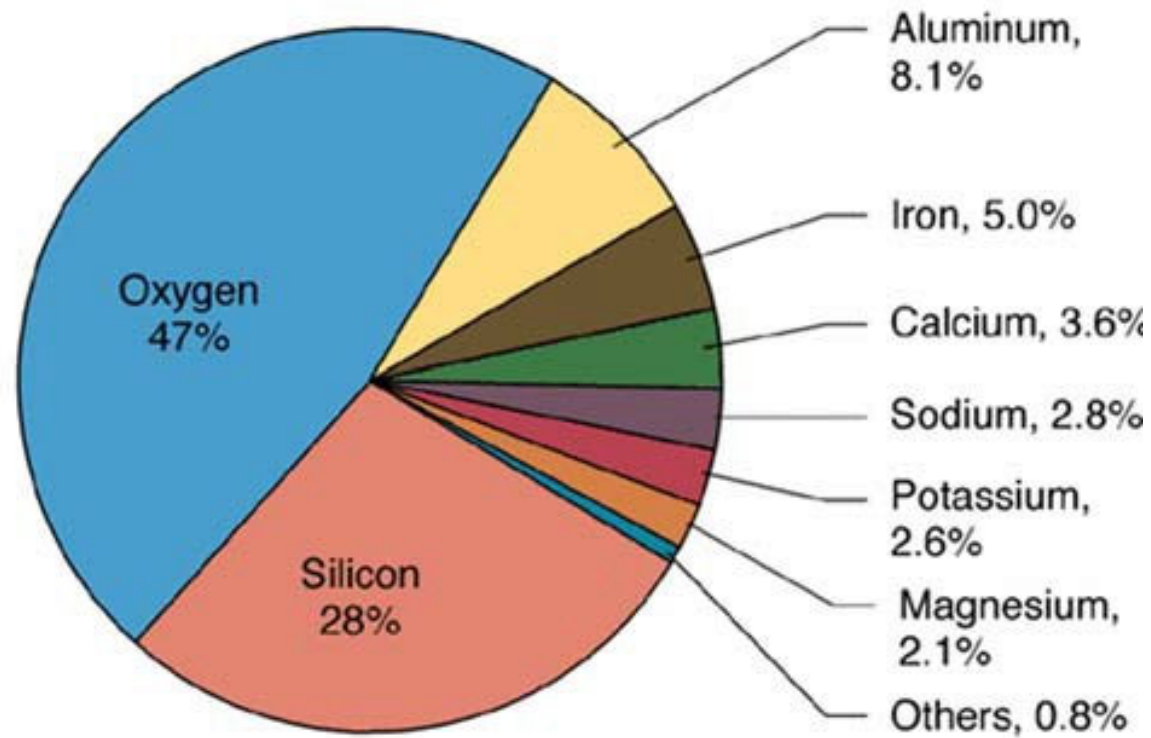
Only the α and β phases are useful alloys; the others are too hard and/or brittle.



B. Aluminum alloys

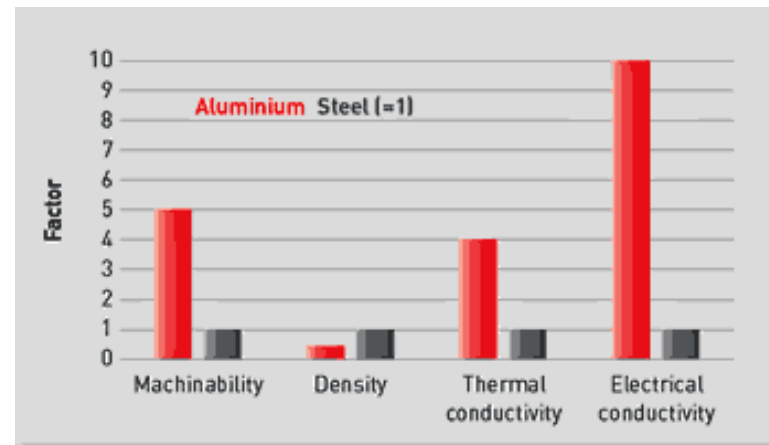
Another widely used metal. Why?

Abundant element of 8.3 wt% on earth crust and normally found in oxide forms (Al_2O_3), i.e., bauxite, kaolinite, nepheline and alunite.



Properties

- High corrosion resistance
- Nonsparking
- Excellent machining properties
- Light weight
- High thermal/electrical conductivity
- High ductility/easily deformable



Comparison of Al and steel



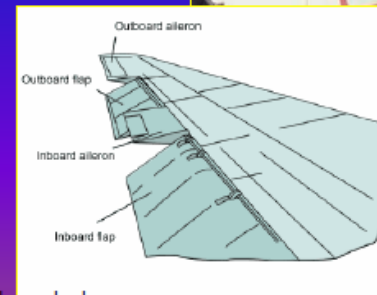
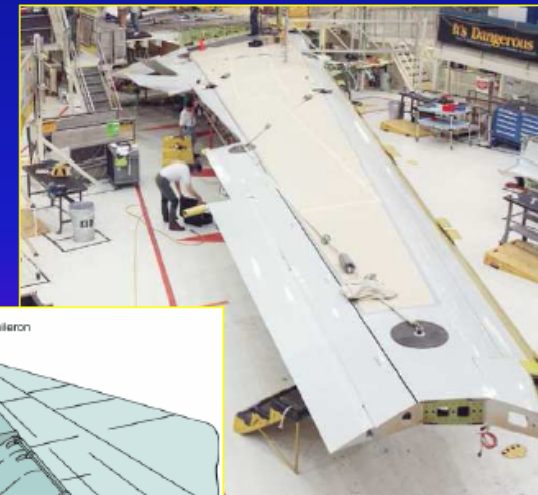
**Construction
& Equipment**



**Containers
& Packaging**



Automotives



Aerospace



- Mg, Cu, Zn and Si are the most commonly used alloying elements in aluminum, which have sufficient solid solubility.
- Cr, Mn and Zr are used primarily to form compounds which control grain structure.

Annealed high-purity aluminum has very low yield strength, but can be strengthened by solid solution hardening.

- Mg is the most effective strengthener on the weight basis due to its high solubility.
- Mn and Cu are the most effective strengtheners at 0.5%, but tend to form Al_6Mn precipitates and insoluble Al-Cu-Fe constituents respectively .



Table 3.5. Types and Properties of Aluminum Alloys

Metal additive	Resultant properties/applications
Cu (<i>ca.</i> 5%)	Heat treatable, high strength to weight ratio, limited corrosion resistance and weldability/autowheels and suspension components, aircraft fuselage, power lines
Mn (<i>ca.</i> 1.2%)	Moderate strength without heat treating, high workability/beverage cans, cooking utensils, heat exchangers, storage tanks, furniture, highway signs, roofing, side panels
Si (<i>ca.</i> $\leq$ 12%)	Low thermal expansion and m.p., high wear resistance/forged engine pistons, welding rod, brazing alloys, architectural products
Mg (<i>ca.</i> 0.3–5%)	Good weldability and strength, good corrosion resistance/ornamental trim, cans, household appliances, boats and ships, bridge railings, race cars
Zn (<i>ca.</i> 3–8%)	Heat treatable, moderate–very high strength/airframe structures, high-strength forgings



Mg-Al alloy

- Mg in Al: solid-state strengthening
- The maximal solubility is 5 wt%, and down to 1.5 wt% at room temperature.
- Fe and Si are deliberately added, which will strengthen the alloy due to the formation of Mg_2Si and Fe_3Si precipitates upon cooling (precipitation hardening).
- In the presence of Mn, the hardening effect is even more pronounced, due to the formation of FeMnAl_6 crystallites.



Precipitation hardening (heat treatment)

- Some of the Al-alloys may not be heat treated, because precipitates will deleteriously form along grain boundaries.
 - pure Al (*i.e.*, containing trace dopants),
Al-Mn,
Al-Si,
Al-Mg alloys.
- Some of the Al-alloys greatly strengthened by heat treatment through precipitation hardening.
 - Cu-Al (CuAl_2 precipitates),
Al-Zn-Mg (MgZn_2 precipitates),
Li-Al alloys (Al_3Li precipitates).
- As with all precipitation-hardening techniques, the size and dispersion of the crystallites must be carefully controlled through the heating/cooling regime.

C. Refractory metals

Refractory metals exhibit low thermal and electrical conductivities and have equally low thermal expansion properties.

Table 3.6. Properties of the Refractory Metals

Metal (Lattice)	Density (kg m ⁻³)	Melting point (°C)	Resistivity (μΩ cm ⁻¹)	Thermal conduct. (W m ⁻¹ K ⁻¹)	CLTE ^a (μm m ⁻¹ K ⁻¹)
Ti (HCP)	4,540	1,668	42.0	21.9	8.35
Zr (HCP)	6,506	1,852	42.1	22.6	5.78
Hf (HCP)	13,310	2,233	35.5	22.3	5.90
V (BCC)	6,110	1,915	25.0	30.7	8.40
Nb (BCC)	8,570	2,230	15.2	53.7	7.10
Ta (BCC)	16,654	2,996	13.2	57.5	6.60
Cr (BCC)	7,140	1,900	13.0	93.9	4.90
Mo (BCC)	10,220	2,610	5.70	139	5.43
W (BCC)	19,300	3,407	5.65	174	4.59
Re (HCP)	21,010	3,270	13.5	48.0	6.70
Ir (FCC)	22,650	2,410	5.30	146	6.40
Os (HCP)	22,590	3,054	8.12	87.6	4.57

^aCoefficient of linear thermal expansion.



Properties

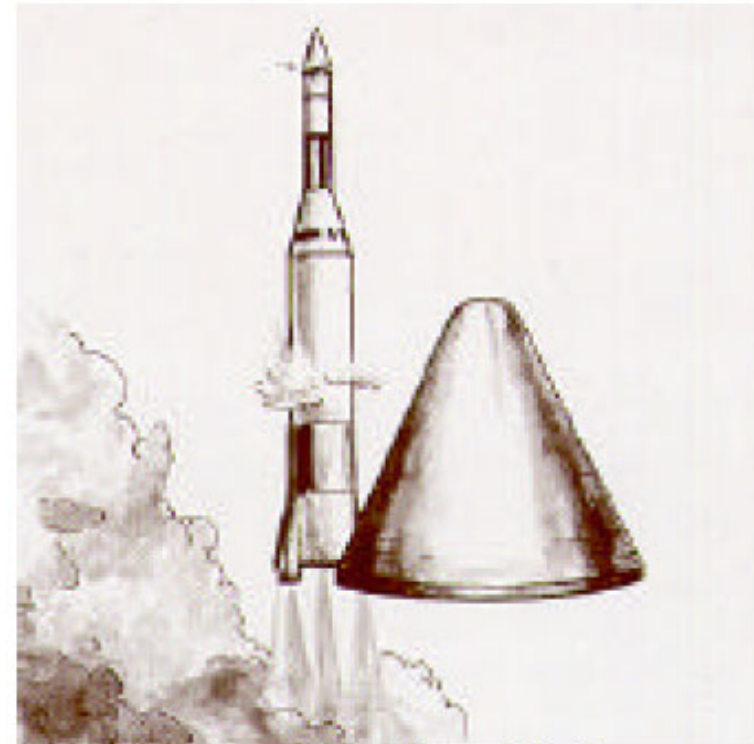
- Extremely high melting points ($> 1650\text{ }^{\circ}\text{C}$)
- Low thermal and electrical conductivities
- Low thermal expansion
- High hardness
 - these metals are all soft and ductile in their pure states. However, the metals are rarely obtained in their pure forms, since they spontaneously react with C, B, O, N, and other nonmetals to form stable interstitial compounds.
- Incompressibility
- High corrosion resistance
 - Since refractories show a high reactivity toward constituent gases of the atmosphere, the metals in their finely divided forms are highly pyrophoric and must be handled within an inert-atmosphere glove box.



Very high melting point

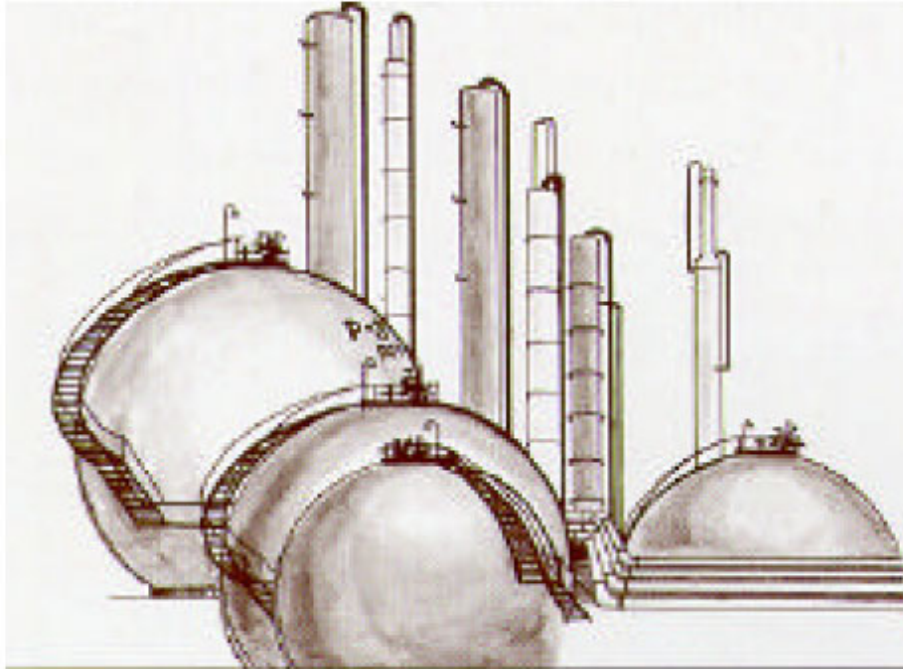
(2468°C/4474°F to 3410°C/6170°F):

High melting points of tungsten, tantalum and molybdenum make them useful in processing molten metals and minerals such as glass making.

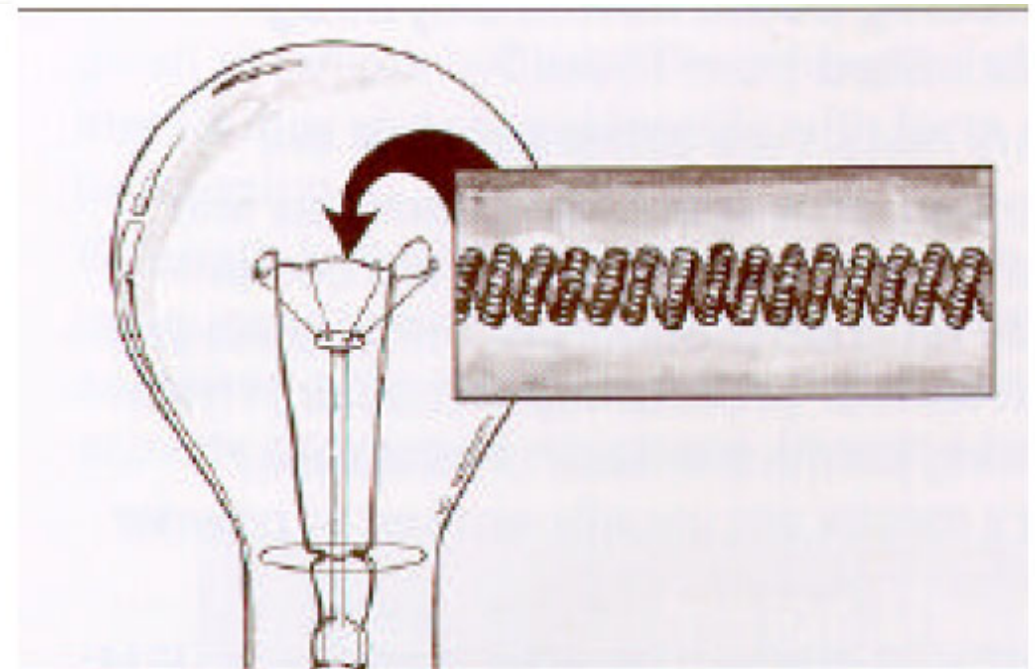


Excellent strength at high temperatures.

Even when heated to 1000°C (1832°F) tungsten rocket nose cones still have twice the tensile strength iron has at room temperature.



Exceptional resistance to corrosion. Stainless steel piping in chemical plants contains refractory metals.



High resistance to thermal shock. The stresses of rapid expansion due to heat would destroy most metal filaments in just a few on-off cycles. A tungsten filament, because of its high melting point and good non-sag characteristic, will withstand thousands of on-off cycles and still remain intact.



Hardness: Cutting tools today are made of tungsten carbide. For cutting, forming steel and other metals...even for drilling oil wells and in mining.

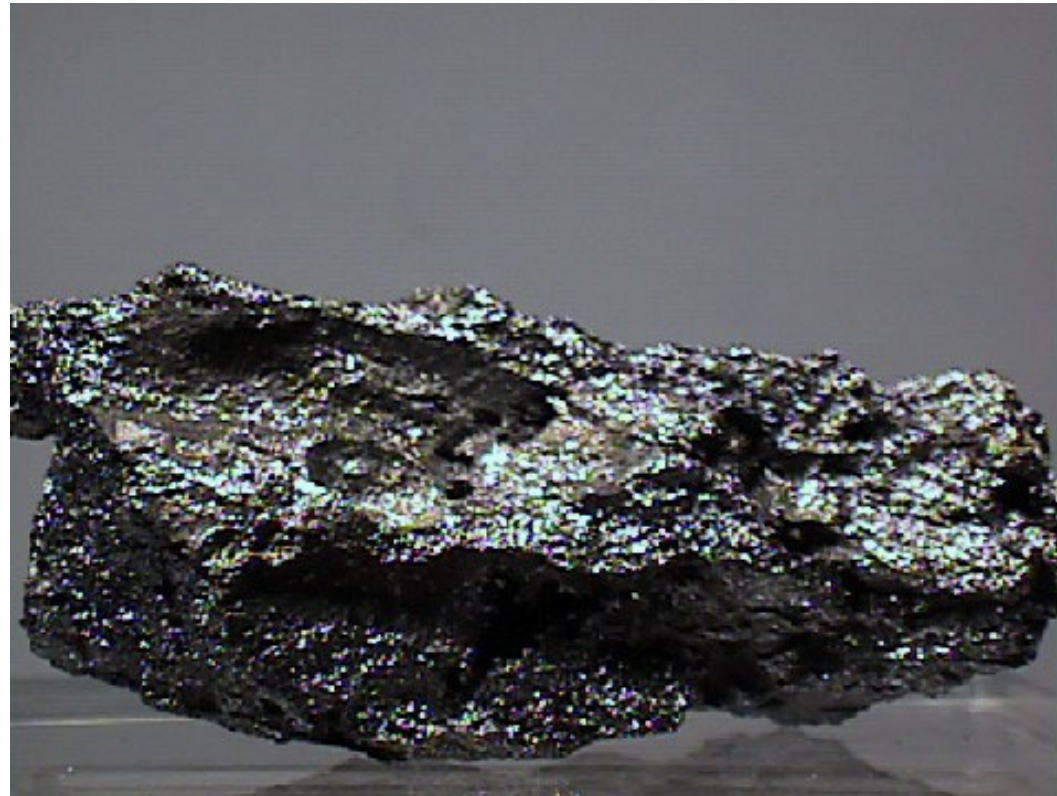


Excellent wear and abrasion resistance: Refractory metals, often in alloy form extend the life of seals, bushings, nozzles, valve seats and many points of high wear. Alloys with gold and silver also make excellent long-life contact points or electronic equipment.



Titanium (Ti)

- High strength/weight ratio
- Superior corrosion resistance





Properties

VERY STRONG – as strong as steel but 45% lighter.

RESISTS CORROSION – more resistant to acids and alkalis than aluminium, iron and magnesium.

LOW DENSITY – so **LIGHT** and **EASY TO SHAPE**

VERY HIGH MELTING POINT 1800°C

CAN WITHSTAND EXTREMES OF TEMPERATURE when
ALLOYED with aluminium and vanadium



- It's **strength**, **light weight** and **resistance to corrosion** make titanium alloy a good material to use in biomedical components e.g. hip replacements

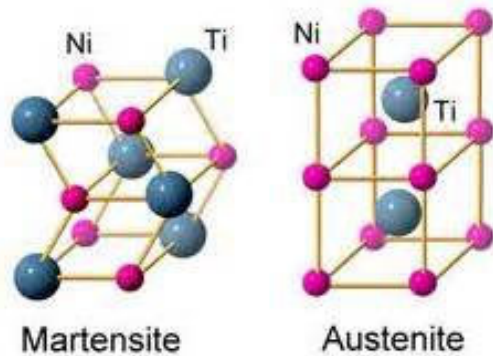


- Because of its **strength** combined with **light weight**, titanium alloy is also used to make: Aerospace and marine.



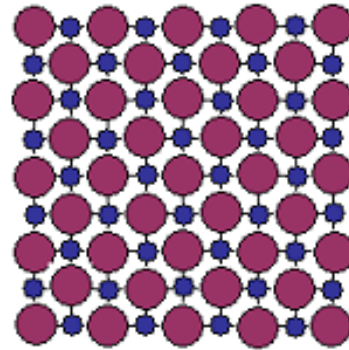
D. Shape-memory alloys

- Shape Memory Alloys (SMAs) are metallic alloys that undergo a solid-to-solid phase transformation which can exhibit large recoverable strains.
- First record of shape memory transformation was found in 1932 (AuCd).
- Shape memory effect was discovered in Nickel-Titanium at the Naval Ordnance Laboratory (NiTiNOL)
- Pseudoelasticity



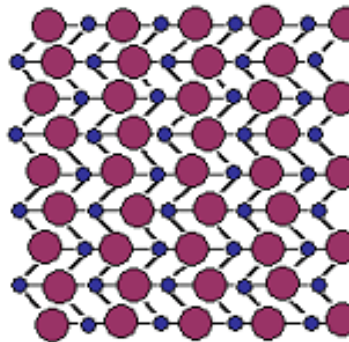
Austenite

- High temperature phase
- Cubic Crystal Structure

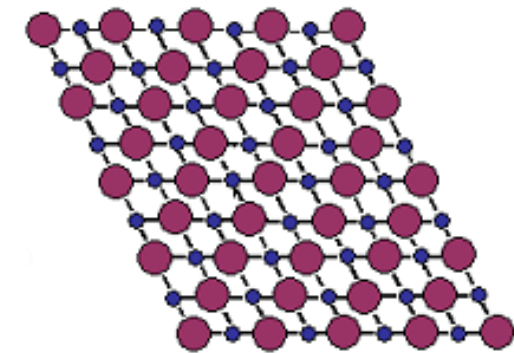


Martensite

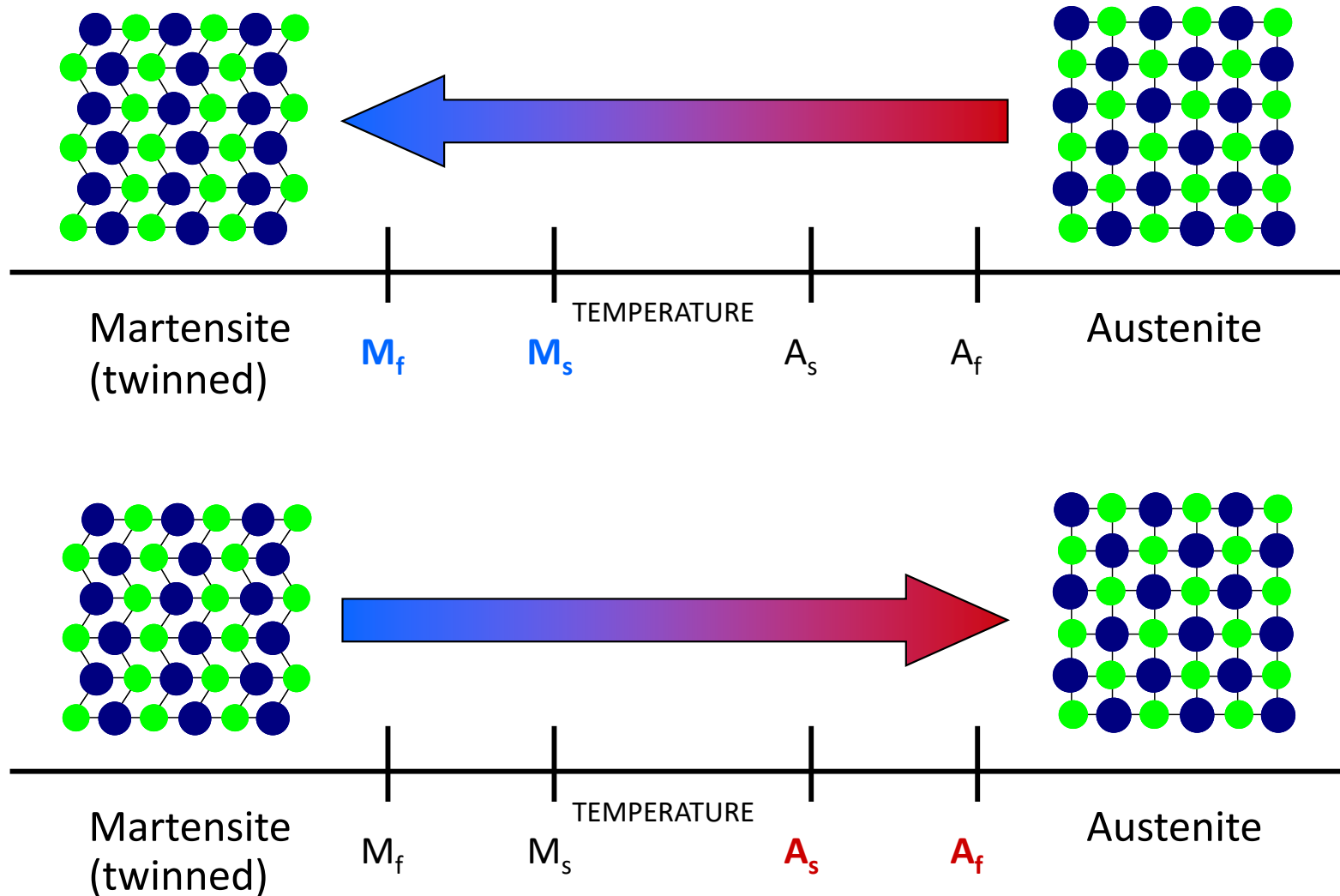
- Low temperature phase
- Monoclinic Crystal Structure



Twinned Martensite



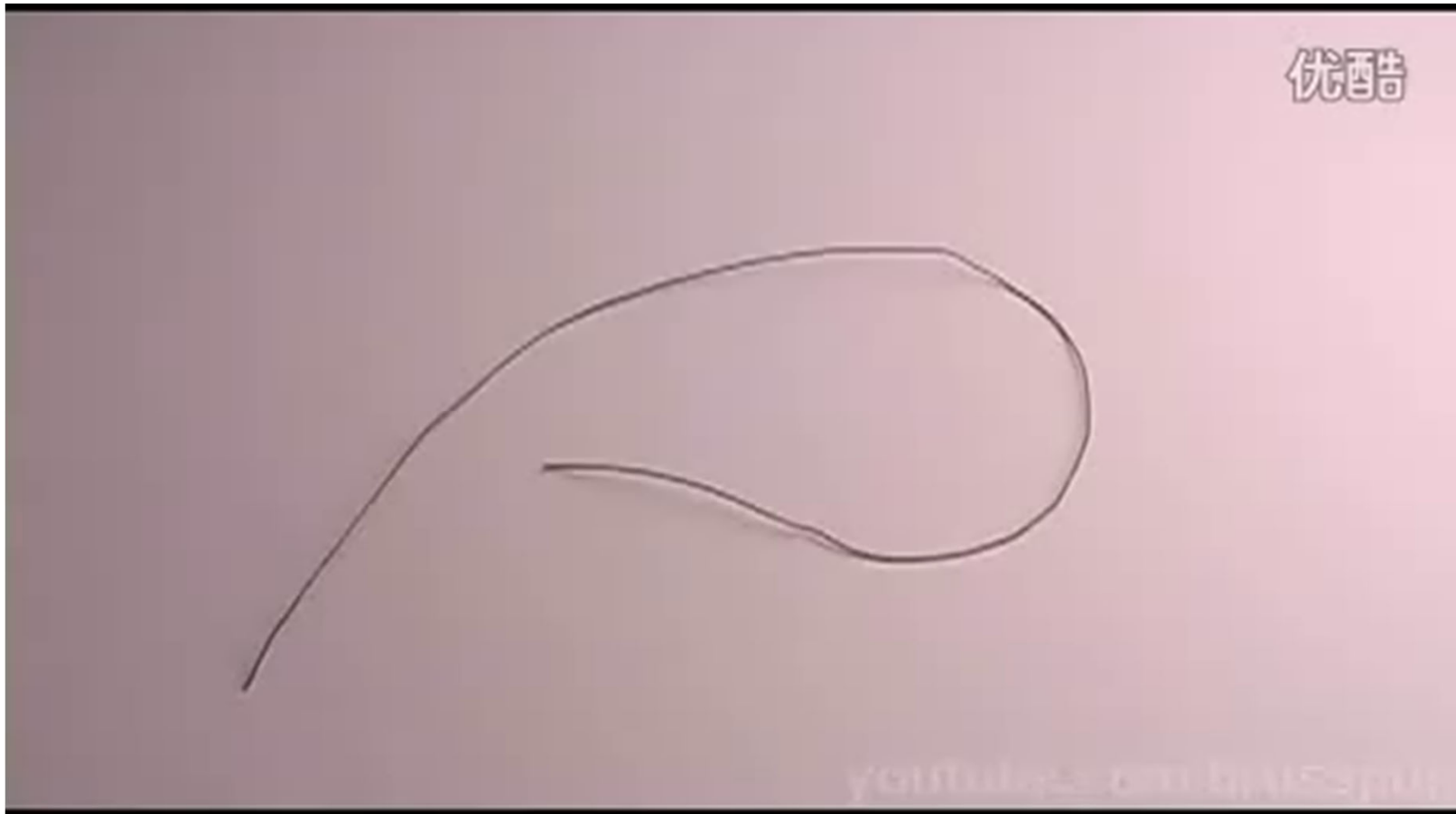
Detwinned Martensite



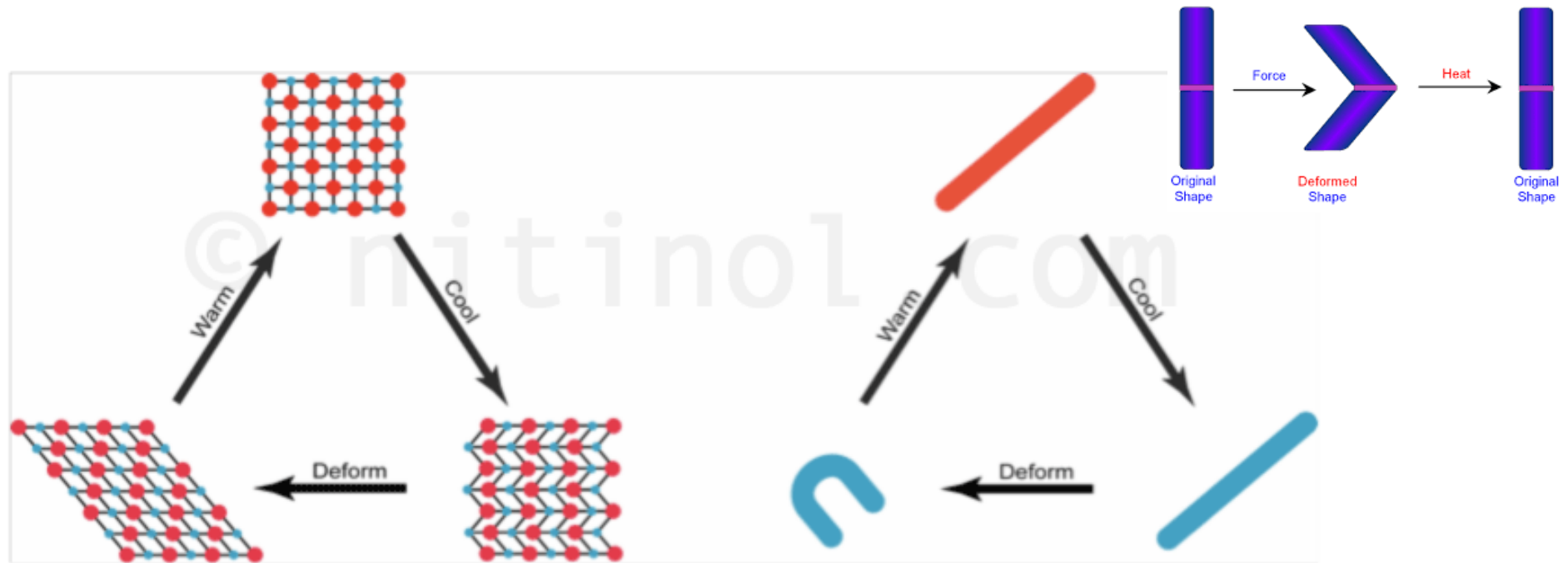
Since both austenite and twinned martensite have the same macroscopic shape/size, reheating the material will not result in any observable shape change.



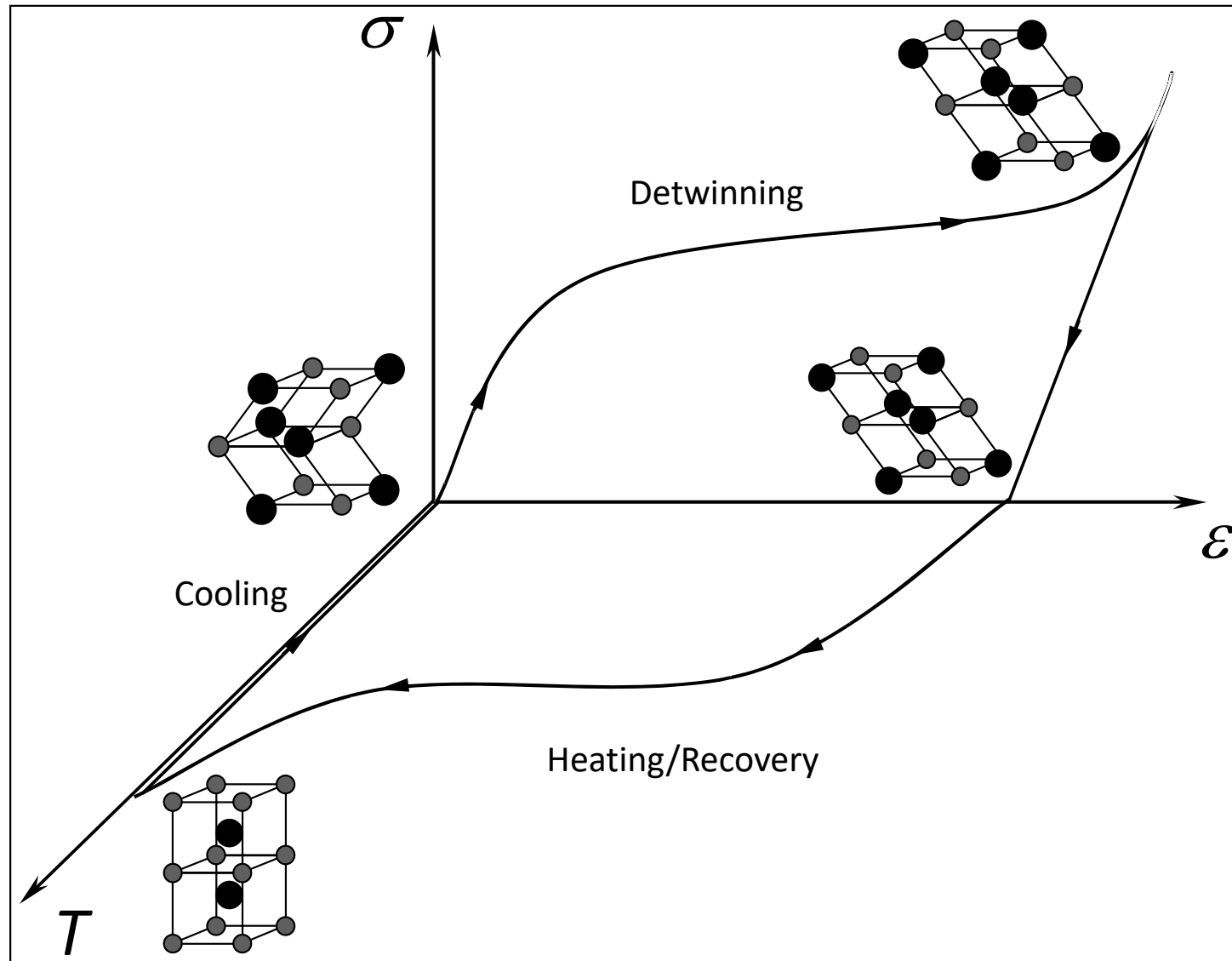
What are Shape Memory Alloys?



Shape Memory Effect:

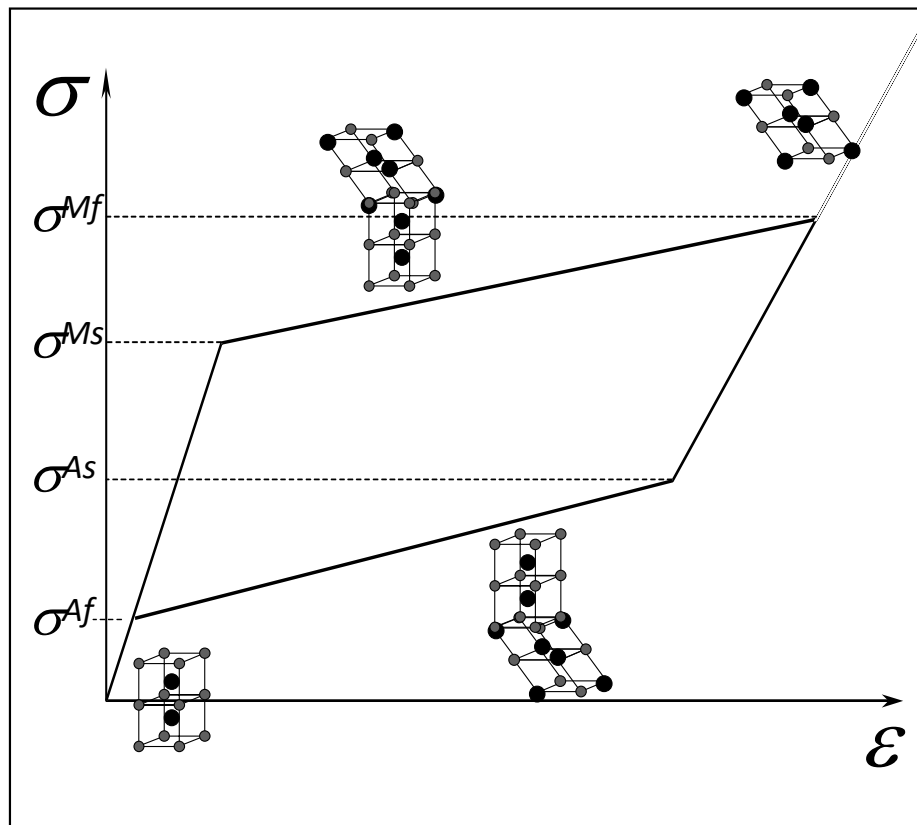


If the material is plastically deformed through bending, *etc.* at low temperature, it will become detwinned and the new shape will prevail.



The Pseudoelastic Effect

What will happen if a stress is applied to the material in its high-temperature austenitic phase?



- When the alloy is above the A_f , a load strong enough will force the austenite into the martensite phase.
- Since the temperature is above A_f , the original shape will be reformed immediately after the load is removed.
- Such an immediate shape change is referred to as pseudoelasticity



Applications: Robotic



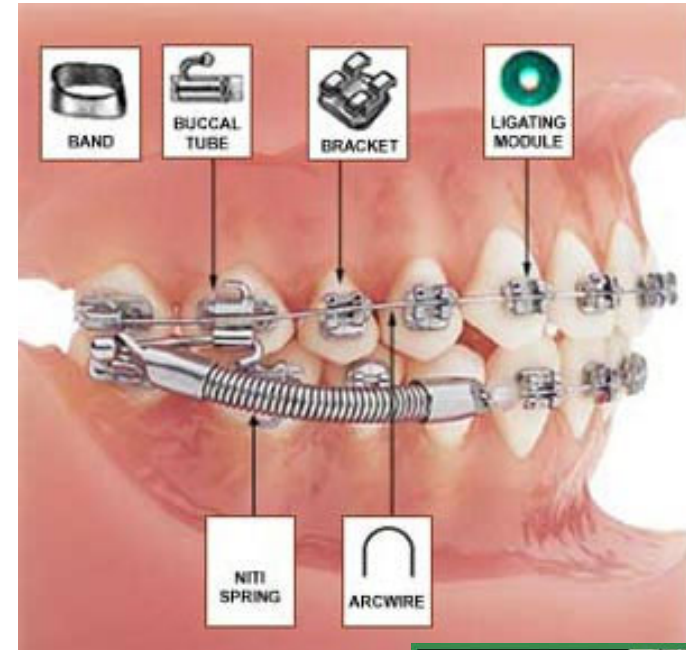
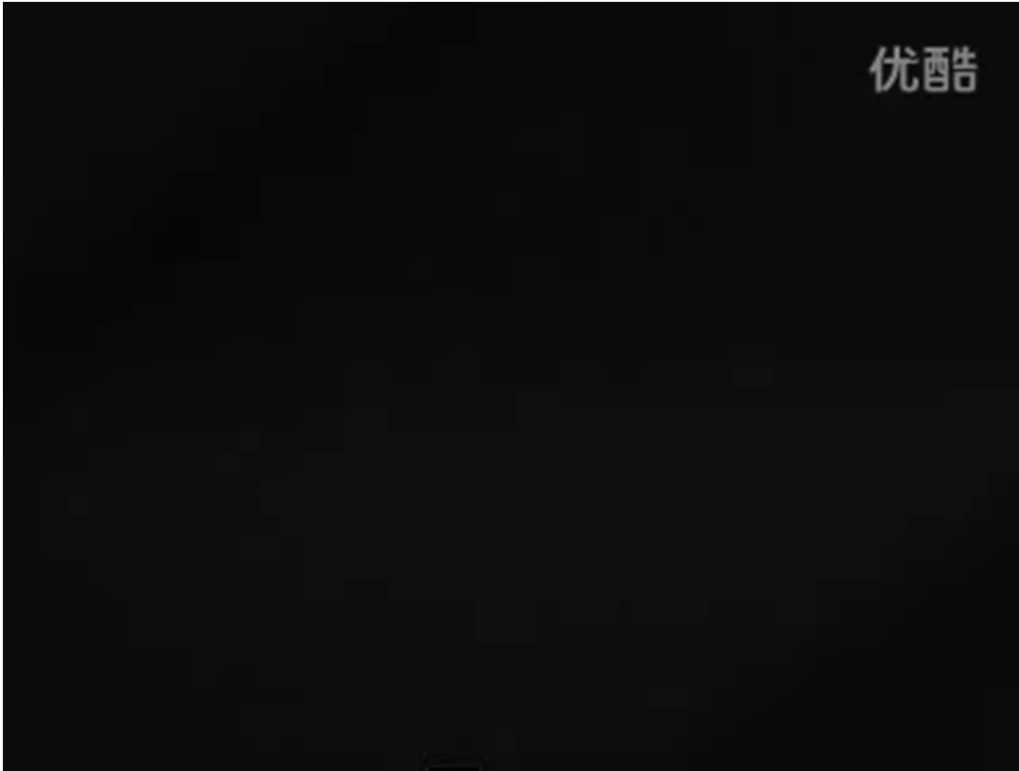
The implementation of SMA wires coupled with a simple DC control system can be used to drive small objects without the addition of relatively heavy motors, gears, or drive mechanisms.



Wires have the ability to flex the robotic muscles according to electric pulses sent through the wire.



Applications: Medical



Medical Stents





Applications: Automobile



An intriguing future application will be for automobile frames. Someday soon, your car may reshape itself in front of your eyes within minutes after an accident!



3.3. Metal surface treatments for corrosion resistance

The primary corrosive processes:

- Oxidizing agents (e.g., moist air, HNO_3 , H_2SO_4)
- Halogenated species (e.g., Cl_2 , HCl , HF , CFCs)
 - These agents degrade the metal by forming oxides, hydroxides, or halides that introduce embrittling grain boundaries on the surface.
- Detrimental gases (e.g., CO and H_2S)
 - Both of these gases exhibit dissociative adsorption on metal surfaces, resulting in carbide or sulfide formation and concomitant embrittlement of the metal.



Three strategies that may be used to protect a metal surface from its environment:

- inert-layer passivation
- sacrificial metal coatings
- organic-based coatings

Pretreatment process is often required for metal surfaces to allow the coatings to be strongly adsorbed to the surface:

- mechanical descaling (*e.g.*, abrasive blast techniques)
- chemical pickling (*i.e.*, acid treatment)

Corrosion resistance technique: **phosphating**

Phosphate coatings are used on steel parts for corrosion resistance, lubricity, or as a foundation for subsequent coatings or painting.

It serves as a conversion coating in which a dilute solution of phosphoric acid and phosphate salts is applied via spraying or immersion and chemically reacts with the surface of the part being coated to form a layer of insoluble, crystalline phosphates.

- Vivianite $[\text{Fe}_3(\text{PO}_4)_2 \cdot 4\text{H}_2\text{O}]$
- Hopeite $[\text{Zn}_3(\text{PO}_4)_2 \cdot 4\text{H}_2\text{O}]$
- Phosphophyllite $[\text{Zn}_2\text{Fe}(\text{PO}_4)_2 \cdot 4\text{H}_2\text{O}]$
- Crystalline pseudophosphophyllite $[(\text{Zn}, \text{M}, \text{Ni})_3(\text{PO}_4)_2 \cdot 4\text{H}_2\text{O}]$, where $\text{M} = \text{Fe}, \text{Al}, \text{Cr}, \text{etc.}$

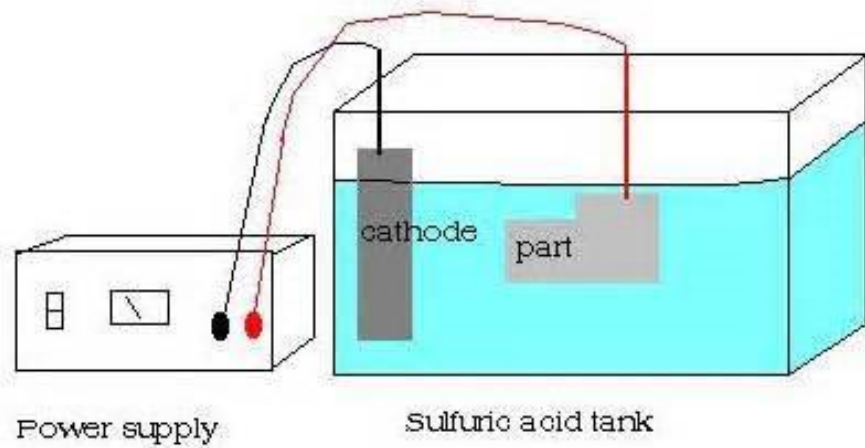




Corrosion resistance technique:

Anodizing or anodic oxidation

- In this method the metallic surface acts as an anode, being oxidized during an electrochemical event.
- The most common metals/alloys are those containing aluminum, magnesium, and zinc. However, it is also possible to anodize other metals such as copper, steel, and cadmium for protective and decorative applications.
- The anodizing electrolytic solution consists of strong acids, generally combinations of chromic, sulfuric, oxalic, or boric acids.





Other nonmetallic coatings

- Common **inorganic-based coatings** include vitreous enamels, ceramics, glass, cements, carbides, and nitrides.
- **Organic-based protectants** are paint coatings, plastic coatings, adhesive tapes, and sheet linings.
- **Refractory coatings** such as carbides (e.g., TiC, B₄C, WC, and WCO), nitrides (e.g., AlN and BN), oxides (e.g., Al₂O₃, BeO, Cr₂O₃, ThO₂, and ZrO₂), silicides (e.g., NbSi₂, WSi₂, and MoSi₂), and borides (e.g., ZrB₂ and TiB₂) impart both corrosion/abrasive wear and temperature resistance to the underlying substrate.



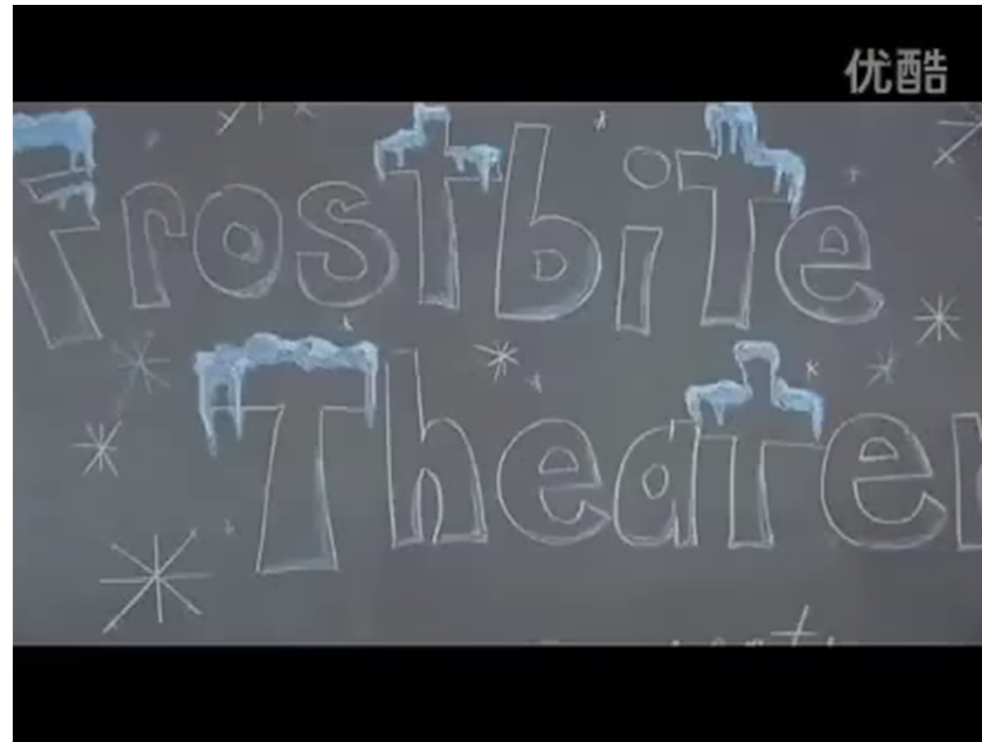
- The **composite coatings**, comprising the above refractory compounds in association with a metal powder (e.g., Cr + ZrB₂, Cr + HfO₂, Al + SiC, Ti + TiB₂, and PtRh+ZrB₂). For these coatings, the ceramic and metal powders are suspended in an aqueous solution with the assistance of surfactants, and sprayed onto the metal surface.



- The **simple painting**:
 - ✓ Paints comprise finely divided solid inorganic or organic pigments that are suspended in association with binder molecules within a volatile solvent.
 - ✓ The volatile medium comprises solvents that are used to solubilize the binder and dilutants whose purpose is to place the paint at a suitable viscosity for its application.
 - ✓ Additives such as antifungal agents, driers, *etc.* are often used to broaden the application of the coating.



3.4. Magnetism in metals and alloys





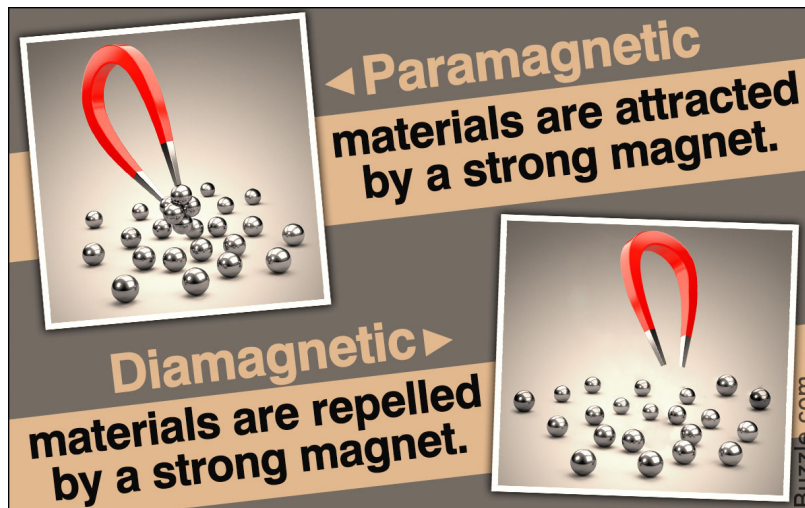
Magnetic susceptibility, χ , is a dimensionless proportionality constant that indicates the degree of magnetization (B) of a material in response to an applied magnetic field (H).

$$\chi = \frac{B}{H}$$

The susceptibility indicates whether a material is attracted into or repelled out of a magnetic field, which in turn has implications for practical applications.



Magnetic Property	Direction of Polarization (I) Relative to External Field	Relative Magnetic Susceptibility (χ) in ppm	Typical Materials
Diamagnetism	Opposite	-10	Water, fat, calcium, most biologic tissues
Paramagnetism	Same	+1	Molecular O ₂ , simple salts and chelates of metals (Gd, Fe, Mn, Cu), organic free radicals
Superparamagnetism	Same	+5000	Ferritin, hemosiderin, SPIO contrast agents
Ferromagnetism	Same	> 10,000	Iron, steel



Types of magnetism

	Diamagnetic	Paramagnetic
Electron pairing	$\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ No unpaired electrons	$\uparrow\downarrow \uparrow\downarrow \uparrow$ At least one unpaired electron
Spin alignment with magnetic field B	$B \rightarrow$ Anti-parallel	$B \rightarrow$ Parallel
Reaction to magnets	 Very weakly repelled	 Attracted
Effect on magnetic field lines	 Field bends slightly away from the material	 Field bends toward the material



- Paramagnetism
- Diamagnetism
- Ferromagnetism

Weak effects, not familiar to most people, but *all* materials exhibit one or the other.

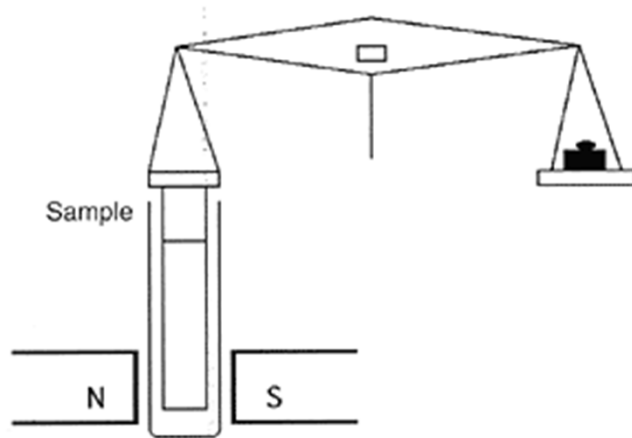
Strong effect:

Responsible for all permanent magnets.

Responsible for the strong magnetic properties of iron.

Only special materials (ferromagnetic).

Very familiar to everybody.



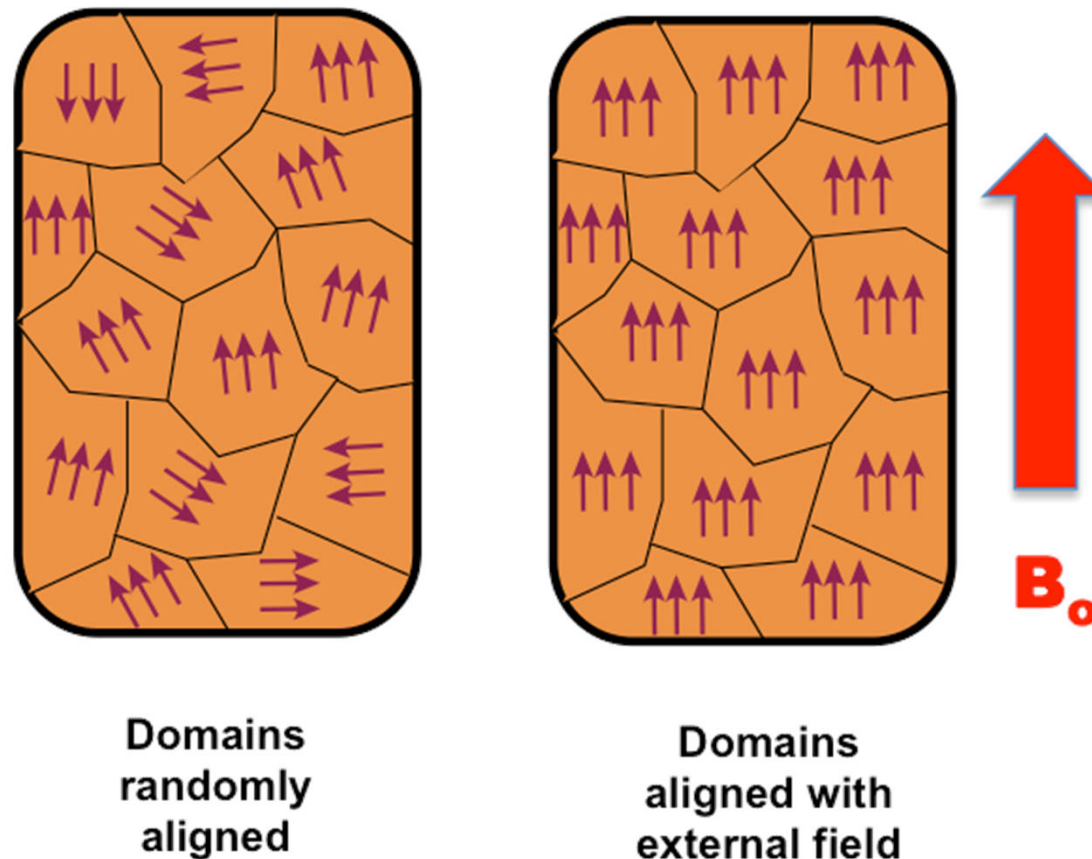
Magnetic susceptibility balance (MSB)

This technique measures the amount of repulsive (for diamagnetic) or attractive (for paramagnetic) force between the sample and a permanent magnet within the instrument.



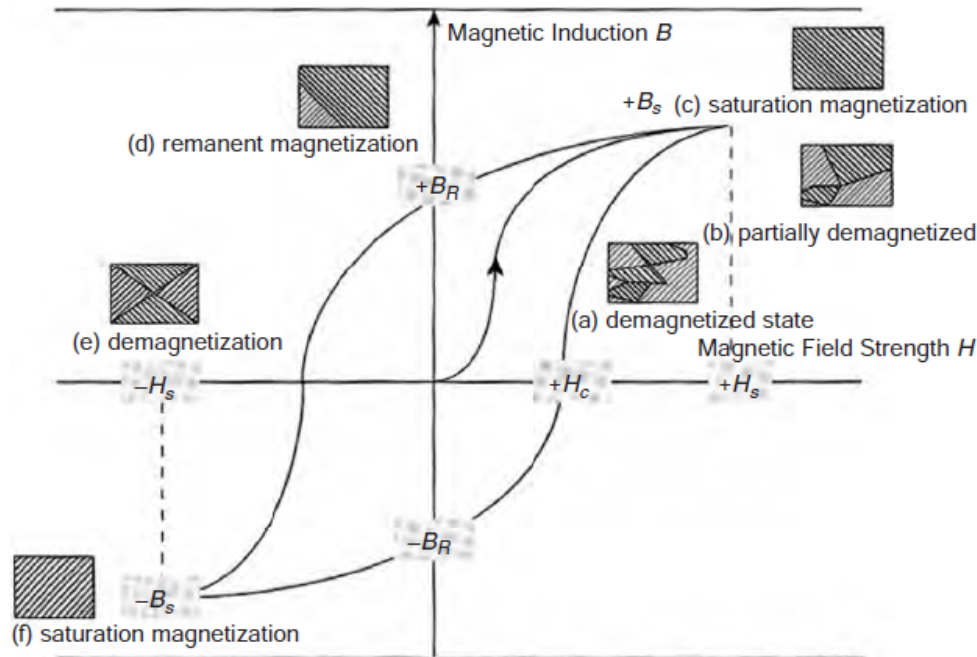
Ferromagnetism

- Ferromagnetism is only possible when atoms are arranged in a lattice and the atomic magnetic moments can interact to align parallel to each other.
- The presence of magnetic domains within the material, which are regions where the atomic magnetic moments are aligned. The movement of these domains determines how the material responds to a magnetic field and as a consequence the susceptibility is a function of applied magnetic field.
- Therefore, ferromagnetic materials are usually compared in terms of saturation magnetisation (magnetisation when all domains are aligned) rather than susceptibility.



Magnetic domains are small, randomly oriented and spontaneously magnetized regions in a ferromagnetic material whose spins are locked into orderly arrays. Application of small external fields moves the walls between them to form larger domains. Higher external fields reorient the domains to face the same direction and increase net magnetization.

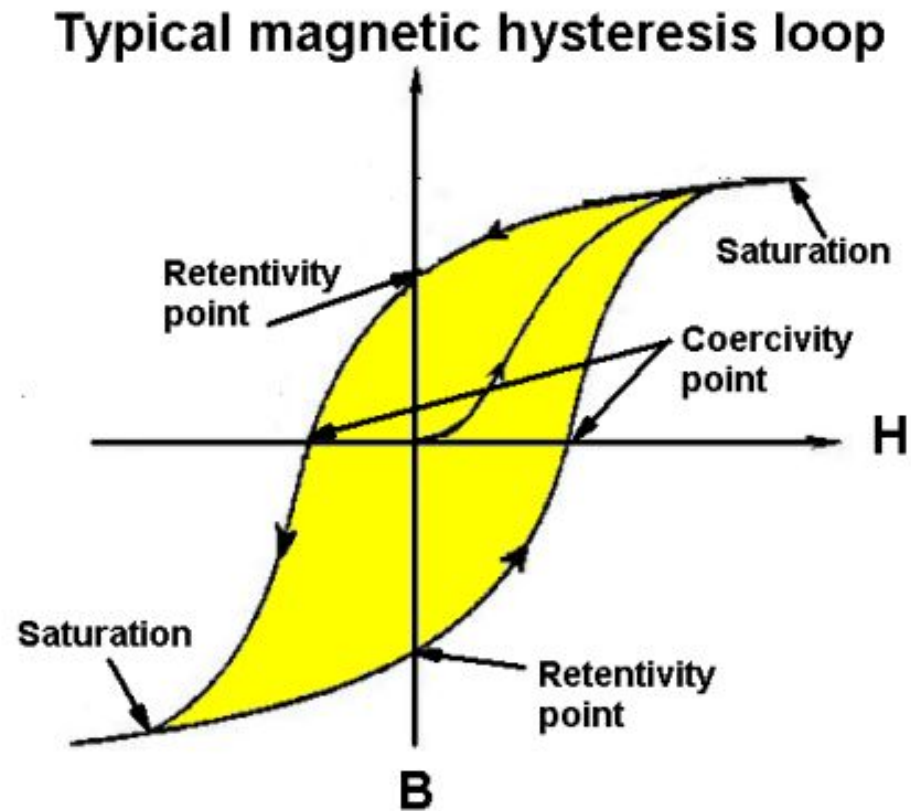
Magnetic hysteresis



H: the applied magnetic fields
B: the induced magnetic fields

- When the external magnetic field reaches a maximum value (H_s), the material will form a single domain with a net saturation magnetization (B_s) in a direction parallel to the applied field.
- Once the external field is removed from the material, the domain walls spring back toward their original positions, and the magnetization decreases to a value referred to as the remanence (B_R).
- In order to remove the induced magnetism of the material, an opposite magnetic field is applied, known as the coercive magnetic field (H_c). If the magnitude of the opposed field is increased to a maximum value ($-H_s$), a saturation magnetization will again be found in the material, parallel to the new direction of the applied field.

Hysteresis

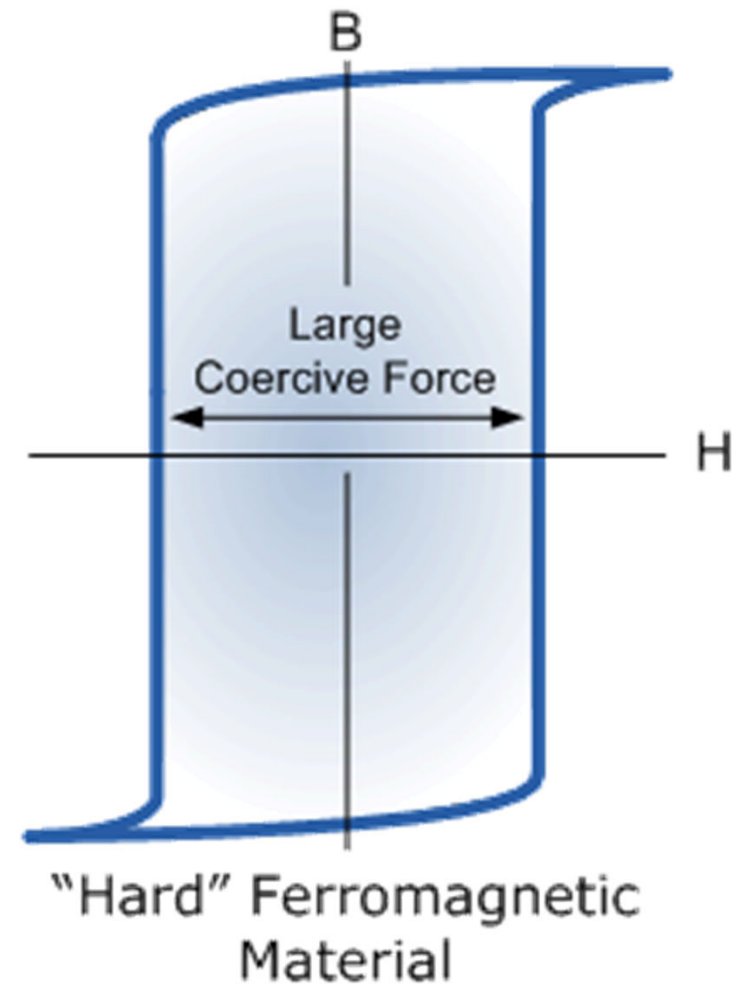
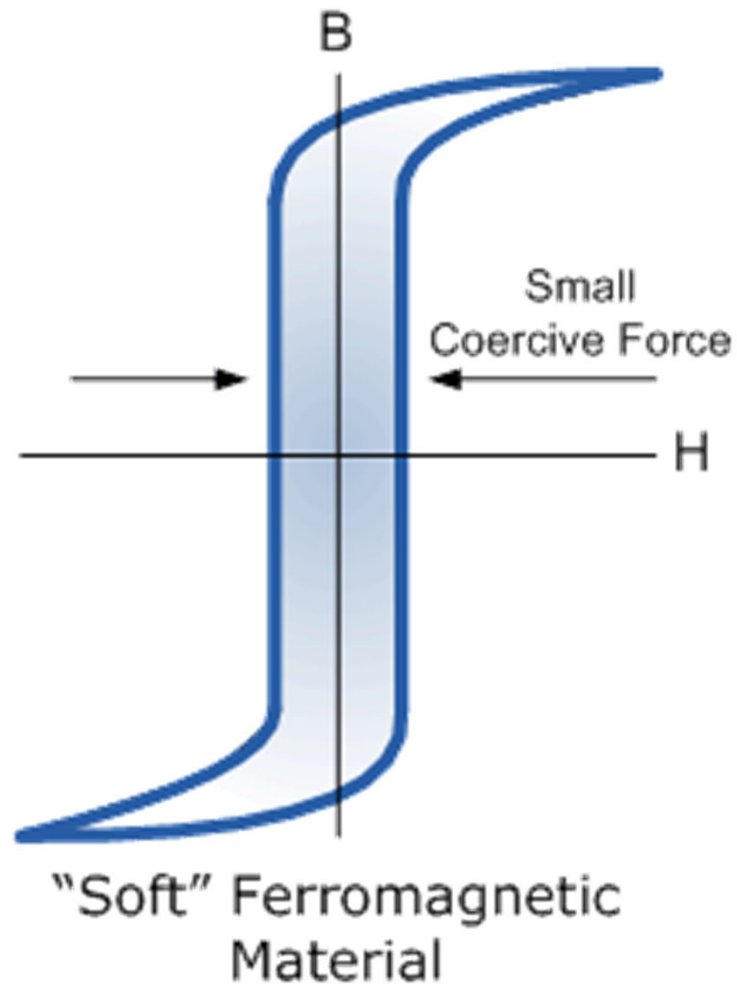


H_s : maximal magnetic field

H_c : coercive magnetic field

B_s : saturation magnetization

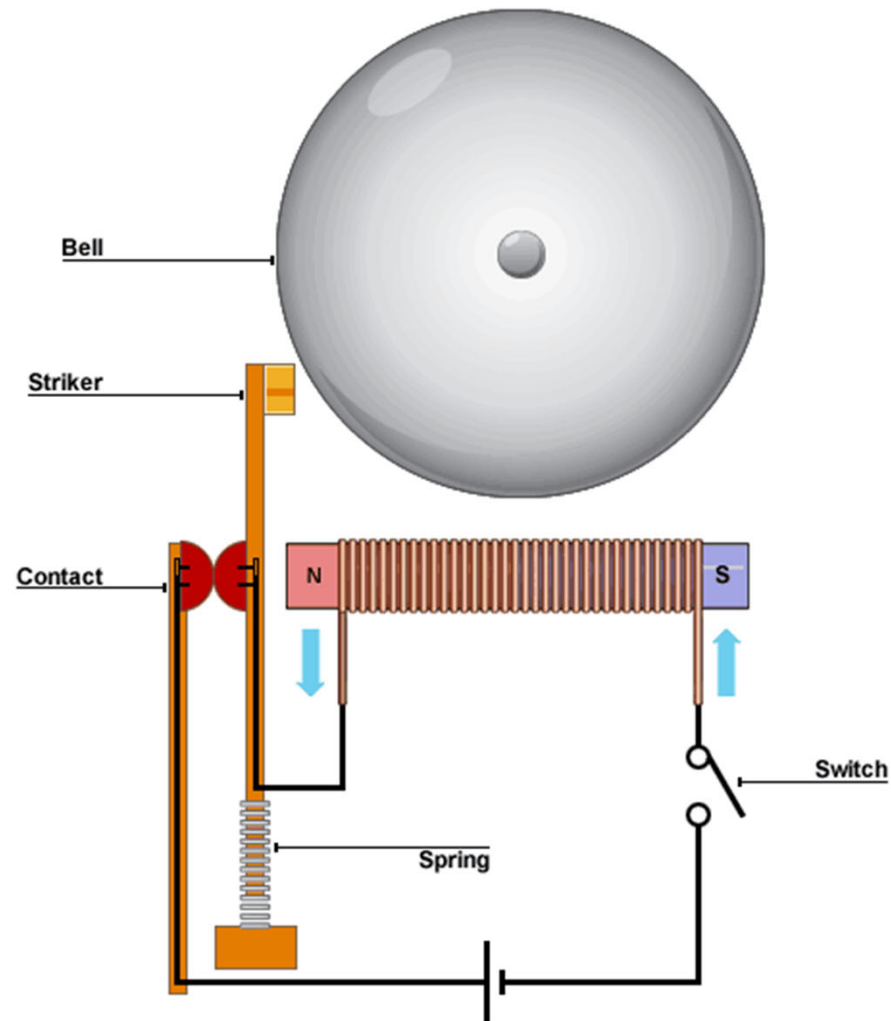
B_R : remanent magnetization





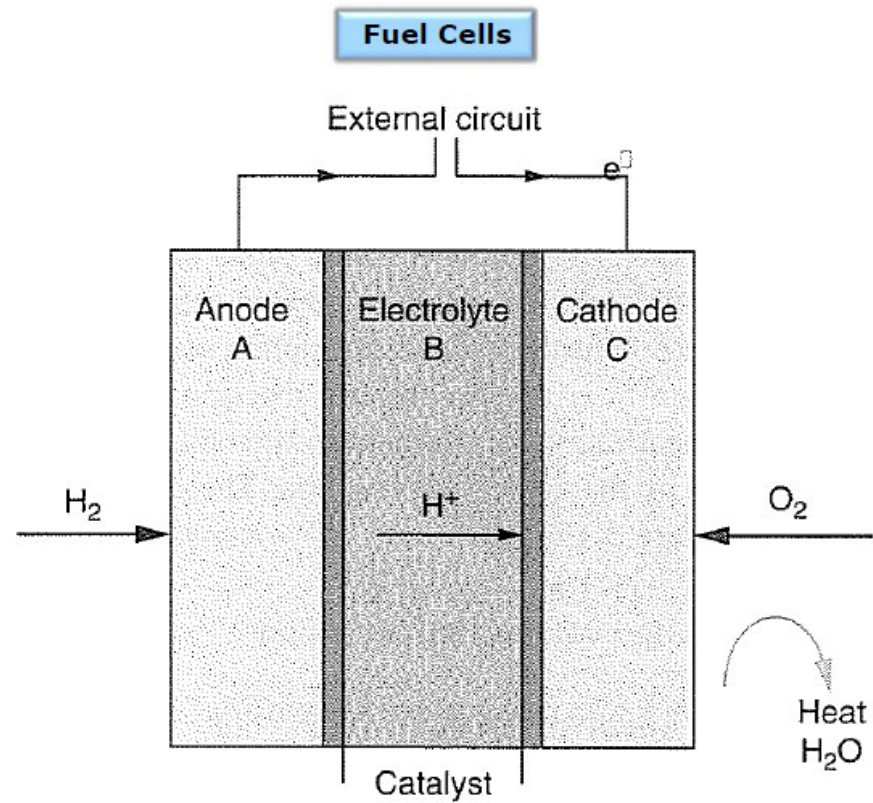
Hard Magnetic Material	Soft Magnetic Material
Coercivity & Retentivity are large	Coercivity & Retentivity are small.
Cannot be easily magnetized & demagnetized	Can be easily magnetized & demagnetized.
Have small values of permeability and susceptibility	Have large values of susceptibility and permeability.
Used to make permanent magnets.	Used to make electromagnets.
Iron-nickel-aluminum alloys, copper-nickel-iron alloys, copper-nickel-cobalt alloys	Iron- silicon alloys, ferrous- nickel alloys, ferrites, garnets.

The electrical bell





3.5. Reversible hydrogen storage



Hydrogen combines with many elements to form binary hydrides, MH_n . There are three general classes of hydrides:

1. Saline or Binary (involving Group 1 and 2 metals; may be envisioned as an ionic lattice consisting of M^{n+} and H^- ions, e.g., LiH , NaH , BaH_2)
2. Covalent (molecules containing covalently bound hydrogen to nonmetals, with individual molecules held together by intermolecular forces, e.g., CH_4 , SiH_4)
3. Metallic/interstitial (hydrogen molecules are contained in vacant interstitial sites of a transition-metal lattice, e.g., $PdH_{0.6}$)



To be successful for hydrogen storage in energy devices, the following five parameters must be met:

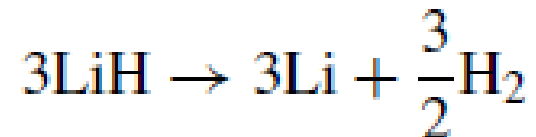
1. The solid material must be able to adsorb/desorb at least 6.5 wt% and 65 g L⁻¹ of hydrogen gas.
2. The decomposition temperature necessary for generation of hydrogen from the material should be in the range of 60–90 °C.
3. The absorption/desorption of hydrogen from the material should be reversible.
4. The material should be low cost, precluding the use of noble-metal alloys.
5. The storage solid should be nontoxic and inert under environmental conditions. That is, the solid should not react with water, oxygen, nitrogen, *etc.*



- To date, no material (neither metals nor nonmetals) has been discovered that satisfies all of the above five constraints. Although hydrides exist for most elements of the Periodic Table, only the light elements (*e.g.*, Li, Mg, Al) are able to meet criterion 1 above.
- It has recently been discovered that decreasing the particle size of metal alloy particles through ball-milling processes will increase the adsorption kinetics by an order of magnitude.



NaAlH_4 is most attractive for hydrogen storage applications due to its relatively low H_2 desorption temperature (80°C)





- There continues to be significant research efforts devoted to the design of suitable catalysts that will improve the relatively slow kinetics associated with the reversible H_2 storage of $MAI H_4$ compounds. Intermetallic compounds doped with Ti and Zr have been successfully used to improve the rate of hydride formation/release in these compounds.
- Sometimes catalytic dopants do not alloy with the host metal; for example, the addition of Nb and V to Mg form heterogeneous mixtures rather than intermetallic compounds.



Summary

1. Extraction of metals
2. Powder metallurgy

Powder production; Blending; Compaction;
Sintering; Finishing operations

3. Phase behavior of iron-carbon alloys

α – *Fe(ferrite)*; γ – *Fe(austenite)*; δ – *Fe*;
cementite (Fe_3C)

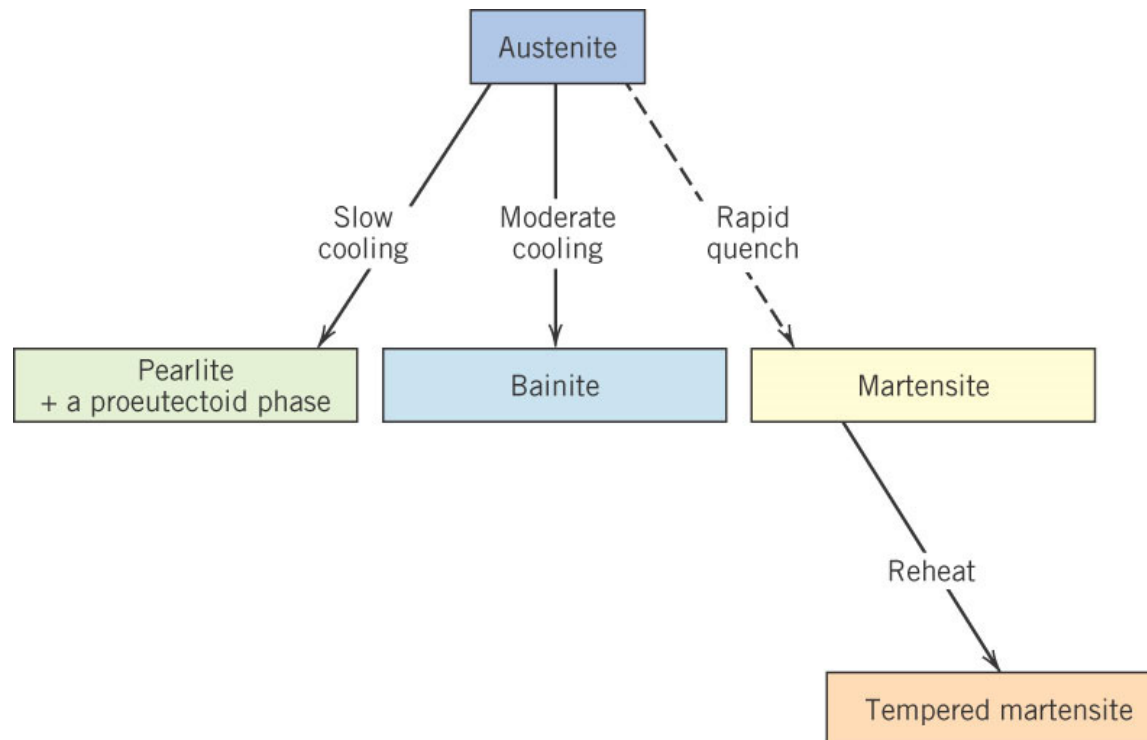
Steel (pearlite)

cast iron (ledeburite)



4. Hardening mechanisms of steels

- grain size reduction,
- solid-solution alloying,
- Work hardening,
- Precipitation hardening,
- Transformation hardening.





5. Stainless steels

Fe C Cr alloy

6. Other metals

---coinage metals

---aluminum alloys

---refractory metals

---shape-memory alloys (Ni-Ti)

7. Corrosion resistance treatment

8. Magnetism in metals and alloys